

# CMS ACCESS Model API

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## Operations Manual – Version 0.9.8 DRAFT

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## General Guidance

This Operations Manual provides general implementation guidance applicable to all ACCESS Model APIs.

### Common Patterns

All ACCESS Model APIs follow consistent asynchronous processing and parameter patterns for ease of implementation. The Alignment API uses an asynchronous request-response pattern to accommodate the processing time required for backend logic. This pattern allows clients to submit requests and subsequently poll for results without maintaining a persistent connection. The pattern uses two sets of operations to support a submit-and-poll workflow:

1. **\$check-eligibility, \$align, \$unalign, \$report-data:** Submits a request to the API
2. **\$submission-status:** Polls for the status of a submitted request

### Why Asynchronous?

ACCESS operations can take some time to complete, making a synchronous request-response pattern impractical for real-world implementations.

Processing these operations may require additional steps, such as:

- Conducting additional data validation specific to the ACCESS Model
- Processing complex composition structures with nested sections
- Integrating with multiple backend systems to evaluate ACCESS Model requirements
- Applying control group randomization algorithms

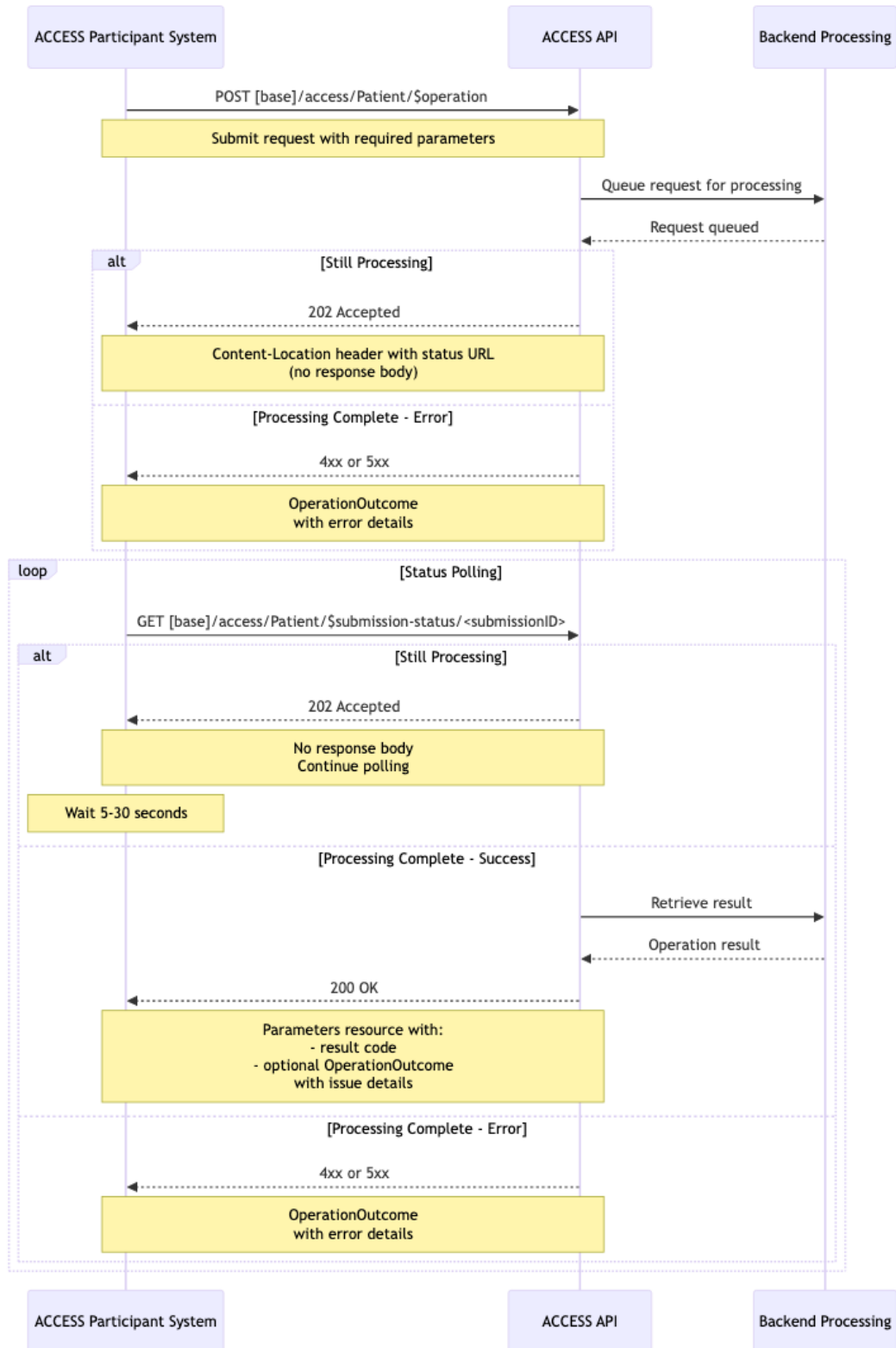
### Workflow Overview

The typical workflow consists of the following steps:

1. **Submission:** The client submits a `$check-eligibility`, `$align`, `$unalign`, `$report-data` operation
2. **Acknowledgment:** The server returns `HTTP 202 Accepted` with a `Content-Location` header containing the URL for retrieving submission status
3. **Polling:** The client periodically polls using the `$submission-status` operation through the submission status URL provided
4. **Completion:** When processing completes, the submission status returns `HTTP 200 OK` with the final result

### Interaction Pattern:

The following sequence diagram illustrates the general asynchronous pattern used by all ACCESS APIs:



## Required Parameters

All primary operations (`$check-eligibility`, `$align`, `$unalign`, `$report-data`) require these common parameters:

- **participantID** - ACCESS participant identifier. This identifier is supplied to the ACCESS participant during the Request for Application (RFA) process and takes the form of "ACCES<5-digit number>" (ACCES00001, ACCES12345)
- **payerID** - The payer ID for the patient. Uses the X12 EDI Payer ID standard (`urn:oid:2.16.840.1.113883.3.221.5`) with [CARIN Blue Button identifier typing] (<http://hl7.org/fhir/us/caribb/CodeSystem/C4BBIdentifierType>). EDI Payer IDs are 5-character alphanumeric codes widely used in healthcare transactions (e.g., eligibility checks, claims, remittance).
- Patient profile instance, containing the `identifier`, `name`, `gender` of the patient - this information conforms with the US Core Patient Profile requirements. The `$check-eligibility`, `$align`, and `$unalign` operations have a patient parameter. The `$report-data` operation requires the Patient profile instance to be included in the `dataBundle` parameter instead of as a separate top-level parameter.
- **track** - One of the ACCESS Model tracks (eCKM, CKM, MSK, BH)

Some operations may require additional parameters. Please consult the documentation in this guide for information about each specific API.

## `$submission-status` Operation

The `$submission-status` operation retrieves the current status of a previously submitted request. This is a **read-only operation** that uses the HTTP GET method. The URL to use to retrieve the submission status is returned by the primary operation (extracted from the `Content-Location` URL).

**Submission Status URL:** GET <submission status URL>

**Output Parameters** (when complete):

- **result** (CodeableConcept, required): Result code with descriptive text indicating the status. Actual values depend on the API in use.
- **issues** (OperationOutcome, optional): This parameter **MAY** be included to provide more detail about the context of the result code.

## Example Response from a Primary Operation:

```
HTTP/1.1 202 Accepted
Content-Location: https://[base]/access/Patient/$submission-
status/sub-123456
```

(No response body - processing asynchronously)

## Implementation Guidance

### Base URL

All primary operations are invoked at the resource level:

```
POST https://[base]/access/[resource]/[$operation]?
      entityId=[participantID]
```

Where [base] is the FHIR server base URL provided during participant registration.

Where [resource] is Patient.

Where [\$operation] is the primary dollar sign operation (\$check-eligibility, \$align, \$unalign, \$report-data)

Where [entityId] is the Participant ID (e.g., of the form ACCES#####)

### Content Types

- **Request:** application/fhir+json
- **Response:** application/fhir+json

### Patient Identification

All APIs require a US Core Patient profile instance containing an **identifier**. Each API requires the patient's member ID (e.g., for CMS's ACCESS Model, the patient's MBI) in the Patient resource **identifier** data element. For CMS's ACCESS Model only, a patient must be a Medicare beneficiary to be eligible and align to a CMS ACCESS participant:

```
{
  "resourceType": "Patient",
  "identifier": [
    {
      "type": {
        "coding": [
          {
            "system": "http://terminology.hl7.org/CodeSystem/v2-
0203",
            "code": "MC"
          }
        ]
      },
      "system": "http://terminology.hl7.org/NamingSystem/cmsMBI",
      "value": "1EG4TE5MK73"
    }
  ],
  "name": [
    {
      "family": "Doe",
      "given": ["John"]
    }
  ],
  "gender": "male",
```

```
    "birthDate": "1950-01-01"  
  }
```

**NOTE:** The ACCESS Model leverages the US Core Patient FHIR profile, which has the following additional requirements beyond the Patient identifier:

**In addition to an identifier, each Patient must have:**

1. Patient name
2. Gender (male, female, other, or unknown)

**Each Patient must support the following data elements and SHOULD include them in submissions:**

1. Birth date

Please see the documentation on US Core Patient profile at <https://hl7.org/fhir/us/core/STU6.1/StructureDefinition-us-core-patient.html> for more information.

**NOTE:** The IG will accept any patient identifier and payer identifier so that this IG can be used by other payers looking to align with the ACCESS Model.

#### Using the Content-Location HTTP Header

ACCESS operations return a Content-Location HTTP header containing the URL to check the status of the submission. Clients should:

1. Extract the Content-Location header value from the HTTP response
2. Use this URL for subsequent \$submission-status calls

**Example:**

```
Content-Location: https://[base]/access/Patient/$submission-  
status/sub-123456
```

#### Polling Strategy

Clients should implement an appropriate polling strategy to balance responsiveness with server load:

- **Initial Poll:** Wait 5-10 seconds after submission before the first status check
- **Subsequent Polls:** Poll every 10-30 seconds while receiving HTTP 202 Accepted
- **Maximum Duration:** Implement a timeout (recommended: 5 minutes) after which the client should stop polling and notify the user; the actual timeout duration should be configured based on operational requirements
- **Exponential Backoff:** Consider implementing exponential backoff to reduce server load for long-running requests

### Polling Logic:

1. Initiate the ACCESS operation (POST \$<operation> with required parameters)
2. Receive HTTP 202 with Content-Location
3. Wait initial delay (5-10 seconds)
4. While (timeout not exceeded):
  - a. Poll status (GET \$submission-status)
  - b. If HTTP 202: wait and continue loop
  - c. If HTTP 200: process result and exit
  - d. If HTTP 4xx/5xx: handle error and exit
5. If timeout exceeded: handle as error

### Interpreting Responses

**HTTP 202 Accepted:** The request is being processed. No result is available yet. Continue polling.

**HTTP 200 OK:** Processing is complete. The response body contains a Parameters resource with the result. For example, here is a sample response from a successful request:

```
{
  "resourceType": "Parameters",
  "parameter": [{
    "name": "result",
    "valueCodeableConcept": {
      "coding": [{
        "system": "https://dsacms.github.io/cmml-access-
model/CodeSystem/ACCESSAlignmentResultCS",
        "code": "aligned",
        "display": "Aligned"
      }],
      "text": "Patient is eligible and has been aligned so the
ACCESS participant can now begin providing services to the patient
under the model."
    }
  ]
}
```

Clients should:

1. Extract `parameter[0].valueCodeableConcept` from the Parameters resource
2. Check `coding[0].code` against known alignment result codes
3. Display text element for user-friendly messaging
4. Take appropriate action based on the result code for the particular API. These actions are covered in the documentation for each API.

### Error Handling

Clients should be prepared to handle the following conditions:

### HTTP Status Code Errors:

- **400 Bad Request:** Malformed request - check Parameters structure and required fields
- **404 Not Found:** Submission ID not found - verify the submission ID is correct
- **500 Internal Server Error:** Server processing error - retry after delay or contact support
- **503 Service Unavailable:** Service temporarily unavailable - implement retry logic

#### Client-Side Errors:

- **Timeout:** No response received after maximum polling duration - log and notify user
- **Network Error:** Connection failure - implement retry logic with exponential backoff

**Best Practice:** Always check HTTP status codes before attempting to parse the response body. Only HTTP 200 responses contain the result from the original request.

#### Security Considerations

- All API interactions **SHALL** use TLS 1.3
- Clients **SHALL** authenticate using approved OAuth 2.0
- Submission identifiers should be treated as sensitive information and protected accordingly
- Clients should implement appropriate audit logging for all requests
- The `Content-Location` HTTP header URL may contain sensitive identifiers and should be handled securely

#### Best Practices

1. **Idempotency:** Avoid submitting duplicate requests for the same patient
2. **Resource Validation:** Validate input resources against FHIR profiles before submission
3. **MBI Handling:** Ensure Medicare Beneficiary Identifiers (MBIs) are handled according to CMS data sharing policies
4. **Result Code Handling:** Implement robust parsing for all possible result codes
5. **Submission ID Management:** Store submission IDs securely and maintain their association with original requests

#### OperationOutcome

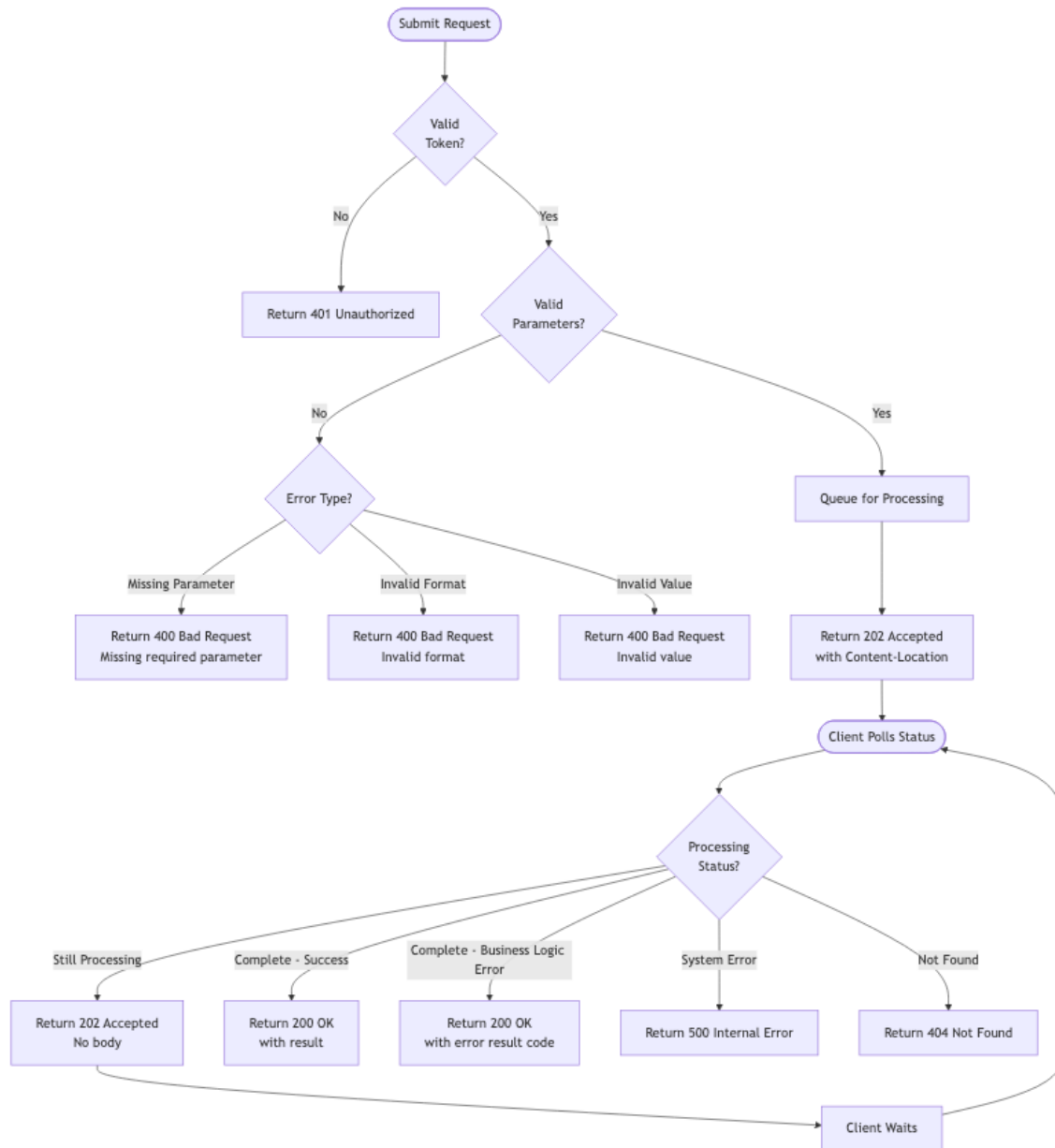
Errors are returned using FHIR OperationOutcome resources:

```
{
  "resourceType": "OperationOutcome",
  "issue": [
    {
      "severity": "error",
      "code": "invalid",
      "details": {
        "text": "Missing required parameter: participantID"
      }
    }
  ]
}
```



### Error Handling Flow Diagram

The following diagram illustrates the complete error handling flow for ACCESS API interactions:



### Key Decision Points:

1. **Authentication:** First check validates the OAuth token
2. **Parameter Validation:** Validates required parameters and their formats

3. **Asynchronous Processing:** Valid requests are queued and return HTTP 202 (Accepted)
4. **Status Polling:** Client repeatedly checks status until completion (returns something other than HTTP 202)
5. **Result Interpretation:**
  - HTTP 200 with success result code = operation succeeded
  - HTTP 200 with issue result code = business logic prevented operation (e.g., not-eligible, already-aligned)
  - HTTP 4xx/5xx = technical error occurred

## Common Error Scenarios

### Submission Errors (400 Bad Request)

#### Missing Required Parameter:

POST https://[base]/access/Patient/\$check-eligibility?entityID=ACCES12345

HTTP/1.1 400 Bad Request

Content-Type: application/fhir+json

```
{
  "resourceType": "OperationOutcome",
  "issue": [
    {
      "severity": "error",
      "code": "required",
      "details": { "text": "Missing required parameter: patient" }
    }
  ]
}
```

#### Invalid MBI Format:

HTTP/1.1 400 Bad Request

Content-Type: application/fhir+json

```
{
  "resourceType": "OperationOutcome",
  "issue": [
    {
      "severity": "error",
      "code": "invalid",
      "details": { "text": "Invalid Medicare Beneficiary Identifier (MBI)
format" }
    }
  ]
}
```

#### Invalid Track Code:

HTTP/1.1 400 Bad Request

Content-Type: application/fhir+json

```
{
  "resourceType": "OperationOutcome",
  "issue": [
    {
      "severity": "error",
      "code": "code-invalid",
      "details": {
        "text": "Invalid track code. Must be one of: eCKM, CKM, MSK, BH"
      }
    }
  ]
}
```

### **Malformed Parameters Resource:**

HTTP/1.1 400 Bad Request

Content-Type: application/fhir+json

```
{
  "resourceType": "OperationOutcome",
  "issue": [
    {
      "severity": "error",
      "code": "structure",
      "details": {
        "text": "Parameters resource structure is invalid"
      },
      "diagnostics": "Expected Parameters.parameter array but received malformed JSON"
    }
  ]
}
```

### **Invalid Participant ID (400):**

HTTP/1.1 400 Bad Request

Content-Type: application/fhir+json

```
{
  "resourceType": "OperationOutcome",
  "issue": [
    {
      "severity": "error",
      "code": "invalid",
      "details": {
        "text": "Participant ID is not valid"
      }
    }
  ]
}
```

### **FHIR Validation Failure:**

HTTP/1.1 400 Bad Request

Content-Type: application/fhir+json

```
{
  "resourceType": "OperationOutcome",
```

```

"issue": [
  {
    "severity": "error",
    "code": "invalid",
    "details": {
      "text": "FHIR validation failed: Patient.identifier.system is
required when Patient.identifier.value is present"
    },
    "diagnostics": "The Patient resource does not conform to the US
Core Patient profile. MBI identifier must include both 'system'
(http://terminology.hl7.org/NamingSystem/cmsMBI) and 'value' elements."
  }
]
}

```

### **Status Check Errors**

#### **Submission ID Not Found (404):**

GET https://[base]/access/Patient/\$submission-status/invalid-sub-id

HTTP/1.1 404 Not Found

Content-Type: application/fhir+json

```

{
  "resourceType": "OperationOutcome",
  "issue": [
    {
      "severity": "error",
      "code": "not-found",
      "details": { "text": "Submission ID 'invalid-sub-id' not found" }
    }
  ]
}

```

### **Authentication and Authorization Errors**

#### **Missing or Invalid Token (401):**

HTTP/1.1 401 Unauthorized

WWW-Authenticate: Bearer error="invalid\_token"

Content-Type: application/fhir+json

```

{
  "resourceType": "OperationOutcome",
  "issue": [
    {
      "severity": "error",
      "code": "security",
      "details": {
        "text": "Invalid or expired access token"
      }
    }
  ]
}

```

#### **Insufficient Scope (403):**

HTTP/1.1 403 Forbidden  
Content-Type: application/fhir+json

```
{
  "resourceType": "OperationOutcome",
  "issue": [
    {
      "severity": "error",
      "code": "forbidden",
      "details": {
        "text": "Insufficient scope for requested operation."
      }
    }
  ]
}
```

### **Server Errors**

#### **Service Temporarily Unavailable (503):**

HTTP/1.1 503 Service Unavailable  
Retry-After: 60  
Content-Type: application/fhir+json

```
{
  "resourceType": "OperationOutcome",
  "issue": [
    {
      "severity": "error",
      "code": "transient",
      "details": {
        "text": "Service temporarily unavailable. Please retry after 60
seconds."
      }
    }
  ]
}
```

#### **Internal Server Error (500):**

HTTP/1.1 500 Internal Server Error  
Content-Type: application/fhir+json

```
{
  "resourceType": "OperationOutcome",
  "issue": [
    {
      "severity": "error",
      "code": "exception",
      "details": {
        "text": "An unexpected error occurred processing your request.
Please contact support with submission ID if this persists."
      }
    }
  ]
}
```

#### **Example Status Check Request (Processing Complete - Aligned and Switch Approved):**

GET https://[base]/access/Patient/\$submission-status/sub-123456

### Response:

HTTP/1.1 200 OK

Content-Type: application/fhir+json

```
{
  "resourceType": "Parameters",
  "parameter": [
    {
      "name": "result",
      "valueCodeableConcept": {
        "coding": [
          {
            "system": "https://dsacms.github.io/cmmi-access-
model/CodeSystem/ACCESSAlignmentResultCS",
            "code": "aligned-switch-approved",
            "display": "Aligned and switch approved"
          }
        ],
        "text": "The request to switch the patient's alignment from a
different participant after the 90-day lock in period is accepted and the
patient is considered switched and now re-aligned."
      }
    }
  ]
}
```

There are other types of errors specific to each API that may be returned. Please see the section specific to that API in this manual to get more information and examples for those errors.

### Standards Alignment

This asynchronous pattern aligns with:

- **FHIR Asynchronous Request Pattern:** Follows FHIR's general guidance for asynchronous operations using HTTP 202 Accepted
- **HTTP Standards:** Uses appropriate HTTP status codes (202 Accepted, 200 OK, 4xx/5xx for errors)
- **FHIR Parameters Resource:** Uses standardized Parameters resources for operation input and output
- **FHIR Operations Framework:** Implements custom operations with proper input/output handling
- **FHIR OperationOutcome:** Uses OperationOutcome for error responses following FHIR conventions
- **US Core Profiles:** Aligns with US Core Patient and Condition profiles for patient data

For additional information about FHIR patterns, see:

- FHIR Asynchronous Request Pattern (<https://hl7.org/fhir/R4/async.html>)
- FHIR Parameters Resource (<https://hl7.org/fhir/R4/parameters.html>)
- FHIR Operations Framework (<https://hl7.org/fhir/R4/operations.html>)
- FHIR OperationOutcome (<https://hl7.org/fhir/R4/operationoutcome.html>)
- US Core Implementation Guide (<https://hl7.org/fhir/us/core/STU6.1/>)

## Implementation Notes and Edge Cases

### *Handling Network Interruptions*

**Scenario:** Network failure occurs during polling

#### **Implementation Guidance:**

- Store submission IDs and associated request context before starting to poll
- Implement resume logic that can continue polling after network restoration
- Use exponential backoff when reconnecting after network failure
- Set reasonable timeout limits (5 minutes recommended) to avoid infinite polling loops

### *Duplicate Submission Prevention*

**Scenario:** User clicks submit button multiple times or system retries unnecessarily

#### **Implementation Guidance:**

- Implement client-side button disabling after first click
- Store submission IDs locally to detect duplicates
- Check if a recent submission exists for the same patient/operation before submitting again
- Use idempotency tokens if retrying after uncertain failures

### *Token Expiration During Long Operations*

**Scenario:** OAuth token expires while waiting for async operation to complete

#### **Implementation Guidance:**

- Refresh tokens proactively before expiration (5 minutes before recommended)
- Store token expiration time and check before each status poll
- Implement automatic token refresh in polling loop
- Handle HTTP 401 responses by refreshing token and retrying once

### *Multiple Diagnoses Across Tracks*

**Scenario:** Patient has conditions qualifying for multiple tracks

#### **Implementation Guidance:**

- Patient can be aligned to multiple tracks simultaneously\*
- Each track requires a separate alignment request
- Track alignments are independent - success/failure in one does not affect others
- Data reporting is track-specific - submit separate bundles for each track

- Patient can be aligned to different participants or the same participant in different tracks

\* **The one exception** to this rule applies to the eCKM and CKM tracks. Patients cannot be enrolled in both tracks at the same time. If a patient is already aligned to the eCKM track and elects to enroll in the CKM track, or vice versa, they can change tracks immediately. In this case the 90-day lock-in period does not apply.

#### Next Steps and support

- Review the Authentication guidance
- Consult API-specific guidance pages under the API Guidance menu
- Reference the Artifacts page in the ACCESS Model FHIR IG for technical specifications
- Contact the ACCESS Model support team if you have any questions ([ACCESSModelTeam@cms.hhs.gov](mailto:ACCESSModelTeam@cms.hhs.gov))



## Authentication

### Authentication Process for ACCESS APIs

This section describes the authentication and authorization process required to access all ACCESS Model API endpoints.

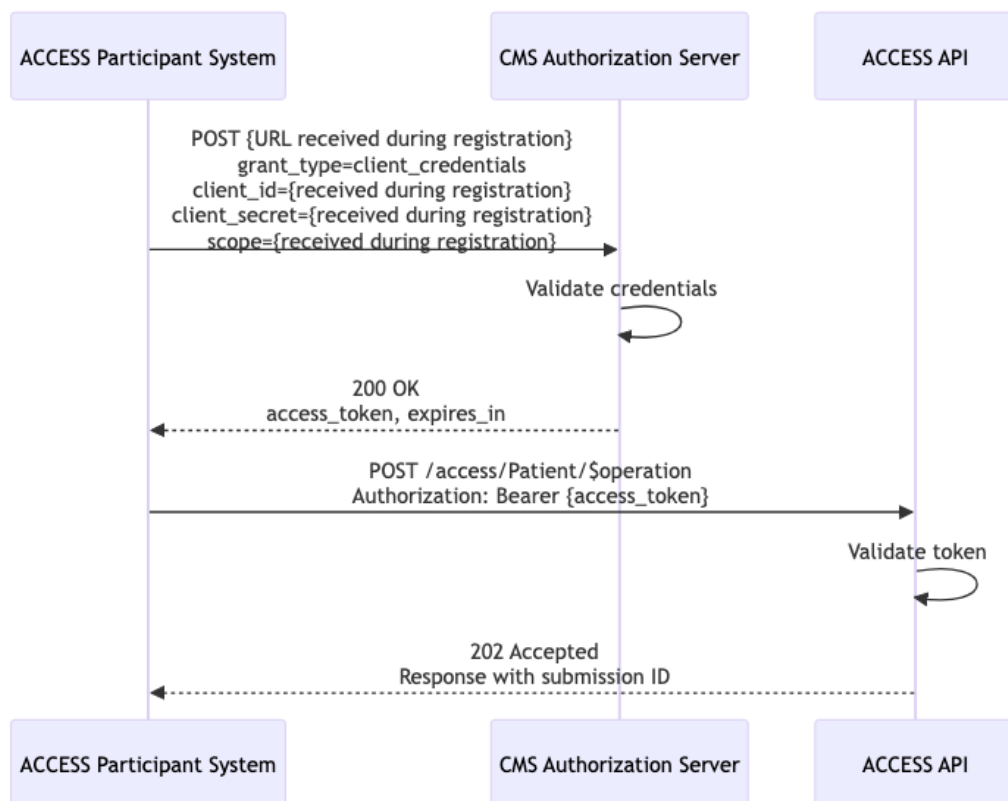
#### Overview

The ACCESS APIs use OAuth 2.0 for authentication and authorization. All ACCESS participants must complete a registration process and obtain appropriate credentials before accessing any API.

#### OAuth 2.0 Client Credentials Flow

The ACCESS APIs use the OAuth 2.0 Client Credentials grant type, which is appropriate for server-to-server authentication where the client (ACCESS participant system) is acting on its own behalf.

#### Authentication Flow



#### Step 1: Obtain Access Token

**Token Endpoint:** `POST {URL received during registration}`

**Request Headers:**

Content-Type: application/x-www-form-urlencoded

**Request Body (form-encoded):**

```
grant_type=client_credentials
client_id={received during registration}
client_secret={received during registration}
scope={received during registration}
```

**Example cURL Request:**

```
curl -X POST {URL received during registration} \
  -H "Content-Type: application/x-www-form-urlencoded" \
  -d "grant_type=client_credentials" \
  -d "client_id={received during registration}" \
  -d "client_secret={received during registration}" \
  -d "scope={received during registration}"
```

**Successful Response:**

```
{
  "access_token": "eyJhbGciOiJSUzI1NiIsInR5cCI6IkpXVCJ9...",
  "token_type": "Bearer",
  "expires_in": 300,
  "scope": "{allowed scope}"
}
```

It is **strongly** recommended to get a new access token for each new request.

**Step 2: Use Access Token in API Requests**

Include the access token in the Authorization header of all API requests:

**Request Headers:**

Authorization: Bearer {access\_token}  
Content-Type: application/fhir+json

**Example Request:**

```
POST https://[base]/access/Patient/$check-eligibility?
      entityId=[participantID]
Authorization: Bearer eyJhbGciOiJSUzI1NiIsInR5cCI6IkpXVCJ9...
Content-Type: application/fhir+json
{
  "resourceType": "Parameters",
  "parameter": [...]
}
```

**Token Management****Token Expiration**

Access tokens have a limited lifetime (typically 5 minutes). Clients must:

1. **Monitor token expiration** - Track the `expires_in` value from the token response
2. **Refresh before expiration** - Request a new token before the current one expires

3. **Handle HTTP 401 responses** - If a token expires mid-use, obtain a new token and retry the request

#### *Token Caching Best Practices*

- Cache tokens to avoid unnecessary authentication requests
- Request a new token 2 minutes before expiration to account for clock skew
- Implement thread-safe token management in multi-threaded applications
- Store credentials and tokens securely (never in source code or logs)

#### *Example Token Management (Pseudocode)*

```
class TokenManager:
    token = null
    expires_at = null

    function getValidToken():
        if token is null or currentTime() >= expires_at - 5 minutes:
            response = requestNewToken()
            token = response.access_token
            expires_at = currentTime() + response.expires_in
        return token

    function requestNewToken():
        return POST to auth_server with client_credentials
```

#### *Required Scopes*

The ACCESS APIs require the OAuth 2.0 scopes provided during the client registration process. Scopes may vary based on which APIs your organization has been approved to access.

#### *Error Handling*

##### *Authentication Errors*

##### **Invalid Credentials (HTTP 401 Unauthorized):**

```
{
  "error": "invalid_client",
  "error_description": "Client authentication failed"
}
```

**Action:** Verify your `client_id` and `client_secret` are correct.

##### **Invalid Token (HTTP 401 Unauthorized):**

```
HTTP/1.1 401 Unauthorized
WWW-Authenticate: Bearer error="invalid_token", error_description="Token
has expired"
```

**Action:** Request a new access token.

##### **Insufficient Scope (HTTP 403 Forbidden):**

```
{
  "resourceType": "OperationOutcome",
  "issue": [
    {
      "severity": "error",
      "code": "forbidden",
      "details": {
        "text": "Insufficient scope for requested operation"
      }
    }
  ]
}
```

**Action:** Ensure your token includes all required scopes for the API operation you are attempting to access.

## Security Requirements

### Transport Security

- **TLS 1.3 Required:** All communication must use TLS 1.3
- **Certificate Validation:** Clients must validate server certificates
- **No Insecure Protocols:** HTTP, TLS 1.0, TLS 1.1, TLS 1.2, and SSL are not permitted

### Credential Security

- **Secure Storage:** Store client credentials in secure key management systems or vaults
- **No Code Storage:** Never store credentials in source code, configuration files, or version control
- **Environment Variables:** Use environment variables or secure configuration management
- **Rotation:** Implement credential rotation policies (recommended: every 90 days)
- **Audit Logging:** Log all authentication attempts (excluding sensitive credential data)

### Token Security

- **Secure Transmission:** Only transmit tokens over TLS connections
- **Limited Lifetime:** Accept token expiration and request new tokens as needed
- **No Sharing:** Never share tokens between different applications or environments
- **Secure Storage:** If caching tokens, use encrypted storage
- **Logging:** Never log tokens in application logs or monitoring systems

### Multi-API Access

A single set of credentials may provide access to multiple ACCESS APIs. The OAuth scopes in your access token will determine which operations you can perform.

## Troubleshooting

### Common Issues

#### Token Repeatedly Expires Mid-Request

- Check system clock synchronization

- Implement proactive token refresh (2 minutes before expiration)
- Verify network latency is not causing delays

### **Authentication Works in Test but Fails in Production**

- Verify you are using production credentials (not test credentials)
- Check that production endpoint URLs are correct
- Confirm your organization has been approved for production access

### **HTTP 403 Forbidden Despite Valid Token**

- Verify your scope includes the operation you are attempting
- Check that your participant ID is active
- Ensure you are using the correct API endpoints

### ***Additional Resources***

- OAuth 2.0 RFC 6749 (<https://tools.ietf.org/html/rfc6749>) - OAuth 2.0 specification
- OAuth 2.0 Client Credentials (<https://tools.ietf.org/html/rfc6749#section-4.4>) - Client credentials grant flow
- JWT RFC 7519 (<https://tools.ietf.org/html/rfc7519>) - JSON Web Token specification
- OAuth 2.0 Security Best Practices (<https://datatracker.ietf.org/doc/html/draft-ietf-oauth-security-topics>) - Security considerations

## Eligibility API Guidance

### How to Use the API

The ACCESS Eligibility API uses an asynchronous pattern for checking patient eligibility for the CMS ACCESS Model. The pattern uses two operations to support a submit-and-poll workflow:

1. **\$check-eligibility**: Submits an eligibility check request
2. **\$submission-status**: Polls for the status of a submitted request

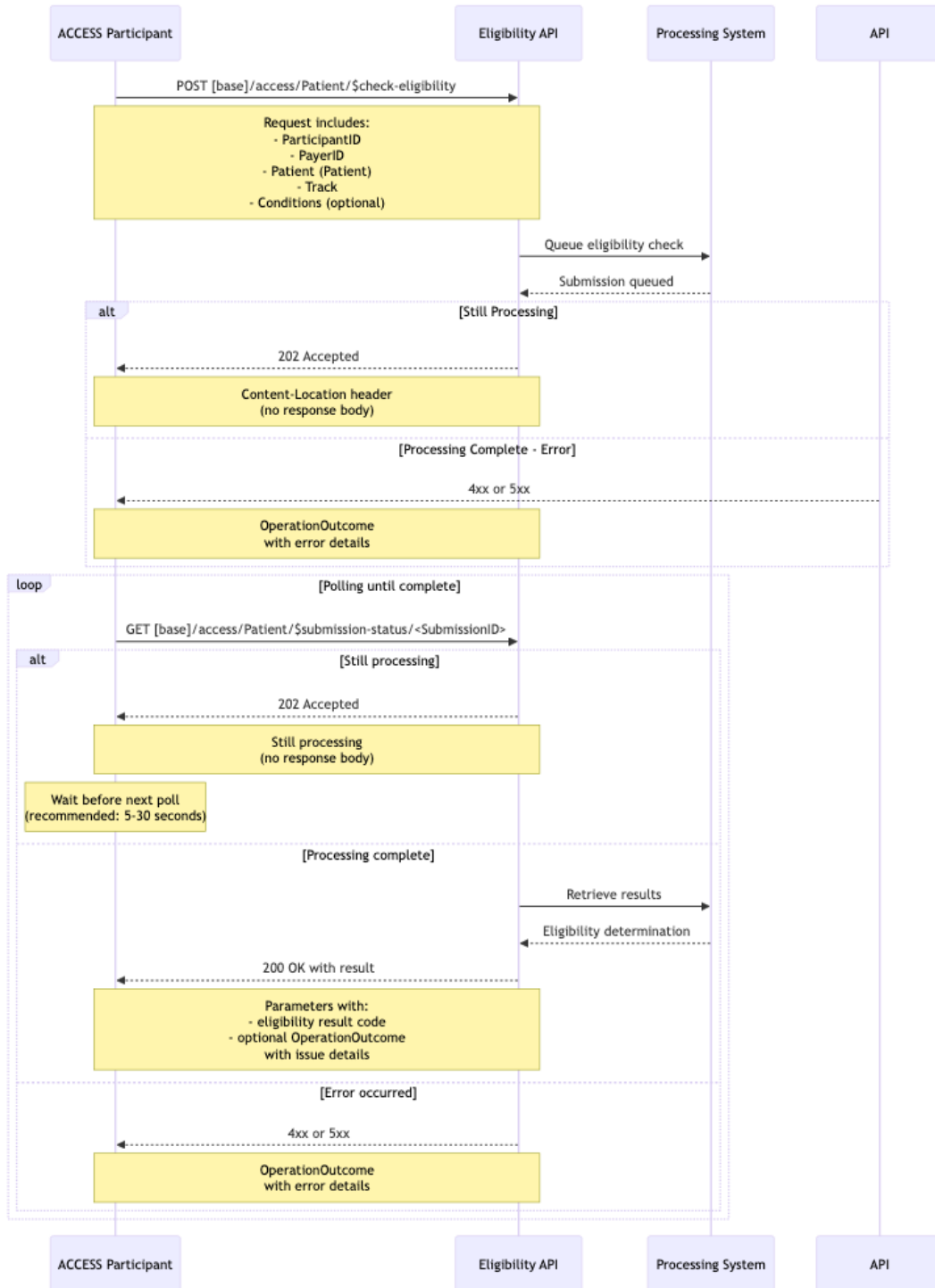
### *Asynchronous Interaction Pattern*

The Eligibility API uses an asynchronous request-response pattern to accommodate the processing time required for eligibility determination. This pattern allows clients to submit eligibility check requests and subsequently poll for results without maintaining a persistent connection.

These operations can take some time to complete, making a synchronous request-response pattern impractical for real-world implementations.

### *Detailed Interaction Sequence*

The following sequence diagram illustrates the complete interaction pattern:



## Operation Details

### \$check-eligibility Operation

The \$check-eligibility operation initiates an asynchronous eligibility determination request.

**Endpoint:** POST [base]/access/Patient/\$check-eligibility?  
entityId=[participantID]

#### Input Parameters:

- **participantID** (Identifier, required): The ACCESS ID for the submitting participant
- **payerID** (Identifier, required): The payer ID for the patient. Uses the X12 EDI Payer ID standard (urn:oid:2.16.840.1.113883.3.221.5) with CARIN Blue Button identifier typing (<http://hl7.org/fhir/us/carin-bb/CodeSystem/C4BBIdentifierType>). EDI Payer IDs are 5-character alphanumeric codes widely used in healthcare transactions (e.g., eligibility checks, claims, remittance).
- **patient** (Patient Resource, required): Patient information conforming to US Core Patient Profile. If the payer is Medicare, it **SHALL** contain the Medicare Beneficiary Identifier (MBI). To be eligible and align to a CMS ACCESS participant, a patient must be a Medicare beneficiary.
- **track** (CodeableConcept, required): The ACCESS Model track to check for eligibility
- **conditions** (Condition Resources, optional): Zero or more Condition resources conforming to the ACCESS Condition Profile describing the patient's health concerns. Implementers **SHOULD** use track-specific profiles (ACCESSeCKMCondition, ACCESSCKMCondition, ACCESSMSKCondition, ACCESSBHCondition) when submitting conditions for a known track. A condition, or diagnosis code, does not need to be included in the eligibility check for it to process. But, if the code is included, it should be an expected code for the track the patient is being considered for.

**NOTE:** The IG will accept any patient identifier and payer identifier so that this IG can be used by other payers looking to align with the ACCESS Model. However, for a patient to be eligible for the CMS ACCESS Model, they need to have Medicare fee-for-service as their primary insurance and you must submit a valid MBI as the patient identifier.

#### Expected Response:

- **Status Code:** 202 Accepted
- **HTTP Headers:** Content-Location header containing the URL to check submission status
- **Response Body:** Empty (no body while processing)

#### Example Request:

```
POST https://[base]/access/Patient/$check-eligibility?entityId=ACCES12345
Content-Type: application/fhir+json
```



```

{
  "resourceType": "Parameters",
  "parameter": [
    {
      "name": "participantID",
      "valueIdentifier": {
        "system": "https://dsacms.github.io/cmml-access-
model/participant-id",
        "value": "ACCES12345"
      }
    },
    {
      "name": "payerID",
      "valueIdentifier": {
        "type": {
          "coding": [{
            "system": "http://hl7.org/fhir/us/carin-
bb/CodeSystem/C4BBIIdentifierType",
            "code": "payerid",
            "display": "Payer ID"
          }]
        },
        "system": "urn:oid:2.16.840.1.113883.3.221.5",
        "value": "12345"
      }
    },
    {
      "name": "patient",
      "resource": {
        "resourceType": "Patient",
        "id": "example",
        "identifier": [
          {
            "type": {
              "coding": [
                {
                  "system": "http://terminology.hl7.org/CodeSystem/v2-
0203",
                  "code": "MC"
                }
              ]
            },
            "system": "http://terminology.hl7.org/NamingSystem/cmsMBI",
            "value": "1234567890A"
          }
        ],
        "name": [
          {
            "family": "Doe",
            "given": ["John"]
          }
        ],
        "gender": "male",
        "birthDate": "1950-01-01"
      }
    }
  ]
}

```

```

    },
    {
      "name": "track",
      "valueCodeableConcept": {
        "coding": [{
          "system": "https://dsacms.github.io/cmmi-access-
model/CodeSystem/ACCESSTrackCS",
          "code": "CKM",
          "display": "Cardio-Kidney-Metabolic track"
        }]
      }
    },
    {
      "name": "condition",
      "resource": {
        "resourceType": "Condition",
        "code": {
          "coding": [
            {
              "system": "http://hl7.org/fhir/sid/icd-10-cm",
              "code": "E11.9",
              "display": "Type 2 diabetes mellitus without complications"
            }
          ]
        },
        "subject": {
          "reference": "Patient/example"
        }
      }
    }
  ]
}

```

### Example Response:

HTTP/1.1 202 Accepted  
 Content-Location: https://[base]/access/Patient/\$submission-status/sub-123456

(No response body - processing asynchronously)

### \$submission-status Operation

The \$submission-status operation retrieves the current status of a previously submitted eligibility check request. This is a **read-only operation** that uses the HTTP GET method.

### Understanding Response Behavior:

- **HTTP 202 Accepted:** Request is still being processed. Response has **no body**. Continue polling.
- **HTTP 200 OK:** Processing is complete. Response **always** contains a Parameters resource with the `result` parameter indicating the outcome.

- **HTTP 4xx/5xx:** An error occurred. Response contains an OperationOutcome with error details (no Parameters resource).

### Eligibility Result Codes:

When processing is complete (HTTP 200), the `result` parameter uses codes from the **ACCESSEligibilityResultVS** value set:

- **eligible:** Patient is eligible for services under the model. This response does not mean the patient has been aligned. It simply means the patient is eligible to be aligned.
- **eligible-pending-diagnosis:** Patient is provisionally eligible for services under the model depending on the diagnosis.
- **eligible-switch-participants:** Patient is eligible to switch participants. This code indicates that the patient is already aligned to another participant, but they are outside of the 90-day lock-in period with that participant.
- **not-eligible-not-medicare:** The patient either: (1) is not enrolled in Medicare Part A and Part B; (2) is not dual eligible for Medicare and Medicaid; or (3) they do not have Medicare as their primary insurance, so they are not eligible for services under the ACCESS Model.
- **not-eligible-services:** The patient is receiving services (including receiving hospice services or dialysis for end stage renal disease (ESRD)) making them ineligible to be part of the ACCESS Model. Patients who are part of the Program of All-Inclusive Care for the Elderly (PACE) Program are also not eligible for the ACCESS Model.
- **not-eligible-diagnoses:** The patient does not have a treating diagnosis that qualifies them for services in the track indicated and therefore cannot get services under the ACCESS Model.
- **not-eligible-control-group:** The patient is technically eligible for the ACCESS Model, but based on the randomized control group algorithm, the patient has been placed in the control group for 12 months and therefore cannot be aligned for 12 months.
- **not-eligible-already-aligned:** The patient is technically eligible, but has already aligned to another participant within the past 90 days and is receiving services under the ACCESS Model in the same track. A patient can only be aligned to one participant in each track.
- **not-eligible-clinical-exclusion:** The patient has a clinical exclusion that prevents them from receiving services under the ACCESS Model.
- **not-eligible-mismatch:** The patient information provided does not match the patient information on record.

### Example Responses:

#### Response:

HTTP/1.1 202 Accepted

(No response body - still processing)

### Example Status Check Request (Processing Complete - Eligible):

GET https://[base]/access/Patient/\$submission-status/sub-123456

#### Response:

HTTP/1.1 200 OK

Content-Type: application/fhir+json

```
{
  "resourceType": "Parameters",
  "parameter": [
    {
      "name": "result",
      "valueCodeableConcept": {
        "coding": [
          {
            "system": "https://dsacms.github.io/cmmi-access-
model/CodeSystem/ACCESSEligibilityResultCS",
            "code": "eligible",
            "display": "Eligible"
          }
        ],
        "text": "Patient is eligible for services under the model. This
response does not mean the patient has been aligned. It simply means the
patient is eligible to be aligned."
      }
    }
  ]
}
```

### Example Status Check Request (Processing Complete - Not Eligible):

GET https://[base]/access/Patient/\$submission-status/sub-123456

#### Response:

HTTP/1.1 200 OK

Content-Type: application/fhir+json

```
{
  "resourceType": "Parameters",
  "parameter": [
    {
      "name": "result",
      "valueCodeableConcept": {
        "coding": [
          {
            "system": "https://dsacms.github.io/cmmi-access-
model/CodeSystem/ACCESSEligibilityResultCS",
            "code": "not-eligible-control-group",
            "display": "Not eligible - assigned to Control Group"
          }
        ],
        "text": "The patient is technically eligible for the ACCESS
Model, but based on the randomized control group algorithm, the patient
has been placed in the control group for 12 months and therefore cannot
be aligned for 12 months."
      }
    }
  ]
}
```

```

    }
  }
]
}

```

### Example Complete Workflow

The following example demonstrates a complete eligibility check workflow:

#### Step 1: Submit Eligibility Check

POST [https://\[base\]/access/Patient/\\$check-eligibility?entityId=ACCES12345](https://[base]/access/Patient/$check-eligibility?entityId=ACCES12345)

Content-Type: application/fhir+json

```

{
  "resourceType": "Parameters",
  "parameter": [
    {
      "name": "participantID",
      "valueIdentifier": {
        "system": "https://dsacms.github.io/cmmi-access-
model/participant-id",
        "value": "ACCES12345"
      }
    },
    {
      "name": "payerID",
      "valueIdentifier": {
        "type": {
          "coding": [{
            "system": "http://hl7.org/fhir/us/carinn-
bb/CodeSystem/C4BBIIdentifierType",
            "code": "payerid",
            "display": "Payer ID"
          }]
        },
        "system": "urn:oid:2.16.840.1.113883.3.221.5",
        "value": "12345"
      }
    },
    {
      "name": "patient",
      "resource": {
        "resourceType": "Patient",
        "id": "example",
        "identifier": [
          {
            "type": {
              "coding": [
                {
                  "system": "http://terminology.hl7.org/CodeSystem/v2-
0203",
                  "code": "MC"
                }
              ]
            }
          }
        ]
      }
    }
  ]
}

```

```

        "system": "http://terminology.hl7.org/NamingSystem/cmsSMBI",
        "value": "1234567890A"
    }
],
"name": [
    {
        "family": "Doe",
        "given": ["John"]
    }
],
"gender" : "male",
"birthDate": "1950-01-01"
}
},
{
    "name": "track",
    "valueCodeableConcept": {
        "coding": [
            {
                "system": "https://dsacms.github.io/cmami-access-
model/CodeSystem/ACCESSTrackCS",
                "code": "CKM",
                "display": "Cardio-Kidney-Metabolic track"
            }
        ]
    }
},
{
    "name": "condition",
    "resource": {
        "resourceType": "Condition",
        "code": {
            "coding": [
                {
                    "system": "http://hl7.org/fhir/sid/icd-10-cm",
                    "code": "E11.9",
                    "display": "Type 2 diabetes mellitus without complications"
                }
            ]
        },
        "subject": {
            "reference": "Patient/example"
        }
    }
}
]
}

```

### Response:

HTTP/1.1 202 Accepted  
 Content-Location: https://[base]/access/Patient/\$submission-status/sub-123456

### Step 2: Extract Submission URL and Wait

- Extract submission URL from Content-Location header
- Wait 5 seconds before first poll

### Step 3: First Status Check (after 5 seconds)

GET https://[base]/access/Patient/\$submission-status/sub-123456

#### Response:

HTTP/1.1 202 Accepted

(Still processing - no body)

### Step 4: Second Status Check (after another 15 seconds)

GET https://[base]/access/Patient/\$submission-status/sub-123456

#### Response:

HTTP/1.1 200 OK

Content-Type: application/fhir+json

```
{
  "resourceType": "Parameters",
  "parameter": [
    {
      "name": "result",
      "valueCodeableConcept": {
        "coding": [
          {
            "system": "https://dsacms.github.io/cmmi-access-
model/CodeSystem/ACCESSEligibilityResultCS",
            "code": "eligible",
            "display": "Eligible"
          }
        ],
        "text": "Patient is eligible for services under the model. This
response does not mean the patient has been aligned. It simply means the
patient is eligible to be aligned."
      }
    }
  ]
}
```

### Step 5: Process Result

- Parse the result code: "eligible"
- Display the text to the user
- Proceed with alignment workflow

### API-Specific Response Scenarios

For general response handling guidance and common scenarios (missing parameters, invalid tokens, server errors, etc.), see the Error Handling section in General Guidance.

The following responses are specific to the Eligibility API and are returned as part of normal processing (HTTP 200 with specific result codes):

#### Not Eligible - Already Aligned

**Scenario:** Patient is already aligned to another provider in the same track.

```
GET https://[base]/access/Patient/$submission-status/sub-123456

HTTP/1.1 200 OK
Content-Type: application/fhir+json

{
  "resourceType": "Parameters",
  "parameter": [
    {
      "name": "result",
      "valueCodeableConcept": {
        "coding": [
          {
            "system": "https://dsacms.github.io/cmml-access-
model/CodeSystem/ACCESSEligibilityResultCS",
            "code": "not-eligible-already-aligned",
            "display": "Not eligible - already aligned to another
participant in the track"
          }
        ],
        "text": "The patient is technically eligible, but is already
aligned to another participant and receiving services under the ACCESS
Model in the same track. A patient can only be aligned to one participant
in each track."
      }
    }
  ]
}
```

#### Not Eligible - Control Group

**Scenario:** Patient has been randomized into the control group.

```
GET https://[base]/access/Patient/$submission-status/sub-123456

HTTP/1.1 200 OK
Content-Type: application/fhir+json

{
  "resourceType": "Parameters",
  "parameter": [
    {
      "name": "result",
      "valueCodeableConcept": {
        "coding": [
```



```

    {
      "system": "https://dsacms.github.io/cmmi-access-
model/CodeSystem/ACCESSEligibilityResultCS",
      "code": "not-eligible-control-group",
      "display": "Not eligible - assigned to Control Group"
    }
  ],
  "text": "The patient is technically eligible for the ACCESS
Model, but based on the randomized control group algorithm, the patient
has been placed in the control group for 12 months and therefore cannot
be aligned for 12 months."
}
}
]
}

```

### Not Eligible - No Qualifying Diagnosis

**Scenario:** Patient lacks the required diagnosis for the specified track.

GET https://[base]/access/Patient/\$submission-status/sub-123456

HTTP/1.1 200 OK

Content-Type: application/fhir+json

```

{
  "resourceType": "Parameters",
  "parameter": [
    {
      "name": "result",
      "valueCodeableConcept": {
        "coding": [
          {
            "system": "https://dsacms.github.io/cmmi-access-
model/CodeSystem/ACCESSEligibilityResultCS",
            "code": "not-eligible-diagnoses",
            "display": "Not eligible - no qualifying diagnosis"
          }
        ],
        "text": "The patient does not have a treating diagnosis that
qualifies them for services in the track indicated and therefore cannot
get services under the ACCESS Model."
      }
    }
  ]
}

```

### Not Eligible - Receiving Exclusionary Services

**Scenario:** Patient is receiving services that prevent ACCESS participation.

GET https://[base]/access/Patient/\$submission-status/sub-123456

HTTP/1.1 200 OK

Content-Type: application/fhir+json

```
{
  "resourceType": "Parameters",
  "parameter": [
    {
      "name": "result",
      "valueCodeableConcept": {
        "coding": [
          {
            "system": "https://dsacms.github.io/cmmi-access-
model/CodeSystem/ACCESSEligibilityResultCS",
            "code": "not-eligible-services",
            "display": "Not eligible - receiving services that prevent
eligibility"
          }
        ],
        "text": "The patient is receiving services (such as hospice, end
stage renal disease (ESRD), etc.) making them ineligible to be part of
the ACCESS Model."
      }
    }
  ]
}
```

### Not Eligible - Not Medicare

**Scenario:** Patient is not receiving Medicare or Medicare is not primary insurance.

GET https://[base]/access/Patient/\$submission-status/sub-123456

HTTP/1.1 200 OK

Content-Type: application/fhir+json

```
{
  "resourceType": "Parameters",
  "parameter": [
    {
      "name": "result",
      "valueCodeableConcept": {
        "coding": [
          {
            "system": "https://dsacms.github.io/cmmi-access-
model/CodeSystem/ACCESSEligibilityResultCS",
            "code": "not-eligible-not-medicare",
            "display": "Not eligible - not receiving Medicare"
          }
        ],
        "text": "The patient either is not enrolled in Medicare Part A
and Part B or dual eligible for Medicare and Medicaid, or they do not
have Medicare as their primary insurance, so they are not eligible for
services under the ACCESS Model."
      }
    }
  ]
}
```

### Not Eligible – Clinical Exclusion

**Scenario:** Patient has a clinical condition that prevents them from receiving services under the ACCESS Model.

GET https://[base]/access/Patient/\$submission-status/sub-123456

HTTP/1.1 200 OK

Content-Type: application/fhir+json

```
{
  "resourceType": "Parameters",
  "parameter": [
    {
      "name": "result",
      "valueCodeableConcept": {
        "coding": [
          {
            "system": "https://dsacms.github.io/cmml-access-
model/CodeSystem/ACCESSEligibilityResultCS",
            "code": "not-eligible-clinical-exclusion",
            "display": "Not eligible - clinical exclusion"
          }
        ],
        "text": "The patient has a clinical exclusion that prevents them
from receiving services under the ACCESS Model."
      }
    ]
  }
}
```

### Not Eligible – Mismatch

**Scenario:** The provided information about the patient does not match the patient information on record.

GET https://[base]/access/Patient/\$submission-status/sub-123456

HTTP/1.1 200 OK

Content-Type: application/fhir+json

```
{
  "resourceType": "Parameters",
  "parameter": [
    {
      "name": "result",
      "valueCodeableConcept": {
        "coding": [
          {
            "system": "https://dsacms.github.io/cmml-access-
model/CodeSystem/ACCESSEligibilityResultCS",
            "code": "not-eligible-mismatch",
            "display": "Not eligible - mismatch"
          }
        ],
        "text": "The patient information provided does not match the"
      }
    ]
  }
}
```

```
patient information on record."
    }
  }
]
```

**Note:** These are successful responses (`HTTP 200`) that contain business logic results. All of these scenarios represent valid processing outcomes for eligibility checks.

## Alignment API Guidance

### How to Use the API

The ACCESS Alignment API uses an asynchronous pattern for aligning patients to a track and participant in the CMS ACCESS Model. The pattern uses two operations to support a submit-and-poll workflow:

1. **\$align**: Submits an alignment request
2. **\$submission-status**: Polls for the status of a submitted request

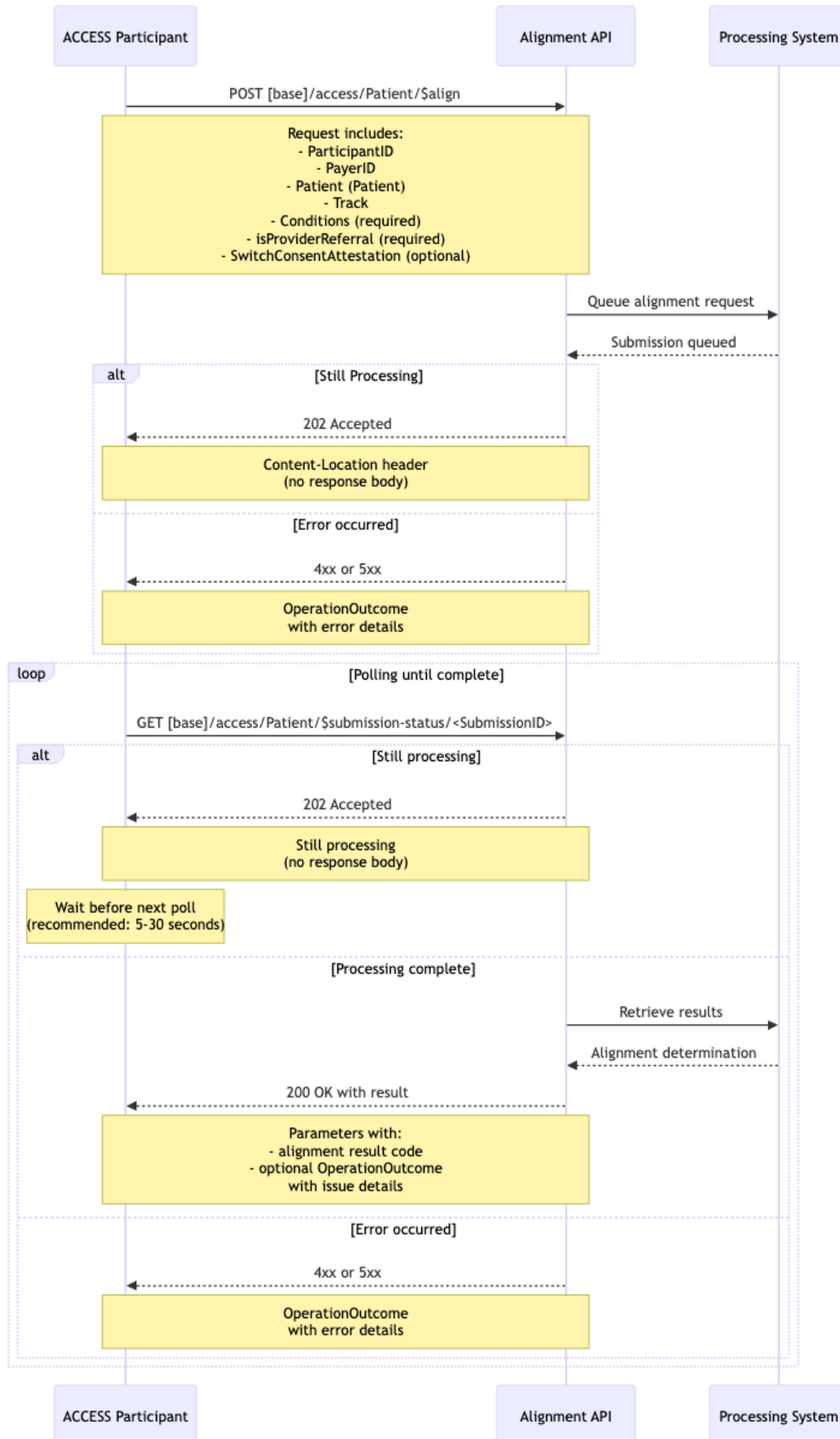
### Asynchronous Interaction Pattern

The Alignment API uses an asynchronous request-response pattern to accommodate the processing time required for alignment determination. This pattern allows clients to submit alignment requests and subsequently poll for results without maintaining a persistent connection.

These operations can take some time to complete, making a synchronous request-response pattern impractical for real-world implementations.

### Detailed Interaction Sequence

The following sequence diagram illustrates the complete interaction pattern:



## Operation Details

### \$align Operation

The \$align operation initiates an asynchronous alignment request to align a patient to a specific track and participant.

**Endpoint:** POST [base]/access/Patient/\$align?  
entityId=[participantID]

#### Input Parameters:

- **participantID** (Identifier, required): The ACCESS ID for the submitting participant
- **payerID** (Identifier, required): The payer ID for the patient. Uses the X12 EDI Payer ID standard (urn:oid:2.16.840.1.113883.3.221.5) with CARIN Blue Button identifier typing (<http://hl7.org/fhir/us/carin-bb/CodeSystem/C4BBIdentifierType>). EDI Payer IDs are 5-character alphanumeric codes widely used in healthcare transactions (e.g., eligibility checks, claims, remittance).
- **patient** (Patient Resource, required): Patient information conforming to US Core Patient Profile. If the payer is Medicare, it **SHALL** contain the Medicare Beneficiary Identifier (MBI). To be eligible and align to a CMS ACCESS participant, a patient must be a Medicare beneficiary.
- **track** (Value set, required): The requested track to align the patient
- **condition** (Condition Resources, required): One or more Condition resources conforming to ACCESS Condition Profile describing the patient's health concerns. Implementers **SHALL** use track-specific profiles (ACCESSeCKMCondition, ACCESSCKMCondition, ACCESSMSKCondition, ACCESSBHKCondition) when aligning to a specific track.
- **isProviderReferral** (boolean, required): Indicates whether the patient was referred to the ACCESS Model by a provider.
- **switchConsentAttestation** (boolean, optional): Boolean flag indicating that the patient has provided consent to switch from another ACCESS participant after the 90-day lock-in period. When `true`, this indicates the participant has obtained and documented the patient's consent to switch participants.

#### Expected Response:

- **Status Code:** 202 Accepted
- **HTTP Headers:** Content-Location header containing the URL to check submission status
- **Response Body:** Empty (no body while processing)

#### Example Request:

```
POST https://[base]/access/Patient/$align?entityId=ACCES12345
Content-Type: application/fhir+json
```

```

{
  "resourceType": "Parameters",
  "parameter": [
    {
      "name": "participantID",
      "valueIdentifier": {
        "system": "https://dsacms.github.io/cmml-access-
model/participant-id",
        "value": "ACCES12345"
      }
    },
    {
      "name": "payerID",
      "valueIdentifier": {
        "type": {
          "coding": [{
            "system": "http://hl7.org/fhir/us/carin-
bb/CodeSystem/C4BBIIdentifierType",
            "code": "payerid",
            "display": "Payer ID"
          }]
        },
        "system": "urn:oid:2.16.840.1.113883.3.221.5",
        "value": "12345"
      }
    },
    {
      "name": "patient",
      "resource": {
        "resourceType": "Patient",
        "id": "example",
        "identifier": [
          {
            "type": {
              "coding": [
                {
                  "system": "http://terminology.hl7.org/CodeSystem/v2-
0203",
                  "code": "MC"
                }
              ]
            },
            "system": "http://terminology.hl7.org/NamingSystem/cmsMBI",
            "value": "1234567890A"
          }
        ],
        "name": [
          {
            "family": "Doe",
            "given": ["John"]
          }
        ],
        "gender": "male",
        "birthDate": "1950-01-01"
      }
    }
  ]
}

```



```

    },
    {
      "name": "track",
      "valueCodeableConcept": {
        "coding": [
          {
            "system": "https://dsacms.github.io/cmml-access-
model/CodeSystem/ACCESSTrackCS",
            "code": "CKM",
            "display": "Cardio-Kidney-Metabolic track"
          }
        ]
      }
    },
    {
      "name": "condition",
      "resource": {
        "resourceType": "Condition",
        "code": {
          "coding": [
            {
              "system": "http://hl7.org/fhir/sid/icd-10-cm",
              "code": "E11.9",
              "display": "Type 2 diabetes mellitus without complications"
            }
          ]
        },
        "subject": {
          "reference": "Patient/example"
        }
      }
    },
    {
      "name": "isProviderReferral",
      "valueBoolean": true
    }
  ]
}

```

### Example Response:

```

HTTP/1.1 202 Accepted
Content-Location: https://[base]/access/Patient/$submission-status/sub-
123456

```

(No response body - processing asynchronously)

### \$submission-status Operation

The \$submission-status operation retrieves the current status of a previously submitted alignment request. This is a **read-only operation** that uses the HTTP GET method.

**Endpoint:** GET [base]/access/Patient/\$submission-status/<submission-id>

**Understanding Response Behavior:**

- **HTTP 202 Accepted:** Request is still being processed asynchronously (no response body)
- **HTTP 200 OK:** Processing complete - response includes Parameters resource with alignment result
- **HTTP 4xx/5xx:** An error occurred. Response contains an OperationOutcome with error details (no Parameters resource)

**Alignment Result Codes:**

When processing is complete (HTTP 200), the `result` parameter uses codes from the **ACCESSAlignmentResultVS** value set:

- **aligned:** Patient is eligible and has been aligned so the participant can now begin providing services to the patient under the ACCESS Model.
- **aligned-switch-approved:** The request to switch the patient's alignment from a different participant after the 90-day lock in period is accepted and the patient is considered switched and now re-aligned.
- **not-aligned-control-group:** The patient is technically eligible, but based on the randomized control group algorithm, the patient has been placed in the control group for 12 months and therefore cannot be aligned for 12 months.
- **not-aligned-already-aligned:** The patient is technically eligible, but is already aligned to another participant and receiving services under the ACCESS Model in the same track. A patient can only be aligned to one participant in each track. If a switch consent attestation is submitted, but the patient is still within the 90-day lock-in period, this response will be received.
- **not-aligned-not-medicare:** The patient either: (1) is not enrolled in Medicare Part A and Part B; (2) is not dual eligible for Medicare and Medicaid; or (3) they do not have Medicare as their primary insurance, so they are not eligible for services under the ACCESS Model.
- **not-aligned-services:** The patient is receiving services (including receiving hospice services or dialysis for end stage renal disease (ESRD)) making them ineligible to be part of the ACCESS Model. Patients who are part of the Program of All-Inclusive Care for the Elderly (PACE) Program are also not eligible for the ACCESS Model.
- **not-aligned-diagnoses:** The patient does not have a treating diagnosis that qualifies them for service in the track indicated and therefore cannot get services under the ACCESS Model.
- **not-aligned-clinical-exclusion:** The patient has a clinical exclusion that prevents them from receiving services under the ACCESS Model.
- **not-aligned-mismatch:** The patient information provided does not match the patient information on record.

- **not-aligned-no-switch-attestation:** Participant switch request did not include switch consent attestation.

### Parameter Encoding for GET:

The submission ID is passed as part of the URL path:

- Path: GET [base]/access/Patient/\$submission-status/sub-123456

**NOTE:** If the \$align operation returns **aligned** or **align-switch-approved**, the **submissionID** from that request is used to provide context for push notifications (e.g., emails) related to the alignment, connecting the participant, patient, and track. That **submissionID** is provided in the Beneficiary Alignment Report for future reference.

### Example Status Check Request (Still Processing):

GET https://[base]/access/Patient/\$submission-status/sub-123456

### Response:

HTTP/1.1 202 Accepted

(No response body - still processing)

### Example Status Check Request (Processing Complete - Aligned):

GET https://[base]/access/Patient/\$submission-status/sub-123456

### Response:

HTTP/1.1 200 OK

Content-Type: application/fhir+json

```
{
  "resourceType": "Parameters",
  "parameter": [
    {
      "name": "result",
      "valueCodeableConcept": {
        "coding": [
          {
            "system": "https://dsacms.github.io/cmml-access-model/CodeSystem/ACCESSAlignmentResultCS",
            "code": "aligned",
            "display": "Aligned"
          }
        ],
        "text": "Patient is eligible and has been aligned so the participant can now begin providing services to the patient under the ACCESS Model."
      }
    }
  ]
}
```

```
]
}
```

### Example Status Check Request (Processing Complete - Not Aligned - No Qualifying Diagnosis):

GET https://[base]/access/Patient/\$submission-status/sub-123456

#### Response:

HTTP/1.1 200 OK

Content-Type: application/fhir+json

```
{
  "resourceType": "Parameters",
  "parameter": [
    {
      "name": "result",
      "valueCodeableConcept": {
        "coding": [
          {
            "system": "https://dsacms.github.io/cmmi-access-
model/CodeSystem/ACCESSAlignmentResultCS",
            "code": "not-aligned-diagnoses",
            "display": "Not aligned - no qualifying diagnosis"
          }
        ],
        "text": "The patient does not have a treating diagnosis that
qualifies them for service in the track indicated and therefore cannot
get services under the ACCESS Model."
      }
    }
  ]
}
```

### Example Status Check Request (Processing Complete - Not Aligned - Control Group):

GET https://[base]/access/Patient/\$submission-status/sub-123456

#### Response:

HTTP/1.1 200 OK

Content-Type: application/fhir+json

```
{
  "resourceType": "Parameters",
  "parameter": [
    {
      "name": "result",
      "valueCodeableConcept": {
        "coding": [
          {
            "system": "https://dsacms.github.io/cmmi-access-
model/CodeSystem/ACCESSAlignmentResultCS",
            "code": "not-aligned-control-group",
            "display": "Not aligned - assigned to Control Group"
          }
        ],

```

```

      "text": "The patient is technically eligible, but based on the
      randomized control group algorithm, the patient has been placed in the
      control group for 12 months and therefore cannot be aligned for 12
      months."
    }
  }
]
}

```

### Example Status Check Request (Processing Complete - Aligned and Switch Approved):

GET https://[base]/access/Patient/\$submission-status/sub-123456

### Response:

HTTP/1.1 200 OK

Content-Type: application/fhir+json

```

{
  "resourceType": "Parameters",
  "parameter": [
    {
      "name": "result",
      "valueCodeableConcept": {
        "coding": [
          {
            "system": "https://dsacms.github.io/cmami-access-
model/CodeSystem/ACCESSAlignmentResultCS",
            "code": "aligned-switch-approved",
            "display": "Aligned and switch approved"
          }
        ],
        "text": "The request to switch the patient's alignment from a
different provider after the 90-day lock in period is accepted and the
patient is considered switched and now re-aligned."
      }
    }
  ]
}

```

### ACCESS Participant Switch Workflow

When a patient wishes to switch ACCESS participants after the 90-day lock-in period:

1. **Verify Eligibility to Switch:** First, use the Eligibility API to check if the patient is eligible to switch participants (result code `eligible-switch-participants`)
2. **Obtain Consent:** Obtain and document the patient's consent to switch participants
3. **Submit Switch Request:** Submit an alignment request with the `switchConsentAttestation` parameter set to `true`
4. **Process Result:** If approved, receive result code `aligned-switch-approved`

### Important Notes:

- All requirements related to "switching" (switch consent, etc.) are only applicable to switching participants. For MSK and BH tracks, this is switching participants within

a track. For the eCKM and CKM tracks this is switching participants within a track or switching participants and moving from eCKM to CKM.

- If a patient is switching participants and moving from the eCKM track to the CKM track, the switch can happen at any time - it is not restricted by the 90-day lock-in period.
- For the other tracks, patients must be aligned for at least 90 days before they can switch participants. If a switch request is submitted before the 90-day lock-in period expires, the result will be `not-aligned-already-aligned`.
- The switch consent attestation must be documented to indicate the patient's explicit consent to change participants.
- When a patient switches participants, the 12-month clock for the period of care restarts with the new alignment date - the date the switch was approved.
- If the patient is staying with the same participant, but moving from the eCKM track to the CKM track due to a change in health status, this will require the participant to first unalign the patient from the eCKM track noting a clinical exclusion. A second API call will then need to be submitted to realign the patient to the CKM track.
- If a participant attempts to submit an alignment request for a patient to the CKM track who is in the eCKM track without first unaligning the patient, they will receive a `not-aligned-already-aligned` response indicating the patient is already aligned.

#### Example Switch Request with Consent Attestation:

POST `https://[base]/access/Patient/$align?entityId=ACCES12345`  
Content-Type: `application/fhir+json`

```
{
  "resourceType": "Parameters",
  "parameter": [
    {
      "name": "participantID",
      "valueIdentifier": {
        "system": "https://dsacms.github.io/cmml-access-
model/participant-id",
        "value": "ACCES12345"
      }
    },
    {
      "name": "payerID",
      "valueIdentifier": {
        "type": {
          "coding": [{
            "system": "http://hl7.org/fhir/us/carin-
bb/CodeSystem/C4BBIdentifierType",
            "code": "payerid",
            "display": "Payer ID"
          }]
        }
      },
      "system": "urn:oid:2.16.840.1.113883.3.221.5",
    }
  ]
}
```

```

        "value": "12345"
    }
},
{
    "name": "patient",
    "resource": {
        "resourceType": "Patient",
        "id": "example",
        "identifier": [
            {
                "type": {
                    "coding": [
                        {
                            "system": "http://terminology.hl7.org/CodeSystem/v2-
0203",
                            "code": "MC"
                        }
                    ]
                },
                "system": "http://terminology.hl7.org/NamingSystem/cmsMBI",
                "value": "1234567890A"
            }
        ],
        "name": [
            {
                "family": "Doe",
                "given": ["John"]
            }
        ],
        "gender": "male",
        "birthDate": "1950-01-01"
    }
},
{
    "name": "track",
    "valueCodeableConcept": {
        "coding": [
            {
                "system": "https://dsacms.github.io/cmml-access-
model/CodeSystem/ACCESSTrackCS",
                "code": "CKM",
                "display": "Cardio-Kidney-Metabolic track"
            }
        ]
    }
},
{
    "name": "condition",
    "resource": {
        "resourceType": "Condition",
        "code": {
            "coding": [
                {
                    "system": "http://hl7.org/fhir/sid/icd-10-cm",
                    "code": "E11.9",

```

```

        "display": "Type 2 diabetes mellitus without complications"
      }
    ]
  },
  "subject": {
    "reference": "Patient/example"
  }
}
},
{
  "name": "isProviderReferral",
  "valueBoolean": false
},
{
  "name": "switchConsentAttestation",
  "valueBoolean": true
}
]
}

```

### Response if Switch Approved:

HTTP/1.1 202 Accepted

Content-Location: [https://\[base\]/access/Patient/\\$submission-status/sub-789012](https://[base]/access/Patient/$submission-status/sub-789012)

Then polling \$submission-status would eventually return:

```

{
  "resourceType": "Parameters",
  "parameter": [
    {
      "name": "result",
      "valueCodeableConcept": {
        "coding": [
          {
            "system": "https://dsacms.github.io/cmml-access-
model/CodeSystem/ACCESSAlignmentResultCS",
            "code": "aligned-switch-approved",
            "display": "Aligned and switch approved"
          }
        ],
        "text": "The request to switch the patient's alignment from a
different participant after the 90-day lock in period is accepted and the
patient is considered switched and now re-aligned."
      }
    }
  ]
}

```

- **not-aligned-no-switch-attestation:** Participant switch request did not include switch consent attestation.



### *Automatic Push Notifications*

When an alignment is successful (result codes `aligned` or `aligned-switch-approved`), the system automatically begins sending push notifications via email about important upcoming events and deadlines. This helps ACCESS participants stay informed about critical dates and required actions.

#### **Automatic Email Notifications are Sent for:**

1. **Provider Lock-In Period Ending** - Notifications before the 90-day lock-in period ends
2. **Control Group Period Ending** - Notifications before the 12-month control group period ends
3. **Alignment Renewal Due** - Notifications when alignment renewal is approaching when a follow-on year is allowed
4. **Baseline Data Reporting Due** - Notifications when the deadline for baseline data submission is approaching
5. **Quarterly Data Reporting Due** - Notifications when the deadline for quarterly data submission is approaching
6. **End-of-Year Data Reporting Due** - Notifications when the deadline for end-of-year data submission is approaching
7. **Unalignment** - Notification for when a patient has been unaligned from the participant and/or track

#### **What This Means for Implementers:**

- **Email Addresses Configured During Onboarding:** Email addresses for receiving notifications are set up during participant registration
- **Automatic Notifications:** Emails are automatically sent when events occur—no additional configuration needed
- **Monitor Email:** Ensure registered email addresses are monitored regularly for time-sensitive notifications
- **Take Timely Action:** Respond to notifications before deadlines to maintain compliance

For complete details on email notification content, event types, and implementation guidance, see the Subscriptions section.

### *Best Practices*

1. **Track Validation:** Verify that the track corresponds to a valid ACCESS track before submission
2. **Diagnosis Validation:** Ensure submitted conditions are relevant to the requested track
3. **Eligibility Pre-Check:** Consider checking eligibility via the Eligibility API before submitting alignment requests

### *Example Complete Workflow*

The following example demonstrates a complete alignment workflow:

#### **Step 1: Submit Alignment Request**

POST https://[base]/access/Patient/\$align?entityId=ACCES12345  
Content-Type: application/fhir+json

```
{
  "resourceType": "Parameters",
  "parameter": [
    {
      "name": "participantID",
      "valueIdentifier": {
        "system": "https://dsacms.github.io/cmmi-access-
model/participant-id",
        "value": "ACCES12345"
      }
    },
    {
      "name": "payerID",
      "valueIdentifier": {
        "type": {
          "coding": [{
            "system": "http://hl7.org/fhir/us/carine-
bb/CodeSystem/C4BBIdentifierType",
            "code": "payerid",
            "display": "Payer ID"
          }]
        },
        "system": "urn:oid:2.16.840.1.113883.3.221.5",
        "value": "12345"
      }
    },
    {
      "name": "patient",
      "resource": {
        "resourceType": "Patient",
        "id": "example",
        "identifier": [
          {
            "type": {
              "coding": [
                {
                  "system": "http://terminology.hl7.org/CodeSystem/v2-
0203",
                  "code": "MC"
                }
              ]
            },
            "system": "http://terminology.hl7.org/NamingSystem/cmsMBI",
            "value": "1234567890A"
          }
        ],
        "name": [
          {
            "family": "Doe",
            "given": ["John"]
          }
        ]
      }
    }
  ]
}
```

```

        "gender" : "male",
        "birthDate": "1950-01-01"
    }
},
{
    "name": "track",
    "valueCodeableConcept": {
        "coding": [
            {
                "system": "https://dsacms.github.io/cmml-access-
model/CodeSystem/ACCESSTrackCS",
                "code": "CKM",
                "display": "Cardio-Kidney-Metabolic track"
            }
        ]
    }
},
{
    "name": "condition",
    "resource": {
        "resourceType": "Condition",
        "code": {
            "coding": [
                {
                    "system": "http://hl7.org/fhir/sid/icd-10-cm",
                    "code": "E11.9",
                    "display": "Type 2 diabetes mellitus without complications"
                }
            ]
        },
        "subject": {
            "reference": "Patient/example"
        }
    }
},
{
    "name": "condition",
    "resource": {
        "resourceType": "Condition",
        "code": {
            "coding": [
                {
                    "system": "http://hl7.org/fhir/sid/icd-10-cm",
                    "code": "I70.0",
                    "display": "Atherosclerosis of aorta"
                }
            ]
        },
        "subject": {
            "reference": "Patient/example"
        }
    }
},
{
    "name": "isProviderReferral",

```

```

        "valueBoolean": true
    }
]
}

```

### Response:

```

HTTP/1.1 202 Accepted
Content-Location: https://[base]/access/Patient/$submission-status/sub-123456

```

### Step 2: Extract Submission URL and Wait

- Extract submission URL from Content-Location header
- Wait 5 seconds before first poll

### Step 3: First Status Check (after 5 seconds)

```
GET https://[base]/access/Patient/$submission-status/sub-123456
```

### Response:

```
HTTP/1.1 202 Accepted
```

(Still processing - no body)

### Step 4: Second Status Check (after another 15 seconds)

```
GET https://[base]/access/Patient/$submission-status/sub-123456
```

### Response:

```

HTTP/1.1 200 OK
Content-Type: application/fhir+json

```

```

{
  "resourceType": "Parameters",
  "parameter": [
    {
      "name": "result",
      "valueCodeableConcept": {
        "coding": [
          {
            "system": "https://dsacms.github.io/cmami-access-model/CodeSystem/ACCESSAlignmentResultCS",
            "code": "aligned",
            "display": "Aligned"
          }
        ]
      },
      "text": "Patient is eligible and has been aligned so the

```

```

participant can now begin providing services to the patient under the
ACCESS Model."
    }
  }
]
}

```

## Step 5: Process Result

- Parse the result code: "aligned"
- Display the text to the user
- Begin providing services to the patient under the ACCESS Model

### API-Specific Response Scenarios

For general response handling guidance and common scenarios (missing parameters, invalid tokens, server errors, etc.), see the Error Handling section in General Guidance.

The following responses are specific to the Alignment API and are returned as part of normal processing:

#### Not Aligned - Already Aligned (Lock-In Period Active)

**Scenario:** Patient is already aligned to another provider and is still within the 90-day lock-in period.

```
GET https://[base]/access/Patient/$submission-status/sub-123456
```

```
HTTP/1.1 200 OK
```

```
Content-Type: application/fhir+json
```

```

{
  "resourceType": "Parameters",
  "parameter": [
    {
      "name": "result",
      "valueCodeableConcept": {
        "coding": [
          {
            "system": "https://dsacms.github.io/cmml-access-
model/CodeSystem/ACCESSAlignmentResultCS",
            "code": "not-aligned-already-aligned",
            "display": "Not aligned - already aligned to another
participant in the track"
          }
        ],
        "text": "The patient is technically eligible, but is already
aligned to another participant and receiving services under the ACCESS
Model in the same track. A patient can only be aligned to one participant
in each track. If a switch consent attestation is submitted, but the
patient is still within the 90-day lock-in period, this response will be
received."
      }
    }
  ]
}

```

```

    }
  }
]
}

```

#### Not Aligned - Missing Required Consent for Switch

**Scenario:** Patient is attempting to switch providers after lock-in period but consent attestation is missing.

HTTP/1.1 400 Bad Request

Content-Type: application/fhir+json

```

{
  "resourceType": "OperationOutcome",
  "issue": [
    {
      "severity": "error",
      "code": "required",
      "details": {
        "text": "Switch consent attestation is required when patient is
switching participants after lock-in period"
      },
      "diagnostics": "Patient is eligible to switch participants but the
'switchConsentAttestation' parameter must be provided."
    }
  ]
}

```

#### Not Aligned - No Qualifying Diagnosis

**Scenario:** Patient does not have a diagnosis that qualifies for the requested track.

HTTP/1.1 200 OK

Content-Type: application/fhir+json

```

{
  "resourceType": "Parameters",
  "parameter": [
    {
      "name": "result",
      "valueCodeableConcept": {
        "coding": [
          {
            "system": "https://dsacms.github.io/cmami-access-
model/CodeSystem/ACCESSAlignmentResultCS",
            "code": "not-aligned-diagnoses",
            "display": "Not aligned - no qualifying diagnosis"
          }
        ],
        "text": "The patient does not have a treating diagnosis that
qualifies them for service in the track indicated and therefore cannot
get services under the ACCESS Model."
      }
    }
  ]
}

```

```
]
}
```

### Not Aligned – Clinical Exclusion

**Scenario:** The patient has a clinical exclusion that prevents them from receiving services under the ACCESS Model.

HTTP/1.1 200 OK

Content-Type: application/fhir+json

```
{
  "resourceType": "Parameters",
  "parameter": [
    {
      "name": "result",
      "valueCodeableConcept": {
        "coding": [
          {
            "system": "https://dsacms.github.io/cmami-access-
model/CodeSystem/ACCESSAlignmentResultCS",
            "code": "not-aligned-clinical-exclusion",
            "display": "Not aligned - clinical exclusion"
          }
        ],
        "text": "The patient has a clinical exclusion that prevents them
from receiving services under the ACCESS Model."
      }
    ]
  }
}
```

### Not Aligned – Mismatch

**Scenario:** The patient information provided does not match the patient information on record.

HTTP/1.1 200 OK

Content-Type: application/fhir+json

```
{
  "resourceType": "Parameters",
  "parameter": [
    {
      "name": "result",
      "valueCodeableConcept": {
        "coding": [
          {
            "system": "https://dsacms.github.io/cmami-access-
model/CodeSystem/ACCESSAlignmentResultCS",
            "code": "not-aligned-mismatch",
            "display": "Not aligned - mismatch"
          }
        ],
        "text": "The patient information provided does not match the
patient information on record."
      }
    ]
  }
}
```

```

    }
  }
]
}

```

### Missing Required Conditions Parameter

**Scenario:** Alignment request submitted without required conditions.

POST https://[base]/access/Patient/\$align?entityId=ACCES12345

HTTP/1.1 400 Bad Request

Content-Type: application/fhir+json

```

{
  "resourceType": "OperationOutcome",
  "issue": [
    {
      "severity": "error",
      "code": "required",
      "details": {
        "text": "Missing required parameter: conditions"
      },
      "diagnostics": "At least one Condition resource must be provided
for alignment requests. Conditions must conform to the US Core Condition
profile and include ICD-10-CM codes."
    }
  ]
}

```

**Note:** These scenarios represent a mix of business logic results (HTTP 200 with specific result codes) and technical validation errors (HTTP 400). Always check the HTTP status code first to determine the type of response.

## Implementation Notes and Edge Cases

### *Patient Switches Between Participants*

**Scenario:** Patient aligned to ACCESS Participant A wants to switch to Participant B

#### **Implementation Guidance:**

1. Participant B checks eligibility (may return `not-eligible-already-aligned` if within 90-day lock-in period)
2. Wait until 90-day lock-in period expires
3. Patient provides consent to switch and Participant documents the consent
4. Participant B submits alignment with `switchConsentAttestation = true`.
5. If approved, patient is automatically unaligned from Participant A
6. Participant A receives unalignment notification automatically



**Important Notes:**

- Participant A does NOT need to manually unalign the patient
- Participant B handles the entire switch process
- System ensures only one active alignment per track
- Lock-in period prevents frequent participant switching

## Unalignment API Guidance

### How to Use the API

The ACCESS Unalignment API uses an asynchronous pattern for processing manual unalignment requests for patients in the ACCESS Model. The pattern uses two operations to support a submit-and-poll workflow:

1. **\$unalign**: Submits an unalignment request
2. **\$submission-status**: Polls for the status of a submitted request

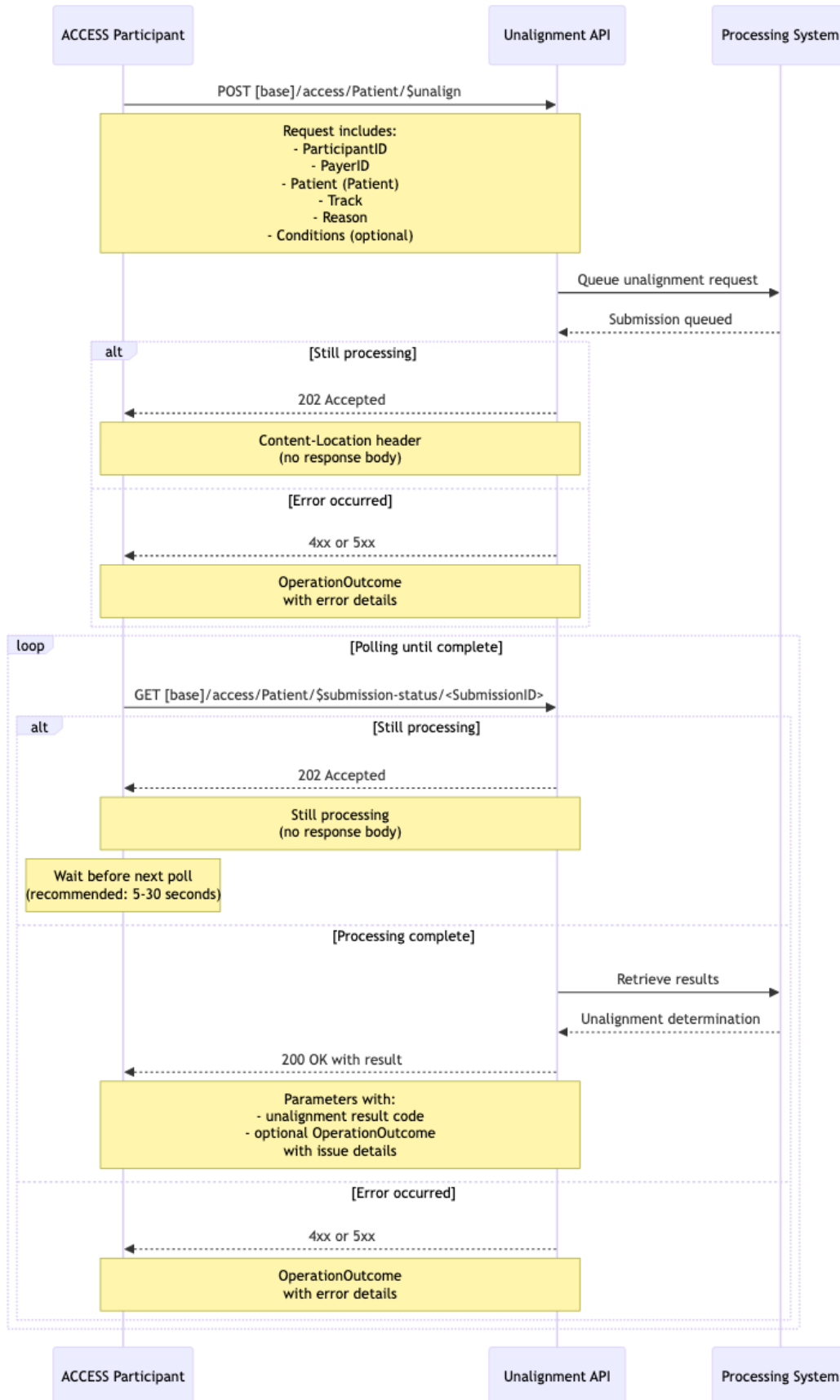
### Asynchronous Interaction Pattern

The Unalignment API uses an asynchronous request-response pattern to accommodate the processing time required for unalignment determination. This pattern allows clients to submit unalignment requests and subsequently poll for results without maintaining a persistent connection.

These operations can take some time to complete, making a synchronous request-response pattern impractical for real-world implementations.

### Detailed Interaction Sequence

The following sequence diagram illustrates the complete interaction pattern:



## Operation Details

### \$unalign Operation

The \$unalign operation initiates an asynchronous manual unalignment request.

**Endpoint:** POST [base]/access/Patient/\$unalign?  
entityId=[participantID]

#### Input Parameters:

- **participantID** (Identifier, required): The ACCESS ID for the submitting participant
- **payerID** (Identifier, required): The payer ID for the patient. Uses the X12 EDI Payer ID standard (urn:oid:2.16.840.1.113883.3.221.5) with CARIN Blue Button identifier typing (<http://hl7.org/fhir/us/carin-bb/CodeSystem/C4BBIdentifierType>). EDI Payer IDs are 5-character alphanumeric codes widely used in healthcare transactions (e.g., eligibility checks, claims, remittance).
- **patient** (Patient Resource, required): Patient information conforming to US Core Patient Profile. **SHOULD** contain the Medicare Beneficiary Identifier (MBI)
- **track** (Identifier, required): The track from which the patient should be unaligned
- **reason** (CodeableConcept, required): The reason for the manual unalignment request. Value from **ACCESSUnalignmentReasonVS** value set
- **conditions** (Condition Resources, conditional): Zero or more Condition resources conforming to the ACCESS Condition Profile describing the patient's health concerns. **NOTE: This parameter is required** when reason is no-longer-clinically-eligible to document the disqualifying diagnosis (enforced by invariant access-unalign-condition-required)

#### Expected Response:

- **Status Code:** 202 Accepted
- **HTTP Headers:** Content-Location header containing the URL to check submission status
- **Response Body:** Empty (no body while processing)

#### Example Request:

POST https://[base]/access/Patient/\$unalign?entityId=ACCES12345  
Content-Type: application/fhir+json

```
{
  "resourceType": "Parameters",
  "parameter": [
    {
      "name": "participantID",
      "valueIdentifier": {
        "system": "https://dsacms.github.io/cmmi-access-
```

```

model/participant-id",
  "value": "ACCESS12345"
},
{
  "name": "payerID",
  "valueIdentifier": {
    "type": {
      "coding": [{
        "system": "http://hl7.org/fhir/us/carin-
bb/CodeSystem/C4BBIdentifierType",
        "code": "payerid",
        "display": "Payer ID"
      }]
    },
    "system": "urn:oid:2.16.840.1.113883.3.221.5",
    "value": "12345"
  }
},
{
  "name": "patient",
  "resource": {
    "resourceType": "Patient",
    "identifier": [
      {
        "type": {
          "coding": [
            {
              "system": "http://terminology.hl7.org/CodeSystem/v2-
0203",
              "code": "MC"
            }
          ]
        },
        "system": "http://terminology.hl7.org/NamingSystem/cmsMBI",
        "value": "1234567890A"
      }
    ],
    "name": [
      {
        "family": "Doe",
        "given": ["John"]
      }
    ],
    "gender": "male",
    "birthDate": "1950-01-01"
  }
},
{
  "name": "track",
  "valueCodeableConcept": {
    "coding": [
      {
        "system": "https://dsacms.github.io/cmmi-access-
model/CodeSystem/ACCESSTrackCS",

```

```

        "code": "CKM",
        "display": "Cardio-Kidney-Metabolic track"
      }
    ]
  },
  {
    "name": "reason",
    "valueCodeableConcept": {
      "coding": [
        {
          "system": "https://dsacms.github.io/cmml-access-
model/CodeSystem/ACCESSUnalignmentReasonCS",
          "code": "loss-of-contact",
          "display": "Loss of contact"
        }
      ]
    }
  }
]
}

```

### Example Response:

HTTP/1.1 202 Accepted  
 Content-Location: https://[base]/access/Patient/\$submission-status/sub-123456

(No response body - processing asynchronously)

### Example Request (No Longer Clinically Eligible):

This example demonstrates unalignment when a patient has developed a new condition that makes them ineligible for the ACCESS Model. In this case, the patient was aligned to the CKM (Cardio-kidney-metabolic) track for diabetes management, but has now developed end-stage renal disease (ESRD) requiring dialysis, which makes them ineligible for the ACCESS Model per program requirements.

POST https://[base]/access/Patient/\$unalign?entityId=ACCES12345  
 Content-Type: application/fhir+json

```

{
  "resourceType": "Parameters",
  "parameter": [
    {
      "name": "participantID",
      "valueIdentifier": {
        "system": "https://dsacms.github.io/cmml-access-
model/participant-id",
        "value": "ACCES12345"
      }
    }
  ],
  {

```

```

    "name": "payerID",
    "valueIdentifier": {
      "type": {
        "coding": [{
          "system": "http://hl7.org/fhir/us/carin-
bb/CodeSystem/C4BBIdentifierType",
          "code": "payerid",
          "display": "Payer ID"
        }]
      },
      "system": "urn:oid:2.16.840.1.113883.3.221.5",
      "value": "12345"
    }
  },
  {
    "name": "patient",
    "resource": {
      "resourceType": "Patient",
      "id": "example",
      "identifier": [
        {
          "type": {
            "coding": [
              {
                "system": "http://terminology.hl7.org/CodeSystem/v2-
0203",
                "code": "MC"
              }
            ]
          },
          "system": "http://terminology.hl7.org/NamingSystem/cmsMBI",
          "value": "9876543210B"
        }
      ],
      "name": [
        {
          "family": "Smith",
          "given": ["Jane"]
        }
      ],
      "gender": "female",
      "birthDate": "1955-06-15"
    }
  },
  {
    "name": "track",
    "valueCodeableConcept": {
      "coding": [
        {
          "system": "https://dsacms.github.io/cmml-access-
model/CodeSystem/ACCESSTrackCS",
          "code": "CKM",
          "display": "Cardio-Kidney-Metabolic track"
        }
      ]
    }
  }
]

```

```

    }
  },
  {
    "name": "reason",
    "valueCodeableConcept": {
      "coding": [
        {
          "system": "https://dsacms.github.io/cmmi-access-
model/CodeSystem/ACCESSUnalignmentReasonCS",
          "code": "no-longer-clinically-eligible",
          "display": "No longer clinically eligible"
        }
      ],
      "text": "Patient has developed end-stage renal disease (ESRD)
requiring dialysis, which makes them ineligible for the ACCESS Model."
    }
  },
  {
    "name": "condition",
    "resource": {
      "id": "ESRDConditionExample",
      "meta": {
        "profile": [
          "https://dsacms.github.io/cmmi-access-
model/StructureDefinition/access-clinical-exclusion-condition"
        ]
      },
      "text": {
        "status": "generated",
        "div": "<div xmlns=\"http://www.w3.org/1999/xhtml\">End stage
renal disease (ESRD)</div>"
      },
      "resourceType": "Condition",
      "clinicalStatus": {
        "coding": [
          {
            "system": "http://terminology.hl7.org/CodeSystem/condition-
clinical",
            "code": "active",
            "display": "Active"
          }
        ]
      },
      "verificationStatus": {
        "coding": [
          {
            "system": "http://terminology.hl7.org/CodeSystem/condition-
ver-status",
            "code": "confirmed",
            "display": "Confirmed"
          }
        ]
      },
      "category": [
        {

```



```

        "coding": [
          {
            "system":
"http://terminology.hl7.org/CodeSystem/condition-category",
            "code": "problem-list-item",
            "display": "Problem List Item"
          }
        ]
      },
      "code": {
        "coding": [
          {
            "system": "http://hl7.org/fhir/sid/icd-10-cm",
            "code": "N18.6",
            "display": "End stage renal disease"
          }
        ]
      },
      "subject": {
        "reference": "Patient/example"
      },
      "onsetDateTime": "2026-01-15"
    }
  ]
}

```

### Example Response:

```

HTTP/1.1 202 Accepted
Content-Location: https://[base]/access/Patient/$submission-status/sub-
789012

```

(No response body - processing asynchronously)

### \$submission-status Operation

The \$submission-status operation retrieves the current status of a previously submitted unalignment request. This is a **read-only operation** that uses the HTTP GET method.

**Endpoint:** GET [base]/access/Patient/\$submission-status/<submission-id>

### Output Parameters (when complete):

- **result** (CodeableConcept, required): Result code with descriptive text indicating unalignment status. Value from **ACCESSUnalignmentResultVS** value set.

### HTTP Status Codes:

- **202 Accepted:** Request is still being processed asynchronously (no response body)

- **200 OK:** Processing complete - response includes Parameters resource with unalignment result
- **400 Bad Request:** Invalid request (returns OperationOutcome)
- **404 Not Found:** Submission ID not found (returns OperationOutcome)
- **500 Internal Server Error:** Server error during processing (returns OperationOutcome)

#### Unalignment Result Codes:

When processing is complete (HTTP 200), the `result` parameter uses codes from the **ACCESSUnalignmentResultVS** value set:

- **unaligned:** The request to unalign the patient has been accepted and the patient has been successfully unaligned.
- **unalignment-pending:** Additional review is needed and you will receive further information once the manual review of the unalignment request is complete.
- **patient-not-aligned:** Patient is not currently aligned to this participant in the specified track and therefore cannot be unaligned.
- **cannot-unalign-during-lock-in:** Patient is within the 90-day lock-in period and cannot be unaligned.

#### Unalignment Reason Codes:

The `reason` parameter in the `$unalign` operation uses codes from the **ACCESSUnalignmentReasonVS** value set:

- **geographic-relocated:** Patient has relocated outside of the geographic area in which the aligned participant is licensed to provide services.
- **loss-of-contact:** Despite good faith efforts (defined as making three or more outreach attempts over 30 or more days), contact with the patient has been lost and therefore no longer able to engage with them.
- **no-longer-clinically-eligible:** Patient's conditions have changed such that they are no longer eligible for the ACCESS Model.
- **patient-initiated:** Patient no longer wants to participate in the ACCESS Model after the initial 90-day lock-in period.

#### Parameter Encoding for GET:

The submission ID is passed as part of the URL path:

- **Path:** GET `https://[base]/access/Patient/$submission-status/sub-123456`

#### Example Status Check Request (Still Processing):

GET `https://[base]/access/Patient/$submission-status/sub-123456`

#### Response:

HTTP/1.1 202 Accepted

(No response body - still processing)

### Example Status Check Request (Processing Complete - Unaligned):

GET https://[base]/access/Patient/\$submission-status/sub-123456

#### Response:

HTTP/1.1 200 OK

Content-Type: application/fhir+json

```
{
  "resourceType": "Parameters",
  "parameter": [
    {
      "name": "result",
      "valueCodeableConcept": {
        "coding": [
          {
            "system": "https://dsacms.github.io/cmami-access-
model/CodeSystem/ACCESSUnalignmentResultCS",
            "code": "unaligned",
            "display": "Unaligned"
          }
        ],
        "text": "The request to unalign the patient has been accepted and
the patient has been successfully unaligned."
      }
    }
  ]
}
```

### Example Status Check Request (Pending Manual Review):

GET https://[base]/access/Patient/\$submission-status/sub-123456

#### Response:

HTTP/1.1 200 OK

Content-Type: application/fhir+json

```
{
  "resourceType": "Parameters",
  "parameter": [
    {
      "name": "result",
      "valueCodeableConcept": {
        "coding": [
          {
            "system": "https://dsacms.github.io/cmami-access-
model/CodeSystem/ACCESSUnalignmentResultCS",
            "code": "unalignment-pending",
            "display": "Unalignment pending further review"
          }
        ]
      }
    }
  ]
}
```

```

        }
      ],
      "text": "Additional review is needed and you will receive further
information once the manual review of the unalignment request is
complete."
    }
  }
]
}

```

### Example Status Check Request (Patient Not Aligned):

GET https://[base]/access/Patient/\$submission-status/sub-123456

#### Response:

HTTP/1.1 200 OK

Content-Type: application/fhir+json

```

{
  "resourceType": "Parameters",
  "parameter": [
    {
      "name": "result",
      "valueCodeableConcept": {
        "coding": [
          {
            "system": "https://dsacms.github.io/cmmi-access-
model/CodeSystem/ACCESSUnalignmentResultCS",
            "code": "patient-not-aligned",
            "display": "Patient not aligned"
          }
        ],
        "text": "Patient is not currently aligned to this participant in
the specified track and therefore cannot be unaligned."
      }
    }
  ]
}

```

### Example Status Check Request (Cannot Unalign During Lock-in):

GET https://[base]/access/Patient/\$submission-status/sub-123456

#### Response:

HTTP/1.1 200 OK

Content-Type: application/fhir+json

```

{
  "resourceType": "Parameters",
  "parameter": [
    {
      "name": "result",
      "valueCodeableConcept": {
        "coding": [
          {

```

```

        "system": "https://dsacms.github.io/cmmi-access-
model/CodeSystem/ACCESSUnalignmentResultCS",
        "code": "cannot-unalign-during-lock-in",
        "display": "Cannot unalign during lock-in"
    }
],
    "text": "Patient is within the 90-day lock-in period and cannot
be unaligned."
}
}
]
}

```

### **Automatic Notification Cleanup**

When a patient is successfully unaligned (result code `unaligned`), the system automatically stops sending email notifications about that patient. This ensures participants only receive notifications for currently aligned patients.

**NOTE:** Subscriptions will be maintained until all model obligations are completed for the patient. Some obligations will occur post unalignment.

### **Unalignment Notification Process:**

When the `$unalign` operation completes successfully, the following occurs automatically:

1. **Unalignment Email Sent:** The system sends an email notification to the participant later that day informing them that the patient has been unaligned and the alignment relationship has ended.
2. **Future Notifications Stopped:** The system automatically stops sending future email notifications about events for that patient (e.g., reporting deadlines, renewal reminders) after all model obligations are complete for the patient.

### **Important Notes:**

- **Automatic Process:** Both the unalignment notification delivery and stopping of future notifications happen automatically as part of the unalignment workflow. No additional action is required by the ACCESS participant.
- **Notification Sent First:** An unalignment notification email is sent to inform the participant of the alignment termination before future notifications are stopped.
- **Prevents Future Notifications:** Stopping notifications ensures that participants do not receive future emails about patients who are no longer aligned.
- **Pending Unalignments:** If the unalignment result is `unalignment-pending` (requiring manual review), email notifications continue and no unalignment notification is sent until the unalignment is finalized with an `unaligned` result. Additional information can also be found in the Beneficiary Alignment Report for these cases.
- **Other Patients Unaffected:** Participant will continue to receive notifications for all other aligned patients.

### **Cross-Reference:**

For complete details on the email notifications that are sent during alignment, see the Subscriptions section and the Alignment API documentation.

### *Example Complete Workflow*

The following example demonstrates a complete unalignment request workflow:

#### **Step 1: Submit Unalignment Request**

POST [https://\[base\]/access/Patient/\\$unalign?entityId=ACCES12345](https://[base]/access/Patient/$unalign?entityId=ACCES12345)  
Content-Type: application/fhir+json

```
{
  "resourceType": "Parameters",
  "parameter": [
    {
      "name": "participantID",
      "valueIdentifier": {
        "system": "https://dsacms.github.io/cmml-access-
model/participant-id",
        "value": "ACCES12345"
      }
    },
    {
      "name": "payerID",
      "valueIdentifier": {
        "type": {
          "coding": [{
            "system": "http://hl7.org/fhir/us/carin-
bb/CodeSystem/C4BBIdentifierType",
            "code": "payerid",
            "display": "Payer ID"
          }]
        },
        "system": "urn:oid:2.16.840.1.113883.3.221.5",
        "value": "12345"
      }
    },
    {
      "name": "patient",
      "resource": {
        "resourceType": "Patient",
        "identifier": [
          {
            "type": {
              "coding": [
                {
                  "system": "http://terminology.hl7.org/CodeSystem/v2-
0203",
                  "code": "MC"
                }
              ]
            }
          }
        ]
      }
    }
  ]
}
```

```

        "system": "http://terminology.hl7.org/NamingSystem/cmsMBI",
        "value": "1234567890A"
    }
],
"name": [
    {
        "family": "Doe",
        "given": ["John"]
    }
],
"gender" : "male",
"birthDate": "1950-01-01"
}
},
{
    "name": "track",
    "valueCodeableConcept": {
        "coding": [
            {
                "system": "https://dsacms.github.io/cmml-access-
model/CodeSystem/ACCESSTrackCS",
                "code": "CKM",
                "display": "Cardio-Kidney-Metabolic track"
            }
        ]
    }
},
{
    "name": "reason",
    "valueCodeableConcept": {
        "coding": [
            {
                "system": "https://dsacms.github.io/cmml-access-
model/CodeSystem/ACCESSUnalignmentReasonCS",
                "code": "geographic-relocated",
                "display": "Geographic relocated"
            }
        ]
    },
    "text": "Patient has relocated outside of the geographic area in
which the aligned participant is licensed to provide services."
}
]
}

```

### Response:

HTTP/1.1 202 Accepted  
 Content-Location: https://[base]/access/Patient/\$submission-status/sub-123456

### Step 2: Extract Submission URL and Wait

- Extract submission URL "sub-123456" from Content-Location header
- Wait 5 seconds before first poll

### Step 3: First Status Check (after 5 seconds)

GET https://[base]/access/Patient/\$submission-status/sub-123456

#### Response:

HTTP/1.1 202 Accepted

(Still processing - no body)

### Step 4: Second Status Check (after another 15 seconds)

GET https://[base]/access/Patient/\$submission-status/sub-123456

#### Response:

HTTP/1.1 200 OK

Content-Type: application/fhir+json

```
{
  "resourceType": "Parameters",
  "parameter": [
    {
      "name": "result",
      "valueCodeableConcept": {
        "coding": [
          {
            "system": "https://dsacms.github.io/cmml-access-
model/CodeSystem/ACCESSUnalignmentResultCS",
            "code": "unaligned",
            "display": "Unaligned"
          }
        ],
        "text": "The request to unalign the patient has been accepted and
the patient has been successfully unaligned."
      }
    }
  ]
}
```

### Step 5: Process Result

- Parse the result code: "unaligned"
- Display the text to the user
- Complete unalignment workflow and update internal systems



### API-Specific Response Scenarios

For general response handling guidance and common scenarios (missing parameters, invalid tokens, server errors, etc.), see the Error Handling section in General Guidance.

The following responses are specific to the Unalignment API:

#### Not Unaligned - Patient Not Currently Aligned

**Scenario:** Attempting to unalign a patient who is not currently aligned to this ACCESS participant.

```
GET https://[base]/access/Patient/$submission-status/sub-123456

HTTP/1.1 200 OK
Content-Type: application/fhir+json

{
  "resourceType": "Parameters",
  "parameter": [
    {
      "name": "result",
      "valueCodeableConcept": {
        "coding": [
          {
            "system": "https://dsacms.github.io/cmami-access-
model/CodeSystem/ACCESSUnalignmentResultCS",
            "code": "patient-not-aligned",
            "display": "Patient not aligned"
          }
        ],
        "text": "Patient is not currently aligned to this participant in
the specified track and therefore cannot be unaligned."
      }
    }
  ]
}
```

#### Invalid Unalignment Reason

**Scenario:** Unalignment reason code is not recognized or invalid for the situation.

```
HTTP/1.1 400 Bad Request
Content-Type: application/fhir+json

{
  "resourceType": "OperationOutcome",
  "issue": [
    {
      "severity": "error",
      "code": "code-invalid",
      "details": {
        "text": "Invalid unalignment reason code"
      },
      "diagnostics": "The reason code 'invalid-reason' is not a valid
value from the ACCESSUnalignmentReasonVS value set. Valid reasons
include: geographic-relocated, loss-of-contact, no-longer-clinically-
eligible, patient-initiated."
    }
  ]
}
```

```

    }
  }
]
}

```

### Missing Required Reason Parameter

**Scenario:** Unalignment request submitted without the required reason parameter.

POST https://[base]/access/Patient/\$unalign?entityId=ACCES12345

HTTP/1.1 400 Bad Request

Content-Type: application/fhir+json

```

{
  "resourceType": "OperationOutcome",
  "issue": [
    {
      "severity": "error",
      "code": "required",
      "details": {
        "text": "Missing required parameter: reason"
      },
      "diagnostics": "The 'reason' parameter is required for all
unalignment requests and must contain a CodeableConcept from the
ACCESSUnalignmentReasonVS value set."
    }
  ]
}

```

**Note:** These scenarios represent a mix of business logic results (HTTP 200 with specific result codes) and technical validation errors (HTTP 400). Always check the HTTP status code first to determine the type of response.

## Data Reporting API Guidance

### How to Use the API

The ACCESS Data Reporting API uses an asynchronous pattern for submitting clinical data for patients enrolled in the CMS ACCESS Model. For more information about the data reporting requirements, please see the ACCESS Model Payment Amounts and Performance Targets (<https://www.cms.gov/priorities/innovation/files/access-payments-amts-perf-targets.pdf>).

The pattern uses two operations to support a submit-and-poll workflow:

1. **\$report-data**: Submits a data reporting bundle for an aligned patient
2. **\$submission-status**: Polls for the status of a submitted data report

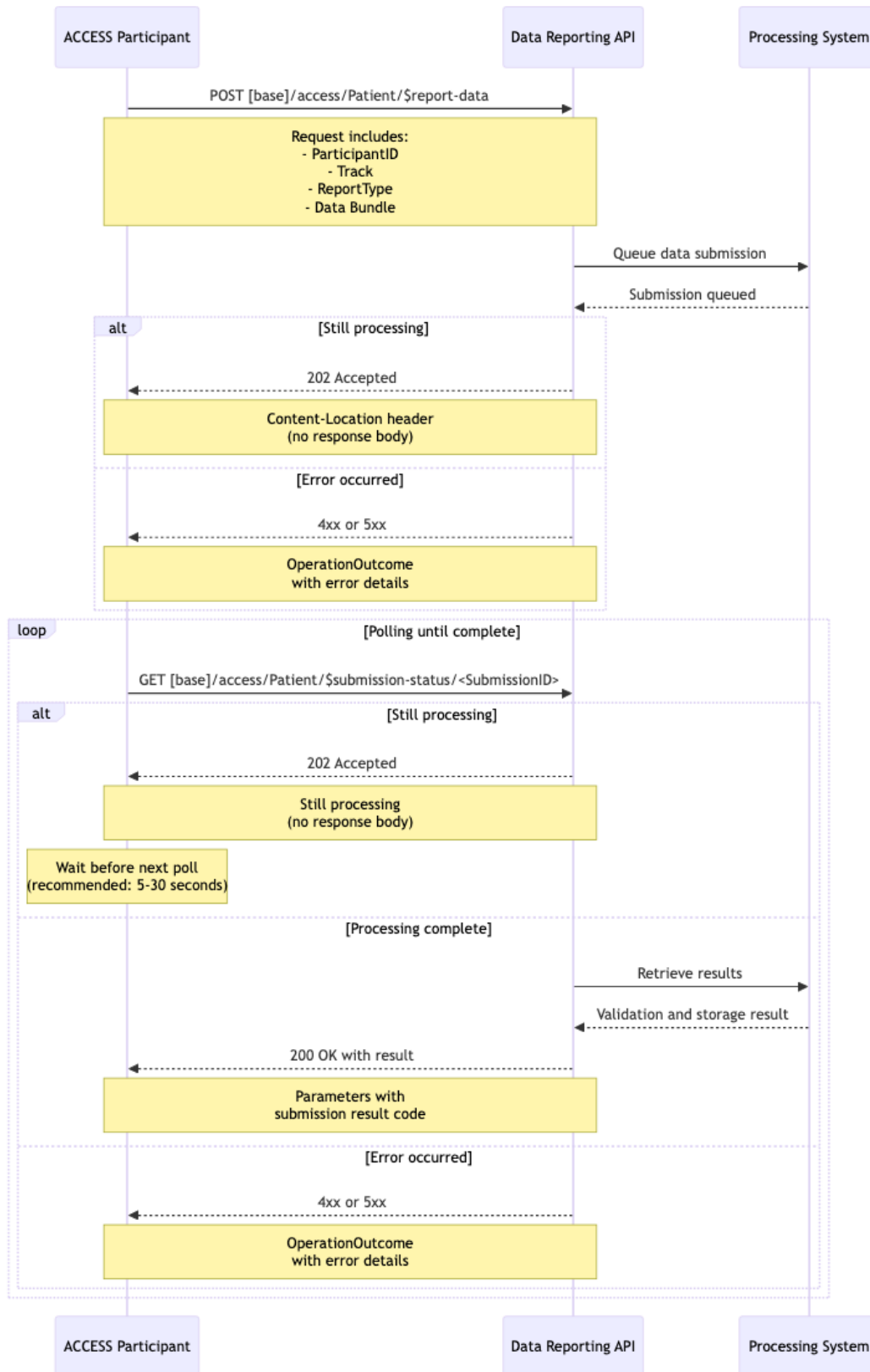
### Asynchronous Interaction Pattern

The Data Reporting API uses an asynchronous request-response pattern to accommodate the processing time required for data validation and storage. This pattern allows clients to submit data reports and subsequently poll for results without maintaining a persistent connection.

These operations can take some time to complete, making a synchronous request-response pattern impractical for real-world implementations.

### Detailed Interaction Sequence

The following sequence diagram illustrates the complete interaction pattern:



## Operation Details

### \$report-data Operation

The `$report-data` operation initiates an asynchronous data reporting submission for an aligned patient.

**Endpoint:** POST `[base]/access/Patient/$report-data?entityId=[participantID]`

#### Input Parameters:

- **participantID** (Identifier, required): The ACCESS ID for the submitting participant
- **payerID** (Identifier, required): The payer ID for the patient. Uses the X12 EDI Payer ID standard (`urn:oid:2.16.840.1.113883.3.221.5`) with CARIN Blue Button identifier typing (<http://hl7.org/fhir/us/carin-bb/CodeSystem/C4BBIdentifierType>). EDI Payer IDs are 5-character alphanumeric codes widely used in healthcare transactions (e.g., eligibility checks, claims, remittance).
- **track** (CodeableConcept, required): The ACCESS track for this submission (CKM, eCKM, MSK, or BH)
- **reportType** (CodeableConcept, required): The type of report being submitted (baseline, quarterly, end-of-period)
- **dataBundle** (Bundle Resource, required): A document Bundle conforming to `ACCESSDataReportingBundle` profile containing the data reporting composition and all referenced resources

**NOTE:** If participants submit the end-of-period data using `reportType = end-of-period` and resubmit the same data as the baseline for the follow-on period using `reportType = baseline`, it will not be considered a duplicate submission.

**NOTE:** Unlike the other ACCESS operations, the `Patient` resource instance is included within the `dataBundle` instead of being a separate parameter for the `$report-data` operation. The `Patient` resource instance is central to and referenced by the other data resource instances (e.g., `Observations` and `QuestionnaireResponses`) and therefore must be included within the `dataBundle`.

#### Expected Response:

- **Status Code:** 202 Accepted
- **HTTP Headers:** `Content-Location` header containing the URL to check submission status
- **Response Body:** Empty (no body while processing)

#### Example Request:

POST https://[base]/access/Patient/\$report-data?entityId=ACCES12345  
Content-Type: application/fhir+json

```
{
  "resourceType": "Parameters",
  "id": "CKMReportDataRequestExample",
  "meta": {
    "profile": [
      "https://dsacms.github.io/cmml-access-
model/StructureDefinition/access-report-data-in"
    ]
  },
  "language": "en",
  "parameter": [
    {
      "name": "participantID",
      "valueIdentifier": {
        "system": "https://dsacms.github.io/cmml-access-
model/participant-id",
        "value": "ACCES12345"
      }
    },
    {
      "name": "payerID",
      "valueIdentifier": {
        "type": {
          "coding": [
            {
              "system": "http://hl7.org/fhir/us/carin-
bb/CodeSystem/C4BBIdentifierType",
              "code": "payerid",
              "display": "Payer ID"
            }
          ]
        },
        "system": "urn:oid:2.16.840.1.113883.3.221.5",
        "value": "12345"
      }
    },
    {
      "name": "track",
      "valueCodeableConcept": {
        "coding": [
          {
            "system": "https://dsacms.github.io/cmml-access-
model/CodeSystem/ACCESSTrackCS",
            "code": "CKM",
            "display": "Cardio-Kidney-Metabolic track"
          }
        ]
      }
    },
    {
      "name": "reportType",
      "valueCodeableConcept": {
```

```

        "coding": [
            {
                "system": "https://dsacms.github.io/cmmi-access-
model/CodeSystem/ACCESSReportTypeCS",
                "code": "baseline",
                "display": "Baseline Data Report"
            }
        ]
    },
    {
        "name": "dataBundle",
        "resource": {
            "resourceType": "Bundle",
            "id": "CKMReportDataBundleExample",
            "meta": {
                "profile": [
                    "https://dsacms.github.io/cmmi-access-
model/StructureDefinition/access-data-reporting-bundle"
                ]
            },
            "language": "en",
            "identifier": {
                "system": "http://example.org/data-bundle-id",
                "value": "bundle-ckm-001"
            },
            "type": "document",
            "timestamp": "2026-01-26T12:00:00Z",
            "entry": [
                {
                    "fullUrl":
"http://example.org/Composition/CKMReportDataCompositionExample",
                    "resource": {
                        "resourceType": "Composition",
                        "id": "CKMReportDataCompositionExample",
                        "meta": {
                            "profile": [
                                "https://dsacms.github.io/cmmi-access-
model/StructureDefinition/access-data-reporting-composition"
                            ]
                        },
                        "language": "en",
                        "text": {
                            "status": "generated",
                            "div": "\u003Cdiv xml:lang=\"en\"
xmlns=\"http://www.w3.org/1999/xhtml\"
lang=\"en\"\u003E\u003Ch3\u003EACCESS CKM Track Data
Report\u003C/h3\u003E\u003Cp\u003EPatient: John Doe (MBI:
1EG4TE5MK73)\u003C/p\u003E\u003Cp\u003EReport Date: 2026-01-
26\u003C/p\u003E\u003Cp\u003EAuthor: Jane
Smith\u003C/p\u003E\u003Cdiv\u003E"
                        },
                        "status": "final",
                        "type": {
                            "coding": [

```

```

        {
            "system": "http://loinc.org",
            "code": "74465-6",
            "display": "Questionnaire response Document"
        }
    ]
},
"subject": {
    "reference": "Patient/PatientExample"
},
"date": "2026-01-26T12:00:00Z",
"author": [
    {
        "reference": "Practitioner/PractitionerExample"
    }
],
"title": "ACCESS CKM Track Data Report",
"custodian": {
    "reference": "Organization/OrganizationExample"
},
"section": [
    {
        "title": "Data reporting for CKM track",
        "code": {
            "coding": [
                {
                    "system": "https://dsacms.github.io/cmmi-access-
model/CodeSystem/ACCESSTrackCS",
                    "code": "CKM"
                }
            ]
        },
        "text": {
            "status": "generated",
            "div": "\u003Cdiv xml:lang=\"en\"
xmlns=\"http://www.w3.org/1999/xhtml\"
lang=\"en\"\u003E\u003Cp\u003EComprehensive CKM Track Data Reporting
including vital signs (blood pressure, body weight, waist circumference),
and laboratory results (HbA1c, eGFR, uACR, lipid
panel).\u003C/p\u003E\u003C/div\u003E"
        },
        "section": [
            {
                "title": "Blood Pressure",
                "code": {
                    "coding": [
                        {
                            "system": "http://loinc.org",
                            "code": "85354-9"
                        }
                    ]
                },
                "entry": [
                    {
                        "reference": "Observation/BloodPressureExample"
                    }
                ]
            }
        ]
    }
]

```



```

        },
        {
            "reference":
"Observation/BloodPressureExample2"
        },
        {
            "reference":
"Observation/BloodPressureExample3"
        }
    ]
},
{
    "title": "Weight",
    "code": {
        "coding": [
            {
                "system": "http://loinc.org",
                "code": "29463-7"
            }
        ]
    },
    "entry": [
        {
            "reference": "Observation/BodyWeightExample"
        }
    ]
},
{
    "title": "BMI",
    "code": {
        "coding": [
            {
                "system": "http://loinc.org",
                "code": "39156-5"
            }
        ]
    },
    "entry": [
        {
            "reference": "Observation/BMIExample"
        }
    ]
},
{
    "title": "Waist Circumference",
    "code": {
        "coding": [
            {
                "system": "http://loinc.org",
                "code": "8280-0"
            }
        ]
    },
    "entry": [
        {

```

```

        "reference":
"Observation/WaistCircumferenceExample"
    }
    ]
},
{
    "title": "HbA1c",
    "code": {
        "coding": [
            {
                "system": "http://loinc.org",
                "code": "4548-4"
            }
        ]
    },
    "entry": [
        {
            "reference": "Observation/HbA1cExample"
        }
    ]
},
{
    "title": "eGFR",
    "code": {
        "coding": [
            {
                "system": "http://loinc.org",
                "code": "98979-8"
            }
        ]
    },
    "entry": [
        {
            "reference": "Observation/eGFRExample"
        }
    ]
},
{
    "title": "uACR",
    "code": {
        "coding": [
            {
                "system": "http://loinc.org",
                "code": "14959-1"
            }
        ]
    },
    "entry": [
        {
            "reference": "Observation/uACRExample"
        }
    ]
},
{
    "title": "Lipids",

```

```

        "code": {
          "coding": [
            {
              "system": "http://loinc.org",
              "code": "18262-6"
            }
          ]
        },
        "entry": [
          {
            "reference":
"Observation/LDLCholesterolExample"
          }
        ]
      }
    ]
  }
},
{
  "fullUrl": "http://example.org/Patient/PatientExample",
  "resource": {
    "resourceType": "Patient",
    "id": "PatientExample",
    "meta": {
      "profile": [
        "http://hl7.org/fhir/us/core/StructureDefinition/us-
core-patient|6.1.0"
      ]
    },
    "language": "en",
    "text": {
      "status": "generated",
      "div": "\u003Cdiv xml:lang=\"en\"
xmlns=\"http://www.w3.org/1999/xhtml\" lang=\"en\" \u003EPatient John Doe,
male, born 1950-01-01, Medicare ID: 1EG4TE5MK73\u003C/div\u003E"
    },
    "identifier": [
      {
        "type": {
          "coding": [
            {
              "system":
"http://terminology.hl7.org/CodeSystem/v2-0203",
              "code": "MC"
            }
          ]
        },
        "system":
"http://terminology.hl7.org/NamingSystem/cmsMBI",
        "value": "1EG4TE5MK73"
      }
    ],
    "name": [

```

```

        {
            "family": "Doe",
            "given": [
                "John"
            ]
        }
    ],
    "gender": "male",
    "birthDate": "1950-01-01"
}
},
{
    "fullUrl":
"http://example.org/Practitioner/PractitionerExample",
    "resource": {
        "resourceType": "Practitioner",
        "id": "PractitionerExample",
        "meta": {
            "profile": [
                "http://hl7.org/fhir/us/core/StructureDefinition/us-
core-practitioner|6.1.0"
            ]
        },
        "language": "en",
        "text": {
            "status": "generated",
            "div": "\u003Cdiv xml:lang=\"en\"
xmlns=\"http://www.w3.org/1999/xhtml\" lang=\"en\" \u003EPractitioner Jane
Smith, NPI: 1234567893\u003C/div\u003E"
        },
        "identifier": [
            {
                "system": "http://hl7.org/fhir/sid/us-npi",
                "value": "1234567893"
            }
        ],
        "name": [
            {
                "family": "Smith",
                "given": [
                    "Jane"
                ]
            }
        ]
    }
},
{
    "fullUrl":
"http://example.org/Organization/OrganizationExample",
    "resource": {
        "resourceType": "Organization",
        "id": "OrganizationExample",
        "meta": {
            "profile": [

```

```

        "http://hl7.org/fhir/us/core/StructureDefinition/us-
core-organization|6.1.0"
    ]
},
"language": "en",
"text": {
    "status": "generated",
    "div": "\u003Cdiv xml:lang=\"en\"
xmlns=\"http://www.w3.org/1999/xhtml\" lang=\"en\" \u003EExample
Healthcare Organization, ACCESS Participant ID:
ACCES12345\u003C/div\u003E"
},
"identifier": [
    {
        "system": "https://dsacms.github.io/cmmi-access-
model/participant-id",
        "value": "ACCES12345"
    }
],
"active": true,
"name": "Example Healthcare Organization"
}
},
{
    "fullUrl":
"http://example.org/Observation/BloodPressureExample",
    "resource": {
        "resourceType": "Observation",
        "id": "BloodPressureExample",
        "meta": {
            "profile": [
                "http://hl7.org/fhir/us/core/StructureDefinition/us-
core-blood-pressure|6.1.0"
            ]
        },
        "language": "en",
        "text": {
            "status": "generated",
            "div": "\u003Cdiv xml:lang=\"en\"
xmlns=\"http://www.w3.org/1999/xhtml\" lang=\"en\" \u003EBlood Pressure:
138/88 mm[Hg] measured on 2026-01-15\u003C/div\u003E"
        },
        "status": "final",
        "category": [
            {
                "coding": [
                    {
                        "system":
"http://terminology.hl7.org/CodeSystem/observation-category",
                        "code": "vital-signs",
                        "display": "Vital Signs"
                    }
                ]
            }
        ]
    }
],
1,

```

```

optional"
    "code": {
      "coding": [
        {
          "system": "http://loinc.org",
          "code": "85354-9",
          "display": "Blood pressure panel with all children
        }
      ],
      "text": "Blood Pressure"
    },
    "subject": {
      "reference": "Patient/PatientExample"
    },
    "effectiveDateTime": "2026-01-15T10:30:00Z",
    "component": [
      {
        "code": {
          "coding": [
            {
              "system": "http://loinc.org",
              "code": "8480-6",
              "display": "Systolic blood pressure"
            }
          ]
        },
        "valueQuantity": {
          "value": 138,
          "unit": "mm[Hg]",
          "system": "http://unitsofmeasure.org",
          "code": "mm[Hg]"
        }
      },
      {
        "code": {
          "coding": [
            {
              "system": "http://loinc.org",
              "code": "8462-4",
              "display": "Diastolic blood pressure"
            }
          ]
        },
        "valueQuantity": {
          "value": 88,
          "unit": "mm[Hg]",
          "system": "http://unitsofmeasure.org",
          "code": "mm[Hg]"
        }
      }
    ]
  }
},
{

```

```

        "fullUrl":
"http://example.org/Observation/BloodPressureExample2",
        "resource": {
            "resourceType": "Observation",
            "id": "BloodPressureExample2",
            "meta": {
                "profile": [
                    "http://hl7.org/fhir/us/core/StructureDefinition/us-
core-blood-pressure|6.1.0"
                ]
            },
            "language": "en",
            "text": {
                "status": "generated",
                "div": "\u003Cdiv xml:lang=\"en\"
xmlns=\"http://www.w3.org/1999/xhtml\" lang=\"en\" \u003EBlood Pressure:
132/84 mm[Hg] measured on 2026-01-22\u003C/div\u003E"
            },
            "status": "final",
            "category": [
                {
                    "coding": [
                        {
                            "system":
"http://terminology.hl7.org/CodeSystem/observation-category",
                            "code": "vital-signs",
                            "display": "Vital Signs"
                        }
                    ]
                }
            ],
            "code": {
                "coding": [
                    {
                        "system": "http://loinc.org",
                        "code": "85354-9",
                        "display": "Blood pressure panel with all children
optional"
                    }
                ]
            },
            "text": "Blood Pressure"
        },
        "subject": {
            "reference": "Patient/PatientExample"
        },
        "effectiveDateTime": "2026-01-22T10:30:00Z",
        "component": [
            {
                "code": {
                    "coding": [
                        {
                            "system": "http://loinc.org",
                            "code": "8480-6",
                            "display": "Systolic blood pressure"
                        }
                    ]
                }
            }
        ]
    }
}

```

```

    ]
  },
  "valueQuantity": {
    "value": 132,
    "unit": "mm[Hg]",
    "system": "http://unitsofmeasure.org",
    "code": "mm[Hg]"
  }
},
{
  "code": {
    "coding": [
      {
        "system": "http://loinc.org",
        "code": "8462-4",
        "display": "Diastolic blood pressure"
      }
    ]
  },
  "valueQuantity": {
    "value": 84,
    "unit": "mm[Hg]",
    "system": "http://unitsofmeasure.org",
    "code": "mm[Hg]"
  }
}
]
}
},
{
  "fullUrl":
"http://example.org/Observation/BloodPressureExample3",
  "resource": {
    "resourceType": "Observation",
    "id": "BloodPressureExample3",
    "meta": {
      "profile": [
        "http://hl7.org/fhir/us/core/StructureDefinition/us-
core-blood-pressure|6.1.0"
      ]
    },
    "language": "en",
    "text": {
      "status": "generated",
      "div": "\u003Cdiv xml:lang=\"en\"
xmlns=\"http://www.w3.org/1999/xhtml\" lang=\"en\" \u003EBlood Pressure:
140/90 mm[Hg] measured on 2026-02-05\u003C/div\u003E"
    },
    "status": "final",
    "category": [
      {
        "coding": [
          {
            "system":
"http://terminology.hl7.org/CodeSystem/observation-category",

```



```

        "code": "vital-signs",
        "display": "Vital Signs"
    }
  ]
}
],
"code": {
  "coding": [
    {
      "system": "http://loinc.org",
      "code": "85354-9",
      "display": "Blood pressure panel with all children
optional"
    }
  ],
  "text": "Blood Pressure"
},
"subject": {
  "reference": "Patient/PatientExample"
},
"effectiveDateTime": "2026-02-05T10:30:00Z",
"component": [
  {
    "code": {
      "coding": [
        {
          "system": "http://loinc.org",
          "code": "8480-6",
          "display": "Systolic blood pressure"
        }
      ]
    },
    "valueQuantity": {
      "value": 140,
      "unit": "mm[Hg]",
      "system": "http://unitsofmeasure.org",
      "code": "mm[Hg]"
    }
  },
  {
    "code": {
      "coding": [
        {
          "system": "http://loinc.org",
          "code": "8462-4",
          "display": "Diastolic blood pressure"
        }
      ]
    },
    "valueQuantity": {
      "value": 90,
      "unit": "mm[Hg]",
      "system": "http://unitsofmeasure.org",
      "code": "mm[Hg]"
    }
  }
]
}

```

```

    }
  ]
}
},
{
  "fullUrl":
"http://example.org/Observation/BodyWeightExample",
  "resource": {
    "resourceType": "Observation",
    "id": "BodyWeightExample",
    "meta": {
      "profile": [
        "http://hl7.org/fhir/us/core/StructureDefinition/us-
core-body-weight|6.1.0"
      ]
    },
    "language": "en",
    "text": {
      "status": "generated",
      "div": "\u003Cdiv xml:lang=\"en\"
xmlns=\"http://www.w3.org/1999/xhtml\" lang=\"en\" \u003EBody Weight: 85
kg measured on 2026-01-15\u003C/div\u003E"
    },
    "status": "final",
    "category": [
      {
        "coding": [
          {
            "system":
"http://terminology.hl7.org/CodeSystem/observation-category",
            "code": "vital-signs",
            "display": "Vital Signs"
          }
        ]
      }
    ],
    "code": {
      "coding": [
        {
          "system": "http://loinc.org",
          "code": "29463-7",
          "display": "Body weight"
        }
      ]
    },
    "subject": {
      "reference": "Patient/PatientExample"
    },
    "effectiveDateTime": "2026-01-15T10:30:00Z",
    "valueQuantity": {
      "value": 85,
      "unit": "kg",
      "system": "http://unitsofmeasure.org",
      "code": "kg"
    }
  }
}

```

```

    }
  },
  {
    "fullUrl": "http://example.org/Observation/BMIExample",
    "resource": {
      "resourceType": "Observation",
      "id": "BMIExample",
      "meta": {
        "profile": [
          "http://hl7.org/fhir/us/core/StructureDefinition/us-
core-bmi|6.1.0"
        ]
      },
      "language": "en",
      "text": {
        "status": "generated",
        "div": "\u003Cdiv xml:lang=\"en\"
xmlns=\"http://www.w3.org/1999/xhtml\" lang=\"en\" \u003EBMI: 27.5 kg/m2
measured on 2026-01-15\u003C/div\u003E"
      },
      "status": "final",
      "category": [
        {
          "coding": [
            {
              "system":
"http://terminology.hl7.org/CodeSystem/observation-category",
              "code": "vital-signs",
              "display": "Vital Signs"
            }
          ]
        }
      ],
      "code": {
        "coding": [
          {
            "system": "http://loinc.org",
            "code": "39156-5",
            "display": "Body mass index (BMI) [Ratio]"
          }
        ]
      },
      "subject": {
        "reference": "Patient/PatientExample"
      },
      "effectiveDateTime": "2026-01-15T10:30:00Z",
      "valueQuantity": {
        "value": 27.5,
        "unit": "kg/m2",
        "system": "http://unitsofmeasure.org",
        "code": "kg/m2"
      }
    }
  },
  {

```

```

        "fullUrl":
"http://example.org/Observation/WaistCircumferenceExample",
        "resource": {
            "resourceType": "Observation",
            "id": "WaistCircumferenceExample",
            "meta": {
                "profile": [
                    "http://hl7.org/fhir/us/core/StructureDefinition/us-
core-simple-observation|6.1.0"
                ]
            },
            "language": "en",
            "text": {
                "status": "generated",
                "div": "\u003Cdiv xml:lang=\"en\"
xmlns=\"http://www.w3.org/1999/xhtml\" lang=\"en\" \u003E
Waist Circumference: 95 cm measured on 2026-01-15\u003C/div\u003E"
            },
            "status": "final",
            "category": [
                {
                    "coding": [
                        {
                            "system":
"http://terminology.hl7.org/CodeSystem/observation-category",
                            "code": "vital-signs",
                            "display": "Vital Signs"
                        }
                    ]
                }
            ],
            "code": {
                "coding": [
                    {
                        "system": "http://loinc.org",
                        "code": "8280-0",
                        "display": "Waist Circumference at umbilicus by Tape
measure"
                    }
                ]
            },
            "subject": {
                "reference": "Patient/PatientExample"
            },
            "effectiveDateTime": "2026-01-15T10:30:00Z",
            "valueQuantity": {
                "value": 95,
                "unit": "cm",
                "system": "http://unitsofmeasure.org",
                "code": "cm"
            }
        }
    },
    {
        "fullUrl": "http://example.org/Observation/HbA1cExample",

```

```

"resource": {
  "resourceType": "Observation",
  "id": "HbA1cExample",
  "meta": {
    "profile": [
      "http://hl7.org/fhir/us/core/StructureDefinition/us-
core-observation-clinical-result|6.1.0"
    ]
  },
  "language": "en",
  "text": {
    "status": "generated",
    "div": "\u003Cdiv xml:lang=\"en\"
xmlns=\"http://www.w3.org/1999/xhtml\" lang=\"en\" \u003EHemoglobin A1c:
6.5% measured on 2026-01-15\u003C/div\u003E"
  },
  "status": "final",
  "category": [
    {
      "coding": [
        {
          "system":
"http://terminology.hl7.org/CodeSystem/observation-category",
          "code": "laboratory",
          "display": "Laboratory"
        }
      ]
    }
  ],
  "code": {
    "coding": [
      {
        "system": "http://loinc.org",
        "code": "4548-4",
        "display": "Hemoglobin A1c/Hemoglobin.total in Blood"
      }
    ]
  },
  "subject": {
    "reference": "Patient/PatientExample"
  },
  "effectiveDateTime": "2026-01-15T10:30:00Z",
  "valueQuantity": {
    "value": 6.5,
    "unit": "%",
    "system": "http://unitsofmeasure.org",
    "code": "%"
  }
},
{
  "fullUrl": "http://example.org/Observation/eGFRExample",
  "resource": {
    "resourceType": "Observation",
    "id": "eGFRExample",

```

```

    "meta": {
      "profile": [
        "http://hl7.org/fhir/us/core/StructureDefinition/us-
core-observation-clinical-result|6.1.0"
      ]
    },
    "language": "en",
    "text": {
      "status": "generated",
      "div": "\u003Cdiv xml:lang=\"en\"
xmlns=\"http://www.w3.org/1999/xhtml\" lang=\"en\" \u003EeGFR: 75
mL/min/1.73m2 measured on 2026-01-15\u003C/div\u003E"
    },
    "status": "final",
    "category": [
      {
        "coding": [
          {
            "system":
"http://terminology.hl7.org/CodeSystem/observation-category",
            "code": "laboratory",
            "display": "Laboratory"
          }
        ]
      }
    ],
    "code": {
      "coding": [
        {
          "system": "http://loinc.org",
          "code": "98979-8",
          "display": "Glomerular filtration rate [Volume
Rate/Area] in Serum, Plasma or Blood by Creatinine-based formula (CKD-EPI
2021)/1.73 sq M"
        }
      ]
    },
    "subject": {
      "reference": "Patient/PatientExample"
    },
    "effectiveDateTime": "2026-01-15T10:30:00Z",
    "valueQuantity": {
      "value": 75,
      "unit": "mL/min/1.73_m2",
      "system": "http://unitsofmeasure.org",
      "code": "mL/min/{1.73_m2}"
    }
  },
  {
    "fullUrl": "http://example.org/Observation/uACRExample",
    "resource": {
      "resourceType": "Observation",
      "id": "uACRExample",
      "meta": {

```

```

        "profile": [
            "http://hl7.org/fhir/us/core/StructureDefinition/us-
core-observation-clinical-result|6.1.0"
        ]
    },
    "language": "en",
    "text": {
        "status": "generated",
        "div": "\u003Cdiv xml:lang=\"en\"
xmlns=\"http://www.w3.org/1999/xhtml\" lang=\"en\" \u003EUrine Albumin-
Creatinine Ratio: 45 mg/g measured on 2026-01-15\u003C/div\u003E"
    },
    "status": "final",
    "category": [
        {
            "coding": [
                {
                    "system":
"http://terminology.hl7.org/CodeSystem/observation-category",
                    "code": "laboratory",
                    "display": "Laboratory"
                }
            ]
        }
    ],
    "code": {
        "coding": [
            {
                "system": "http://loinc.org",
                "code": "14959-1",
                "display": "Microalbumin/Creatinine [Mass Ratio] in
Urine"
            }
        ]
    },
    "subject": {
        "reference": "Patient/PatientExample"
    },
    "effectiveDateTime": "2026-01-15T10:30:00Z",
    "valueQuantity": {
        "value": 45,
        "unit": "mg/g",
        "system": "http://unitsofmeasure.org",
        "code": "mg/g"
    }
},
{
    "fullUrl":
"http://example.org/Observation/LDLCholesterolExample",
    "resource": {
        "resourceType": "Observation",
        "id": "LDLCholesterolExample",
        "meta": {
            "profile": [

```





organization that completed the observation or questionnaire) when submitting the data. The “example.org” is only used as an example URL.

### Example Response:

```
HTTP/1.1 202 Accepted
Content-Location: https://[base]/access/Patient/$submission-status/sub-123456
```

(No response body - processing asynchronously)

### \$submission-status Operation

The `$submission-status` operation retrieves the current status of a previously submitted data report. This is a **read-only operation** that uses the `HTTP GET` method.

### Data Reporting Result Codes:

When processing is complete (`HTTP 200`), the `result` parameter uses codes from the **ACCESSReportDataResultVS** value set:

- **success:** Data submission was successfully processed and recorded for ACCESS Model evaluation.
- **validation-error:** Data submission failed validation (see `OperationOutcome` for specific errors and details).
- **duplicate:** Data submission is a duplicate of a previously submitted report for this reporting period.
- **patient-not-aligned:** Patient is not currently aligned to this participant in the specified track. Data reporting is only accepted for aligned patients.
- **reporting-period-closed:** The reporting period for this data has closed and no further submissions are accepted.
- **incomplete-data:** Data submission is missing required observations or measurements for the specified track.
- **incorrect-track:** The submitted data does not match the track the patient is aligned to.

### Example Responses:

#### Example Status Check Request (Still Processing):

```
GET https://[base]/access/Patient/$submission-status/sub-123456
```

#### Response:

HTTP/1.1 202 Accepted

(No response body - still processing)

### **Example Status Check Request (Processing Complete - Success):**

GET https://[base]/access/Patient/\$submission-status/sub-123456

#### **Response:**

HTTP/1.1 200 OK

Content-Type: application/fhir+json

```
{
  "resourceType": "Parameters",
  "parameter": [
    {
      "name": "result",
      "valueCodeableConcept": {
        "coding": [
          {
            "system":
"https://globalalliantinc.com/access/CodeSystem/ACCESSReportDataResultCS"
,
            "code": "success",
            "display": "Success"
          }
        ],
        "text": "Data submission was successfully processed and recorded
for ACCESS Model evaluation."
      }
    }
  ]
}
```

### **Example Status Check Request (Incomplete Data with Detailed Issues):**

GET https://[base]/access/Patient/\$submission-status/sub-123456

#### **Response:**

HTTP/1.1 200 OK

Content-Type: application/fhir+json

```
{
  "resourceType": "Parameters",
  "parameter": [
    {
      "name": "result",
```

```

    "valueCodeableConcept": {
      "coding": [
        {
          "system": "https://dsacms.github.io/cmmi-access-
model/CodeSystem/ACCESSReportDataResultCS",
          "code": "incomplete-data",
          "display": "Incomplete data"
        }
      ],
      "text": "Data submission is missing required observations or
measurements for the specified track. When this result is returned, the
response MAY include an 'issues' parameter containing an OperationOutcome
resource with detailed, structured information about specific missing
information and how to correct it."
    }
  },
  {
    "name": "issues",
    "resource": {
      "resourceType": "OperationOutcome",
      "issue": [
        {
          "severity": "warning",
          "code": "required",
          "details": {
            "text": "Missing required observation: Hemoglobin Alc
(HbA1c) is required for CKM track quarterly reporting"
          },
          "expression": [

"Bundle.entry[2].resource.section.where(code.coding.code='CKM').section.w
here(code.coding.code='4548-4') "

        ]
      ]
    }
  ]
}
]
}

```

### Example Status Check Request (Patient Not Aligned):

GET https://[base]/access/Patient/\$submission-status/sub-123456

### Response:

HTTP/1.1 200 OK

Content-Type: application/fhir+json

```

{
  "resourceType": "Parameters",
  "parameter": [
    {

```

```

    "name": "result",
    "valueCodeableConcept": {
      "coding": [
        {
          "system": "https://dsacms.github.io/cmmi-access-model/CodeSystem/ACCESSReportDataResultCS",
          "code": "patient-not-aligned",
          "display": "Patient Not Aligned"
        }
      ],
      "text": "Patient is not currently aligned to this participant in the specified track. Data reporting is only accepted for aligned patients."
    }
  ]
}

```

## Implementation Guidance

### Data Bundle Structure

The data bundle submitted via the `DataBundle` parameter must conform to the **ACCESSDataReportingBundle** profile, which requires:

1. **Bundle Type:** Must be "document"
2. **Composition Entry:** First entry must be an `ACCESSDataReportingComposition`
3. **Patient Entry:** Must include the patient resource referenced by the composition
4. **Clinical Data:** Must include all `Observation` and `QuestionnaireResponse` resources (and any other resources) referenced by the composition sections

### Track-Specific Sections:

Each ACCESS track has specific data requirements organized into composition sections. For more information about what data is required or optional for each of the report types, please see the Model Payment Amounts and Performance Targets (<https://www.cms.gov/priorities/innovation/files/access-payments-amts-perf-targets.pdf>).

### CKM (Cardio-Kidney-Metabolic) Track:

- Blood Pressure observations
- Lipid panel results (LDL-C required, others optional)
- Body weight and BMI
- Waist circumference
- HbA1c results
- eGFR (estimated Glomerular Filtration Rate)
- uACR (urine Albumin-to-Creatinine Ratio)

Although eGFR and uACR are only required to be reported during baseline reporting for each period of care, if a patient will be immediately continuing for a follow-on period, it is recommended that eGFR and uACR data be included in the end-of-period reporting for the initial period as this end-of-period reporting can count as the baseline reporting for the follow-on period. Do make sure the clinical relevant dates align with the period of care.

- If eGFR and uACR are not included in end-of-period reporting for the initial period of care and the patient does continue with a follow-on period, all required CKM measures must be submitted within 60 days of the start of the follow-on period (up to 6 total measures depending on the patient's diagnosis). Please note that eGFR and uACR data cannot be submitted alone even if all other measures were submitted for the initial period end-of-period reporting.

#### **eCKM (Early Cardio-Kidney-Metabolic) Track:**

- Blood Pressure observations
- Lipid panel results (LDL-C required, others optional)
- Body weight and BMI
- Waist circumference
- HbA1c results

#### **MSK (Musculoskeletal) Track:**

- PROMIS Physical Function (PF) Short Form 6b (PROMIS-PF)
- PROMIS Pain Interference (PI) Short Form 6a (PROMIS-PI)
- Oswestry Disability Index (ODI)
- Neck Disability Index (NDI)
- QuickDASH (Quick Disabilities of the Arm, Shoulder and Hand)
- PROMIS Version 2.0 Computer Adaptive Test (CAT)
- Patient's Global Impression of Change (PGIC)
- Knee Injury and Osteoarthritis Outcome Score for Joint Replacement (KOOS JR)
- Hip Dysfunction and Osteoarthritis Outcome Score for Joint Replacement (HOOS JR)
- PROMIS Pain Intensity NRS v1.0 (NRS or PROMIS NRS)

#### **BH (Behavioral Health) Track:**

- Depression screening (PHQ-9) scores
- Anxiety screening (GAD-7) scores
- Overall function (WHODAS 2.0) scores
- Patient's Global Impression of Change (PGIC)

### Including Additional Information

The **ACCESSDataReportingComposition** profile supports an optional **additional-information** section that allows participants to include supplementary resources beyond the required track-specific data elements. This section enables more comprehensive data reporting while maintaining a clear separation between required and optional data.

**IMPORTANT:** The **additional-information** section must appear after the required track section within the **ACCESSDataReportingComposition**.

### Section Details:

- **Section Code:** `ACCESSReportDataCompositionSectionCS#additional-information`
- **Cardinality:** `0..1` (optional)
- **Entry References:** `0..*` references to any FHIR Resource
- **Purpose:** Include supplemental information that provides additional context or detail about the data being reported

### Common Use Cases:

1. **Supporting Documentation:** Include `DiagnosticReport` or `DocumentReference` resources that provide additional clinical context
2. **Care Plan Information:** Include `CarePlan` resources that describe the treatment approach for tracked conditions
3. **Supplementary Observations:** Include additional measurements or assessments not explicitly required by the track but relevant to patient care

### Important Considerations:

- All resources referenced in the **additional-information** section must be included in the Bundle
- The section is optional - only use it when you have meaningful supplementary data to provide
- Validation focuses on required track-specific data; additional information is not validated against track requirements
- The section uses standard `Composition.section` pattern, making it familiar to FHIR implementers

For a complete example, see [MSK Data Reporting Bundle with Additional Information Example](#).

## Data Quality Requirements

To ensure successful data submission:

### Completeness:

- Include all required data elements for the reporting period
- Ensure all referenced resources are included in the bundle
- Provide complete patient demographics including the patient ID

### Accuracy:

- Verify clinical data reflects current patient status
- Use appropriate LOINC codes for observations
- Include proper units of measure (UCUM)

### Conformance:

- Validate bundle against `ACCESSDataReportingBundle` profile
- Ensure composition conforms to the `ACCESSDataReportingComposition` profile
- Use US Core profiles for all clinical resources

### Timeliness:

- Submit data within the required reporting windows
- Follow the reporting schedule in the participant agreement

## Handling Incomplete Data

When an `incomplete-data` result is returned, the response includes an optional `issues` parameter containing an `OperationOutcome` resource with detailed, structured information about what information is missing. This allows implementers to programmatically identify and correct errors.

### Response Structure:

The validation error response contains two parameters:

1. ``result``: `CodeableConcept` with code "validation-error" and descriptive text
2. ``issues`` (optional): `OperationOutcome` resource with detailed error information

### Using the OperationOutcome:

The `OperationOutcome` follows standard FHIR conventions and includes:

- ``issue[].severity``: Error severity (error, warning, information)
- ``issue[].code``: FHIR issue type code (e.g., required, value, structure)
- ``issue[].details.text``: Human-readable description of the issue
- ``issue[].expression``: FHIRPath expression pointing to the problematic element

## Processing Incomplete Data:

1. **Check for the `issues` parameter:** When result code is "incomplete-data", look for the optional `issues` parameter
2. **Parse OperationOutcome:** Extract the `issue` array from the `OperationOutcome` resource
3. **Identify errors:** Review each issue with `severity` of "error"
4. **Locate problems:** Use the `expression` element to find the exact location in your Bundle
5. **Correct and resubmit:** Fix identified issues and submit the corrected data

## Common Validation Errors

- **Incorrect LOINC codes:** Observations use wrong LOINC codes for the data element
- **Invalid values:** Observation values outside valid ranges
- **Missing or invalid patient ID:** Patient resource missing Patient ID or other required identifiers
- **Bundle structure errors:** Bundle not conforming to `ACCESSDataReportingBundle` profile
- **Broken references:** Composition references resources not included in bundle
- **Profile violations:** Resources don't conform to required US Core or ACCESS profiles

## Best Practices:

1. **Validate locally first:** Use a FHIR validator to check your bundle before submission
2. **Parse all issues:** Process all issues in the `OperationOutcome`, not just the first one
3. **Use expression paths:** The `expression` element helps locate problems in complex bundles
4. **Check warnings:** Review `warning` severity issues for optional improvements
5. **Maintain error logs:** Keep records of validation errors for quality improvement

## Example Complete Workflows

### MSK Track Workflow Example

The following example demonstrates a complete data reporting workflow for an MSK track patient:

#### Step 1: Submit MSK Data Report

```
POST https://[base]/access/Patient/$report-data?entityId=ACCES12345
Content-Type: application/fhir+json
```

```
{
  "resourceType": "Parameters",
  "id": "MSKReportDataRequestExample",
  "meta": {
```



```

    "profile": [
      "https://dsacms.github.io/cmmi-access-
model/StructureDefinition/access-report-data-in"
    ]
  },
  "language": "en",
  "parameter": [
    {
      "name": "participantID",
      "valueIdentifier": {
        "system": "https://dsacms.github.io/cmmi-access-
model/participant-id",
        "value": "ACCES12345"
      }
    },
    {
      "name": "payerID",
      "valueIdentifier": {
        "type": {
          "coding": [
            {
              "system": "http://hl7.org/fhir/us/carin-
bb/CodeSystem/C4BBIIdentifierType",
              "code": "payerid",
              "display": "Payer ID"
            }
          ]
        },
        "system": "urn:oid:2.16.840.1.113883.3.221.5",
        "value": "12345"
      }
    },
    {
      "name": "track",
      "valueCodeableConcept": {
        "coding": [
          {
            "system": "https://dsacms.github.io/cmmi-access-
model/CodeSystem/ACCESSTrackCS",
            "code": "MSK",
            "display": "Musculoskeletal track"
          }
        ]
      }
    },
    {
      "name": "reportType",
      "valueCodeableConcept": {
        "coding": [
          {
            "system": "https://dsacms.github.io/cmmi-access-
model/CodeSystem/ACCESSReportTypeCS",
            "code": "quarterly",
            "display": "Quarterly Data Report"
          }
        ]
      }
    }
  ]
}

```

```

    ]
  }
},
{
  "name": "dataBundle",
  "resource": {
    "resourceType": "Bundle",
    "id": "MSKReportDataBundleExample",
    "meta": {
      "profile": [
        "https://dsacms.github.io/cmmi-access-
model/StructureDefinition/access-data-reporting-bundle"
      ]
    },
    "language": "en",
    "identifier": {
      "system": "http://example.org/data-bundle-id",
      "value": "bundle-msk-001"
    },
    "type": "document",
    "timestamp": "2026-01-26T12:00:00Z",
    "entry": [
      {
        "fullUrl":
"http://example.org/Composition/MSKReportDataCompositionExample",
        "resource": {
          "resourceType": "Composition",
          "id": "MSKReportDataCompositionExample",
          "meta": {
            "profile": [
              "https://dsacms.github.io/cmmi-access-
model/StructureDefinition/access-data-reporting-composition"
            ]
          },
          "language": "en",
          "text": {
            "status": "generated",
            "div": "\u003Cdiv xml:lang=\"en\"
xmlns=\"http://www.w3.org/1999/xhtml\"
lang=\"en\"\u003E\u003Ch3\u003EACCESS MSK Track Data
Report\u003C/h3\u003E\u003Cp\u003EPatient: John Doe (MBI:
1EG4TE5MK73)\u003C/p\u003E\u003Cp\u003EReport Date: 2026-01-
26\u003C/p\u003E\u003Cp\u003EAuthor: Jane
Smith\u003C/p\u003E\u003C/div\u003E"
          },
          "status": "final",
          "type": {
            "coding": [
              {
                "system": "http://loinc.org",
                "code": "74465-6",
                "display": "Questionnaire response Document"
              }
            ]
          }
        },
      },
    ],
  },
},

```

```

"subject": {
  "reference": "Patient/PatientExample"
},
"date": "2026-01-26T12:00:00Z",
"author": [
  {
    "reference": "Practitioner/PractitionerExample"
  }
],
"title": "ACCESS MSK Track Data Report",
"custodian": {
  "reference": "Organization/OrganizationExample"
},
"section": [
  {
    "title": "Data reporting for MSK track",
    "code": {
      "coding": [
        {
          "system": "https://dsacms.github.io/cmmi-access-
model/CodeSystem/ACCESSTrackCS",
          "code": "MSK"
        }
      ]
    },
    "text": {
      "status": "generated",
      "div": "\u003Cdiv xml:lang=\\"en\\"
xmlns=\\"http://www.w3.org/1999/xhtml\\"
lang=\\"en\\" \u003E\u003Cp\u003EMusculoskeletal Track Data Reporting
including pain assessments and patient-reported
outcomes.\u003Cp\u003E\u003Cdiv\u003E"
    },
    "section": [
      {
        "title": "PROMIS Physical Function Short Form 6b",
        "code": {
          "coding": [
            {
              "system": "http://loinc.org",
              "code": "76804-4"
            }
          ]
        },
        "entry": [
          {
            "reference":
"QuestionnaireResponse/PROMISPhysicalFunctionExample"
          }
        ]
      },
      {
        "title": "PROMIS Physical Function CAT",
        "code": {
          "coding": [

```

```

        {
            "system": "http://loinc.org",
            "code": "91722-9"
        }
    ],
    },
    "entry": [
        {
            "reference":
"QuestionnaireResponse/PROMISPhysicalFunctionCATExample"
        }
    ]
},
{
    "title": "PROMIS Pain Interference Short Form 6a",
    "code": {
        "coding": [
            {
                "system": "http://loinc.org",
                "code": "90973-9"
            }
        ]
    },
    "entry": [
        {
            "reference":
"QuestionnaireResponse/PROMISPainInterferenceSFExample"
        }
    ]
},
{
    "title": "PROMIS Pain Interference CAT",
    "code": {
        "coding": [
            {
                "system": "http://loinc.org",
                "code": "89923-7"
            }
        ]
    },
    "entry": [
        {
            "reference":
"QuestionnaireResponse/PROMISPainInterferenceCATExample"
        }
    ]
},
{
    "title": "Oswestry Disability Index",
    "code": {
        "coding": [
            {
                "system": "http://loinc.org",
                "code": "97908-8"
            }
        ]
    }
}

```

```

        ]
      },
      "entry": [
        {
          "reference":
"QuestionnaireResponse/OswestryDisabilityIndexExample"
        }
      ]
    },
    {
      "title": "Neck Disability Index",
      "code": {
        "coding": [
          {
            "system": "http://loinc.org",
            "code": "82226-2"
          }
        ]
      },
      "entry": [
        {
          "reference":
"QuestionnaireResponse/NeckDisabilityIndexExample"
        }
      ]
    },
    {
      "title": "QuickDASH",
      "code": {
        "coding": [
          {
            "system": "https://dsacms.github.io/cmmi-
access-model/CodeSystem/ACCESSReportDataCompositionSectionCS",
            "code": "QuickDASH"
          }
        ]
      },
      "entry": [
        {
          "reference":
"QuestionnaireResponse/QuickDASHEXample"
        }
      ]
    },
    {
      "title": "Patient's Global Impression of Change",
      "code": {
        "coding": [
          {
            "system": "https://dsacms.github.io/cmmi-
access-model/CodeSystem/ACCESSReportDataCompositionSectionCS",
            "code": "PGIC"
          }
        ]
      },
    },

```

```

        "entry": [
            {
                "reference":
"QuestionnaireResponse/MSKPGICEExample"
            }
        ]
    },
    {
        "title": "KOOS JR",
        "code": {
            "coding": [
                {
                    "system": "http://loinc.org",
                    "code": "82324-5"
                }
            ]
        },
        "entry": [
            {
                "reference":
"QuestionnaireResponse/KOOSJRExample"
            }
        ]
    },
    {
        "title": "HOOS JR",
        "code": {
            "coding": [
                {
                    "system": "http://loinc.org",
                    "code": "82316-1"
                }
            ]
        },
        "entry": [
            {
                "reference":
"QuestionnaireResponse/HOOSJRExample"
            }
        ]
    },
    {
        "title": "PROMIS Pain Intensity NRS",
        "code": {
            "coding": [
                {
                    "system": "http://loinc.org",
                    "code": "72514-3"
                }
            ]
        },
        "entry": [
            {
                "reference":
"QuestionnaireResponse/PROMISNRSExample"
            }
        ]
    }
]

```

```

        }
      ]
    }
  ]
}
},
{
  "fullUrl": "http://example.org/Patient/PatientExample",
  "resource": {
    "resourceType": "Patient",
    "id": "PatientExample",
    "meta": {
      "profile": [
        "http://hl7.org/fhir/us/core/StructureDefinition/us-
core-patient|6.1.0"
      ]
    },
    "language": "en",
    "text": {
      "status": "generated",
      "div": "\u003Cdiv xml:lang=\"en\"
xmlns=\"http://www.w3.org/1999/xhtml\" lang=\"en\" \u003EPatient John Doe,
male, born 1950-01-01, Medicare ID: 1EG4TE5MK73\u003C/div\u003E"
    },
    "identifier": [
      {
        "type": {
          "coding": [
            {
              "system":
"http://terminology.hl7.org/CodeSystem/v2-0203",
              "code": "MC"
            }
          ]
        },
        "system":
"http://terminology.hl7.org/NamingSystem/cmsMBI",
        "value": "1EG4TE5MK73"
      }
    ],
    "name": [
      {
        "family": "Doe",
        "given": [
          "John"
        ]
      }
    ],
    "gender": "male",
    "birthDate": "1950-01-01"
  }
},
{

```

```

        "fullUrl":
"http://example.org/Practitioner/PractitionerExample",
        "resource": {
            "resourceType": "Practitioner",
            "id": "PractitionerExample",
            "meta": {
                "profile": [
                    "http://hl7.org/fhir/us/core/StructureDefinition/us-
core-practitioner|6.1.0"
                ]
            },
            "language": "en",
            "text": {
                "status": "generated",
                "div": "\u003Cdiv xml:lang=\"en\"
xmlns=\"http://www.w3.org/1999/xhtml\" lang=\"en\" \u003EPractitioner Jane
Smith, NPI: 1234567893\u003C/div\u003E"
            },
            "identifier": [
                {
                    "system": "http://hl7.org/fhir/sid/us-npi",
                    "value": "1234567893"
                }
            ],
            "name": [
                {
                    "family": "Smith",
                    "given": [
                        "Jane"
                    ]
                }
            ]
        }
    },
    {
        "fullUrl":
"http://example.org/Organization/OrganizationExample",
        "resource": {
            "resourceType": "Organization",
            "id": "OrganizationExample",
            "meta": {
                "profile": [
                    "http://hl7.org/fhir/us/core/StructureDefinition/us-
core-organization|6.1.0"
                ]
            },
            "language": "en",
            "text": {
                "status": "generated",
                "div": "\u003Cdiv xml:lang=\"en\"
xmlns=\"http://www.w3.org/1999/xhtml\" lang=\"en\" \u003EExample
Healthcare Organization, ACCESS Participant ID:
ACCES12345\u003C/div\u003E"
            },
            "identifier": [

```



```

        {
            "system": "https://dsacms.github.io/cmmi-access-
model/participant-id",
            "value": "ACCES12345"
        }
    ],
    "active": true,
    "name": "Example Healthcare Organization"
}
},
{
    "fullUrl":
"http://example.org/QuestionnaireResponse/PROMISPhysicalFunctionExample",
    "resource": {
        "resourceType": "QuestionnaireResponse",
        "id": "PROMISPhysicalFunctionExample",
        "meta": {
            "profile": [
                "http://hl7.org/fhir/us/core/StructureDefinition/us-
core-questionnaireresponse|6.1.0"
            ]
        },
        "language": "en",
        "text": {
            "status": "generated",
            "div": "\u003Cdiv xml:lang=\"en\"
xmlns=\"http://www.w3.org/1999/xhtml\"
lang=\"en\" \u003E\u003Cp\u003EPROMIS Physical Function (Short Form 6b)
QuestionnaireResponse for John Doe. T-score:
36.8.\u003Cp\u003E\u003Cdiv\u003E"
        },
        "questionnaire":
"http://example.org/Questionnaire/PROMISPhysicalFunctionQuestionnaire",
        "status": "completed",
        "subject": {
            "reference": "Patient/PatientExample"
        },
        "authored": "2026-01-15T10:30:00Z",
        "author": {
            "reference": "Patient/PatientExample"
        },
        "item": [
            {
                "linkId": "pf6b-1",
                "text": "Are you able to do chores such as vacuuming or
yard work?",
                "answer": [
                    {
                        "valueCoding": {
                            "extension": [
                                {
                                    "url":
"http://hl7.org/fhir/StructureDefinition/ordinalValue",
                                    "valueDecimal": 4
                                }
                            ]
                        }
                    }
                ]
            }
        ]
    }
}

```

```

        ],
        "system": "http://loinc.org",
        "code": "LA13918-0",
        "display": "With a little difficulty"
    }
}
],
},
{
    "linkId": "pf6b-2",
    "text": "Are you able to go up and down stairs at a
normal pace?",
    "answer": [
        {
            "valueCoding": {
                "extension": [
                    {
                        "url":
"http://hl7.org/fhir/StructureDefinition/ordinalValue",
                        "valueDecimal": 3
                    }
                ],
                "system": "http://loinc.org",
                "code": "LA13920-6",
                "display": "With some difficulty"
            }
        }
    ],
},
{
    "linkId": "pf6b-3",
    "text": "Are you able to go for a walk of at least 15
minutes?",
    "answer": [
        {
            "valueCoding": {
                "extension": [
                    {
                        "url":
"http://hl7.org/fhir/StructureDefinition/ordinalValue",
                        "valueDecimal": 3
                    }
                ],
                "system": "http://loinc.org",
                "code": "LA13920-6",
                "display": "With some difficulty"
            }
        }
    ],
},
{
    "linkId": "pf6b-4",
    "text": "Are you able to run errands and shop?",
    "answer": [
        {

```

```

        "valueCoding": {
          "extension": [
            {
              "url":
"http://hl7.org/fhir/StructureDefinition/ordinalValue",
              "valueDecimal": 4
            }
          ],
          "system": "http://loinc.org",
          "code": "LA13918-0",
          "display": "With a little difficulty"
        }
      ]
    },
    {
      "linkId": "pf6b-5",
      "text": "Does your health now limit you in doing two
hours of physical labor?",
      "answer": [
        {
          "valueCoding": {
            "extension": [
              {
                "url":
"http://hl7.org/fhir/StructureDefinition/ordinalValue",
                "valueDecimal": 1
              }
            ],
            "system": "http://loinc.org",
            "code": "LA13868-7",
            "display": "Cannot do"
          }
        }
      ]
    },
    {
      "linkId": "pf6b-6",
      "text": "Does your health now limit you in doing
moderate work around the house like vacuuming, sweeping floors or
carrying in groceries?",
      "answer": [
        {
          "valueCoding": {
            "extension": [
              {
                "url":
"http://hl7.org/fhir/StructureDefinition/ordinalValue",
                "valueDecimal": 2
              }
            ],
            "system": "http://loinc.org",
            "code": "LA11911-7",
            "display": "Quite a lot"
          }
        }
      ]
    }
  ]
}

```

```

        }
      ]
    },
    {
      "linkId": "raw_score",
      "answer": [
        {
          "valueDecimal": 17
        }
      ]
    },
    {
      "linkId": "tscore",
      "answer": [
        {
          "valueDecimal": 36.8
        }
      ]
    }
  ]
}
},
{
  "fullUrl":
"http://example.org/QuestionnaireResponse/PROMISPhysicalFunctionCATExample",
  "resource": {
    "resourceType": "QuestionnaireResponse",
    "id": "PROMISPhysicalFunctionCATExample",
    "meta": {
      "profile": [
        "http://hl7.org/fhir/us/core/StructureDefinition/us-
core-questionnaireresponse|6.1.0"
      ]
    },
    "language": "en",
    "text": {
      "status": "generated",
      "div": "\u003Cdiv xml:lang=\"en\"
xmlns=\"http://www.w3.org/1999/xhtml\"
lang=\"en\"\u003E\u003Cp\u003EPROMIS Physical Function (CAT)
QuestionnaireResponse for John Doe. T-score:
38.\u003C/p\u003E\u003C/div\u003E"
    },
    "questionnaire":
"http://example.org/Questionnaire/PROMISPhysicalFunctionCATQuestionnaire"
  },
  "status": "completed",
  "subject": {
    "reference": "Patient/PatientExample"
  },
  "authored": "2026-01-15T10:30:00Z",
  "author": {
    "reference": "Patient/PatientExample"
  }
},

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      "item": [
        {
          "linkId": "pf-cat-1",
          "text": "Does your health now limit you in doing two
hours of physical labor?",
          "answer": [
            {
              "valueCoding": {
                "extension": [
                  {
                    "url":
"http://hl7.org/fhir/StructureDefinition/ordinalValue",
                    "valueDecimal": 2
                  }
                ],
              "system": "http://loinc.org",
              "code": "LA11911-7",
              "display": "Quite a lot"
            }
          ]
        },
        {
          "linkId": "pf-cat-2",
          "text": "Are you able to do chores such as vacuuming or
yard work?",
          "answer": [
            {
              "valueCoding": {
                "extension": [
                  {
                    "url":
"http://hl7.org/fhir/StructureDefinition/ordinalValue",
                    "valueDecimal": 3
                  }
                ],
              "system": "http://loinc.org",
              "code": "LA13920-6",
              "display": "With some difficulty"
            }
          ]
        },
        {
          "linkId": "pf-cat-3",
          "text": "To what extent are you able to carry out your
everyday physical activities such as walking, climbing stairs, carrying
groceries, or moving a chair?",
          "answer": [
            {
              "valueCoding": {
                "extension": [
                  {
                    "url":
"http://hl7.org/fhir/StructureDefinition/ordinalValue",

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        "valueDecimal": 3
      }
    ],
    "system": "http://loinc.org",
    "code": "LA13939-6",
    "display": "Moderately"
  }
}
],
},
{
  "linkId": "pf-cat-4",
  "text": "Does your health now limit you in walking more
than a mile?",
  "answer": [
    {
      "valueCoding": {
        "extension": [
          {
            "url":
"http://hl7.org/fhir/StructureDefinition/ordinalValue",
            "valueDecimal": 2
          }
        ],
        "system": "http://loinc.org",
        "code": "LA11911-7",
        "display": "Quite a lot"
      }
    }
  ]
},
{
  "linkId": "tscore",
  "answer": [
    {
      "valueDecimal": 38
    }
  ]
}
],
},
{
  "fullUrl":
"http://example.org/QuestionnaireResponse/PROMIS PainInterferenceSFExample",
  "resource": {
    "resourceType": "QuestionnaireResponse",
    "id": "PROMIS PainInterferenceSFExample",
    "meta": {
      "profile": [
        "http://hl7.org/fhir/us/core/StructureDefinition/us-
core-questionnaireresponse|6.1.0"
      ]
    }
  },

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        "language": "en",
        "text": {
            "status": "generated",
            "div": "\u003Cdiv xml:lang=\"en\"
xmlns=\"http://www.w3.org/1999/xhtml\"
lang=\"en\"\u003E\u003Cp\u003EPROMIS Pain Interference (Short Form 6a)
QuestionnaireResponse for John Doe. T-score:
58.6.\u003C/p\u003E\u003C/div\u003E"
        },
        "questionnaire":
"http://example.org/Questionnaire/PROMISPainInterferenceSFQuestionnaire",
        "status": "completed",
        "subject": {
            "reference": "Patient/PatientExample"
        },
        "authored": "2026-01-15T10:30:00Z",
        "author": {
            "reference": "Patient/PatientExample"
        },
        "item": [
            {
                "linkId": "pi6a-1",
                "text": "In the past 7 days: How much did pain
interfere with your day to day activities?",
                "answer": [
                    {
                        "valueCoding": {
                            "extension": [
                                {
                                    "url":
"http://hl7.org/fhir/StructureDefinition/ordinalValue",
                                    "valueDecimal": 2
                                }
                            ],
                            "system": "http://loinc.org",
                            "code": "LA13863-8",
                            "display": "A little bit"
                        }
                    }
                ]
            },
            {
                "linkId": "pi6a-2",
                "text": "In the past 7 days: How much did pain
interfere with work around the home?",
                "answer": [
                    {
                        "valueCoding": {
                            "extension": [
                                {
                                    "url":
"http://hl7.org/fhir/StructureDefinition/ordinalValue",
                                    "valueDecimal": 2
                                }
                            ],

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        "system": "http://loinc.org",
        "code": "LA13863-8",
        "display": "A little bit"
    }
}
],
},
{
    "linkId": "pi6a-3",
    "text": "In the past 7 days: How much did pain
interfere with your ability to participate in social activities?",
    "answer": [
        {
            "valueCoding": {
                "extension": [
                    {
                        "url":
"http://hl7.org/fhir/StructureDefinition/ordinalValue",
                        "valueDecimal": 3
                    }
                ],
                "system": "http://loinc.org",
                "code": "LA13909-9",
                "display": "Somewhat"
            }
        }
    ]
},
{
    "linkId": "pi6a-4",
    "text": "In the past 7 days: How much did pain
interfere with your household chores?",
    "answer": [
        {
            "valueCoding": {
                "extension": [
                    {
                        "url":
"http://hl7.org/fhir/StructureDefinition/ordinalValue",
                        "valueDecimal": 2
                    }
                ],
                "system": "http://loinc.org",
                "code": "LA13863-8",
                "display": "A little bit"
            }
        }
    ]
},
{
    "linkId": "pi6a-5",
    "text": "In the past 7 days: How much did pain
interfere with the things you usually do for fun?",
    "answer": [
        {

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        "valueCoding": {
          "extension": [
            {
              "url":
"http://hl7.org/fhir/StructureDefinition/ordinalValue",
              "valueDecimal": 3
            }
          ],
          "system": "http://loinc.org",
          "code": "LA13909-9",
          "display": "Somewhat"
        }
      ],
    },
    {
      "linkId": "pi6a-6",
      "text": "In the past 7 days: How much did pain
interfere with your enjoyment of social activities?",
      "answer": [
        {
          "valueCoding": {
            "extension": [
              {
                "url":
"http://hl7.org/fhir/StructureDefinition/ordinalValue",
                "valueDecimal": 3
              }
            ],
            "system": "http://loinc.org",
            "code": "LA13909-9",
            "display": "Somewhat"
          }
        }
      ]
    },
    {
      "linkId": "raw_score",
      "answer": [
        {
          "valueDecimal": 15
        }
      ]
    },
    {
      "linkId": "tscore",
      "answer": [
        {
          "valueDecimal": 58.6
        }
      ]
    }
  ]
},
},

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    {
      "fullUrl":
"http://example.org/QuestionnaireResponse/PROMISPainInterferenceCATExample",
      "resource": {
        "resourceType": "QuestionnaireResponse",
        "id": "PROMISPainInterferenceCATExample",
        "meta": {
          "profile": [
            "http://hl7.org/fhir/us/core/StructureDefinition/us-
core-questionnaireresponse|6.1.0"
          ]
        },
        "language": "en",
        "text": {
          "status": "generated",
          "div": "\u003Cdiv xml:lang=\"en\"
xmlns=\"http://www.w3.org/1999/xhtml\"
lang=\"en\"\u003E\u003Cp\u003EPROMIS Pain Interference (CAT)
QuestionnaireResponse for John Doe. T-score:
70.\u003C/p\u003E\u003C/div\u003E"
        },
        "questionnaire":
"http://example.org/Questionnaire/PROMISPainInterferenceCATQuestionnaire"
      },
      "status": "completed",
      "subject": {
        "reference": "Patient/PatientExample"
      },
      "authored": "2026-01-15T10:30:00Z",
      "author": {
        "reference": "Patient/PatientExample"
      },
      "item": [
        {
          "linkId": "pi-cat-1",
          "text": "In the past 7 days: How much did pain
interfere with your day to day activities?",
          "answer": [
            {
              "valueCoding": {
                "extension": [
                  {
                    "url":
"http://hl7.org/fhir/StructureDefinition/ordinalValue",
                    "valueDecimal": 4
                  }
                ],
                "system": "http://loinc.org",
                "code": "LA13902-4",
                "display": "Quite a bit"
              }
            }
          ]
        }
      ]
    },

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    {
      "linkId": "pi-cat-2",
      "text": "In the past 7 days: How much did pain
interfere with your ability to participate in social activities?",
      "answer": [
        {
          "valueCoding": {
            "extension": [
              {
                "url":
"http://hl7.org/fhir/StructureDefinition/ordinalValue",
                "valueDecimal": 5
              }
            ],
            "system": "http://loinc.org",
            "code": "LA13914-9",
            "display": "Very much"
          }
        }
      ],
    },
    {
      "linkId": "pi-cat-3",
      "text": "In the past 7 days: How much did pain
interfere with your enjoyment of social activities?",
      "answer": [
        {
          "valueCoding": {
            "extension": [
              {
                "url":
"http://hl7.org/fhir/StructureDefinition/ordinalValue",
                "valueDecimal": 4
              }
            ],
            "system": "http://loinc.org",
            "code": "LA13902-4",
            "display": "Quite a bit"
          }
        }
      ],
    },
    {
      "linkId": "pi-cat-4",
      "text": "In the past 7 days: How much did pain
interfere with work around the home?",
      "answer": [
        {
          "valueCoding": {
            "extension": [
              {
                "url":
"http://hl7.org/fhir/StructureDefinition/ordinalValue",
                "valueDecimal": 5
              }
            ]
          }
        }
      ]
    }
  ]
}

```

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        ],
        "system": "http://loinc.org",
        "code": "LA13914-9",
        "display": "Very much"
    }
}
],
},
{
    "linkId": "tscore",
    "answer": [
        {
            "valueDecimal": 70
        }
    ]
}
]
}
},
{
    "fullUrl":
"http://example.org/QuestionnaireResponse/OswestryDisabilityIndexExample"
,
    "resource": {
        "resourceType": "QuestionnaireResponse",
        "id": "OswestryDisabilityIndexExample",
        "meta": {
            "profile": [
                "http://hl7.org/fhir/us/core/StructureDefinition/us-
core-questionnaireresponse|6.1.0"
            ]
        },
        "language": "en",
        "text": {
            "status": "generated",
            "div": "\u003Cdiv xml:lang=\"en\"
xmlns=\"http://www.w3.org/1999/xhtml\"
lang=\"en\"\u003E\u003Cp\u003EOswestry Disability Index
QuestionnaireResponse for John Doe. Score:
36.\u003C/p\u003E\u003C/div\u003E"
        },
        "questionnaire":
"http://example.org/Questionnaire/OswestryDisabilityIndexQuestionnaire",
        "status": "completed",
        "subject": {
            "reference": "Patient/PatientExample"
        },
        "authored": "2026-01-15T10:30:00Z",
        "author": {
            "reference": "Patient/PatientExample"
        },
        "item": [
            {
                "linkId": "odi-1",
                "text": "Pain intensity",

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        "answer": [
          {
            "valueCoding": {
              "extension": [
                {
                  "url":
"http://hl7.org/fhir/StructureDefinition/ordinalValue",
                  "valueDecimal": 2
                }
              ],
              "display": "The pain is moderate at the moment."
            }
          }
        ]
      },
      {
        "linkId": "odi-2",
        "text": "Personal care (washing, dressing, etc.)",
        "answer": [
          {
            "valueCoding": {
              "extension": [
                {
                  "url":
"http://hl7.org/fhir/StructureDefinition/ordinalValue",
                  "valueDecimal": 1
                }
              ],
              "display": "I can look after myself normally but
it is very painful."
            }
          }
        ]
      },
      {
        "linkId": "odi-3",
        "text": "Lifting",
        "answer": [
          {
            "valueCoding": {
              "extension": [
                {
                  "url":
"http://hl7.org/fhir/StructureDefinition/ordinalValue",
                  "valueDecimal": 3
                }
              ],
              "display": "Pain prevents me lifting heavy
weights off the floor, but I can manage if they are conveniently placed
e.g. on a table."
            }
          }
        ]
      },
      {

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        "linkId": "odi-4",
        "text": "Walking",
        "answer": [
            {
                "valueCoding": {
                    "extension": [
                        {
                            "url":
"http://hl7.org/fhir/StructureDefinition/ordinalValue",
                            "valueDecimal": 1
                        }
                    ],
                    "display": "Pain prevents me from walking more
than one mile."
                }
            }
        ],
        {
            "linkId": "odi-5",
            "text": "Sitting",
            "answer": [
                {
                    "valueCoding": {
                        "extension": [
                            {
                                "url":
"http://hl7.org/fhir/StructureDefinition/ordinalValue",
                                "valueDecimal": 2
                            }
                        ],
                        "display": "Pain prevents me from sitting for
more than one hour."
                    }
                }
            ],
            {
                "linkId": "odi-6",
                "text": "Standing",
                "answer": [
                    {
                        "valueCoding": {
                            "extension": [
                                {
                                    "url":
"http://hl7.org/fhir/StructureDefinition/ordinalValue",
                                    "valueDecimal": 2
                                }
                            ],
                            "display": "Pain prevents me from standing for
more than 1 hour."
                        }
                    }
                ]
            }
        ]
    }

```

```

    },
    {
      "linkId": "odi-7",
      "text": "Sleeping",
      "answer": [
        {
          "valueCoding": {
            "extension": [
              {
                "url":
"http://hl7.org/fhir/StructureDefinition/ordinalValue",
                "valueDecimal": 1
              }
            ],
            "display": "My sleep is occasionally interrupted
by pain."
          }
        }
      ]
    },
    {
      "linkId": "odi-8",
      "text": "Sex life (if applicable)",
      "answer": [
        {
          "valueCoding": {
            "extension": [
              {
                "url":
"http://hl7.org/fhir/StructureDefinition/ordinalValue",
                "valueDecimal": 2
              }
            ],
            "display": "My sex life is nearly normal but is
very painful."
          }
        }
      ]
    },
    {
      "linkId": "odi-9",
      "text": "Social life",
      "answer": [
        {
          "valueCoding": {
            "extension": [
              {
                "url":
"http://hl7.org/fhir/StructureDefinition/ordinalValue",
                "valueDecimal": 3
              }
            ],
            "display": "Pain has restricted my social life
and I do not go out as often."
          }
        }
      ]
    }
  ]
}

```

```

        }
      ]
    },
    {
      "linkId": "odi-10",
      "text": "Traveling",
      "answer": [
        {
          "valueCoding": {
            "extension": [
              {
                "url":
"http://hl7.org/fhir/StructureDefinition/ordinalValue",
                "valueDecimal": 2
              }
            ],
            "display": "Pain is bad but I am able to manage
trips over two hours."
          }
        }
      ]
    },
    {
      "linkId": "score",
      "answer": [
        {
          "valueDecimal": 36
        }
      ]
    }
  ]
}
},
{
  "fullUrl":
"http://example.org/QuestionnaireResponse/NeckDisabilityIndexExample",
  "resource": {
    "resourceType": "QuestionnaireResponse",
    "id": "NeckDisabilityIndexExample",
    "meta": {
      "profile": [
        "http://hl7.org/fhir/us/core/StructureDefinition/us-
core-questionnaireresponse|6.1.0"
      ]
    },
    "language": "en",
    "text": {
      "status": "generated",
      "div": "\u003Cdiv xml:lang=\"en\"
xmlns=\"http://www.w3.org/1999/xhtml\" lang=\"en\"\u003E\u003Cp\u003ENeck
Disability Index QuestionnaireResponse for John Doe. Total score:
40.\u003Cp\u003E\u003Cdiv\u003E"
    },
    "questionnaire":
"http://example.org/Questionnaire/NeckDisabilityIndexQuestionnaire",

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```

    "status": "completed",
    "subject": {
      "reference": "Patient/PatientExample"
    },
    "authored": "2026-01-15T10:30:00Z",
    "author": {
      "reference": "Patient/PatientExample"
    },
    "item": [
      {
        "linkId": "ndi-1",
        "text": "Pain intensity",
        "answer": [
          {
            "valueCoding": {
              "extension": [
                {
                  "url":
"http://hl7.org/fhir/StructureDefinition/ordinalValue",
                  "valueDecimal": 2
                }
              ],
              "system": "http://loinc.org",
              "code": "LA25855-0",
              "display": "The pain is moderate at the moment."
            }
          ]
        },
        "linkId": "ndi-2",
        "text": "Personal care (washing, dressing, etc.)",
        "answer": [
          {
            "valueCoding": {
              "extension": [
                {
                  "url":
"http://hl7.org/fhir/StructureDefinition/ordinalValue",
                  "valueDecimal": 1
                }
              ],
              "system": "http://loinc.org",
              "code": "LA25860-0",
              "display": "I can look after myself normally but
it causes extra pain."
            }
          ]
        },
        "linkId": "ndi-3",
        "text": "Lifting",
        "answer": [

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```

        "valueCoding": {
          "extension": [
            {
              "url":
"http://hl7.org/fhir/StructureDefinition/ordinalValue",
              "valueDecimal": 2
            }
          ],
          "system": "http://loinc.org",
          "code": "LA25867-5",
          "display": "Pain prevents me lifting heavy
weights off the floor, but I can manage if they are conveniently placed,
for example on a table."
        }
      ]
    },
    {
      "linkId": "ndi-4",
      "text": "Reading",
      "answer": [
        {
          "valueCoding": {
            "extension": [
              {
                "url":
"http://hl7.org/fhir/StructureDefinition/ordinalValue",
                "valueDecimal": 2
              }
            ],
            "system": "http://loinc.org",
            "code": "LA25873-3",
            "display": "I can read as much as I want with
moderate pain in my neck."
          }
        }
      ]
    },
    {
      "linkId": "ndi-5",
      "text": "Headaches",
      "answer": [
        {
          "valueCoding": {
            "extension": [
              {
                "url":
"http://hl7.org/fhir/StructureDefinition/ordinalValue",
                "valueDecimal": 3
              }
            ],
            "system": "http://loinc.org",
            "code": "LA26367-5",
            "display": "I have moderate headaches, which come
frequently."
          }
        }
      ]
    }
  ]
}

```

```

        }
    }
]
},
{
    "linkId": "ndi-6",
    "text": "Concentration",
    "answer": [
        {
            "valueCoding": {
                "extension": [
                    {
                        "url":
"http://hl7.org/fhir/StructureDefinition/ordinalValue",
                        "valueDecimal": 2
                    }
                ],
            "system": "http://loinc.org",
            "code": "LA25880-8",
            "display": "I have a fair degree of difficulty in
concentrating when I want to."
        }
    ]
},
{
    "linkId": "ndi-7",
    "text": "Work",
    "answer": [
        {
            "valueCoding": {
                "extension": [
                    {
                        "url":
"http://hl7.org/fhir/StructureDefinition/ordinalValue",
                        "valueDecimal": 2
                    }
                ],
            "system": "http://loinc.org",
            "code": "LA25886-5",
            "display": "I can do most of my usual work, but
no more."
        }
    ]
},
{
    "linkId": "ndi-8",
    "text": "Driving",
    "answer": [
        {
            "valueCoding": {
                "extension": [
                    {

```

```

        "url":
"http://hl7.org/fhir/StructureDefinition/ordinalValue",
        "valueDecimal": 2
    }
    ],
    "system": "http://loinc.org",
    "code": "LA25892-3",
    "display": "I can drive my car as long as I want
with moderate pain in my neck."
    }
    }
    ],
    },
    {
        "linkId": "ndi-9",
        "text": "Sleeping",
        "answer": [
            {
                "valueCoding": {
                    "extension": [
                        {
                            "url":
"http://hl7.org/fhir/StructureDefinition/ordinalValue",
                            "valueDecimal": 2
                        }
                    ],
                    "system": "http://loinc.org",
                    "code": "LA25898-0",
                    "display": "My sleep is mildly disturbed (1-2 hrs
sleepless)."

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        {
            "linkId": "total-score",
            "answer": [
                {
                    "valueDecimal": 40
                }
            ]
        }
    ]
}
},
{
    "fullUrl":
"http://example.org/QuestionnaireResponse/QuickDASHExample",
    "resource": {
        "resourceType": "QuestionnaireResponse",
        "id": "QuickDASHExample",
        "meta": {
            "profile": [
                "http://hl7.org/fhir/us/core/StructureDefinition/us-
core-questionnaireresponse|6.1.0"
            ]
        },
        "language": "en",
        "text": {
            "status": "generated",
            "div": "\u003Cdiv xml:lang=\"en\"
xmlns=\"http://www.w3.org/1999/xhtml\"
lang=\"en\"\u003E\u003Cp\u003EQuickDASH QuestionnaireResponse for John
Doe. Score: 50.\u003C/p\u003E\u003C/div\u003E"
        },
        "questionnaire":
"http://example.org/Questionnaire/QuickDASHQuestionnaire",
        "status": "completed",
        "subject": {
            "reference": "Patient/PatientExample"
        },
        "authored": "2026-01-15T10:30:00Z",
        "author": {
            "reference": "Patient/PatientExample"
        },
        "item": [
            {
                "linkId": "qd-1",
                "text": "Open a tight or new jar.",
                "answer": [
                    {
                        "valueCoding": {
                            "extension": [
                                {
                                    "url":
"http://hl7.org/fhir/StructureDefinition/ordinalValue",
                                    "valueDecimal": 3
                                }
                            ]
                        }
                    }
                ]
            }
        ],

```

```

        "display": "Moderate difficulty"
      }
    }
  ],
},
{
  "linkId": "qd-2",
  "text": "Do heavy household chores (e.g., wash walls,
wash floors).",
  "answer": [
    {
      "valueCoding": {
        "extension": [
          {
            "url":
"http://hl7.org/fhir/StructureDefinition/ordinalValue",
            "valueDecimal": 4
          }
        ],
        "display": "Severe difficulty"
      }
    }
  ]
},
{
  "linkId": "qd-3",
  "text": "Carry a shopping bag or briefcase.",
  "answer": [
    {
      "valueCoding": {
        "extension": [
          {
            "url":
"http://hl7.org/fhir/StructureDefinition/ordinalValue",
            "valueDecimal": 3
          }
        ],
        "display": "Moderate difficulty"
      }
    }
  ]
},
{
  "linkId": "qd-4",
  "text": "Wash your back.",
  "answer": [
    {
      "valueCoding": {
        "extension": [
          {
            "url":
"http://hl7.org/fhir/StructureDefinition/ordinalValue",
            "valueDecimal": 2
          }
        ]
      }
    ]
  ],

```

```

        "display": "Mild difficulty"
    }
}
],
},
{
    "linkId": "qd-5",
    "text": "Use a knife to cut food.",
    "answer": [
        {
            "valueCoding": {
                "extension": [
                    {
                        "url":
"http://hl7.org/fhir/StructureDefinition/ordinalValue",
                        "valueDecimal": 2
                    }
                ],
                "display": "Mild difficulty"
            }
        }
    ]
},
},
{
    "linkId": "qd-6",
    "text": "Recreational activities in which you take some
force or impact through your arm, shoulder, or hand (e.g., golf,
hammering, tennis, etc.).",
    "answer": [
        {
            "valueCoding": {
                "extension": [
                    {
                        "url":
"http://hl7.org/fhir/StructureDefinition/ordinalValue",
                        "valueDecimal": 4
                    }
                ],
                "display": "Severe difficulty"
            }
        }
    ]
},
},
{
    "linkId": "qd-7",
    "text": "During the past week, to what extent has your
arm, shoulder, or hand problem interfered with your normal social
activities with family, friends, neighbors, or groups?",
    "answer": [
        {
            "valueCoding": {
                "extension": [
                    {
                        "url":
"http://hl7.org/fhir/StructureDefinition/ordinalValue",

```

```

        "valueDecimal": 3
      }
    ],
    "display": "Moderately"
  }
}
],
},
{
  "linkId": "qd-8",
  "text": "During the past week, were you limited in your
work or other regular daily activities as a result of your arm, shoulder,
or hand problem?",
  "answer": [
    {
      "valueCoding": {
        "extension": [
          {
            "url":
"http://hl7.org/fhir/StructureDefinition/ordinalValue",
            "valueDecimal": 4
          }
        ],
        "display": "Very limited"
      }
    }
  ]
},
{
  "linkId": "qd-9",
  "text": "Arm, shoulder, or hand pain",
  "answer": [
    {
      "valueCoding": {
        "extension": [
          {
            "url":
"http://hl7.org/fhir/StructureDefinition/ordinalValue",
            "valueDecimal": 3
          }
        ],
        "display": "Moderate"
      }
    }
  ]
},
{
  "linkId": "qd-10",
  "text": "Tingling (pins and needles) in your arm,
shoulder, or hand",
  "answer": [
    {
      "valueCoding": {
        "extension": [
          {

```



```

        "url":
"http://hl7.org/fhir/StructureDefinition/ordinalValue",
        "valueDecimal": 2
    }
    ],
    "display": "Mild"
}
}
],
},
{
    "linkId": "qd-11",
    "text": "During the past week, how much difficulty have
you had sleeping because of the pain in your arm, shoulder, or hand?",
    "answer": [
        {
            "valueCoding": {
                "extension": [
                    {
                        "url":
"http://hl7.org/fhir/StructureDefinition/ordinalValue",
                        "valueDecimal": 3
                    }
                ],
                "display": "Moderate difficulty"
            }
        }
    ]
},
},
{
    "linkId": "score",
    "answer": [
        {
            "valueDecimal": 50
        }
    ]
}
]
}
},
{
    "fullUrl":
"http://example.org/QuestionnaireResponse/MSKPGICEExample",
    "resource": {
        "resourceType": "QuestionnaireResponse",
        "id": "MSKPGICEExample",
        "meta": {
            "profile": [
                "http://hl7.org/fhir/us/core/StructureDefinition/us-
core-questionnaireresponse|6.1.0"
            ]
        },
        "language": "en",
        "text": {
            "status": "generated",

```

```

        "div": "\u003Cdiv xml:lang=\"en\"
xmlns=\"http://www.w3.org/1999/xhtml\" lang=\"en\" \u003E\u003Cp\u003E
QuestionnaireResponse for John Doe. Response: Very much
improved.\u003C/p\u003E\u003C/div\u003E"
    },
    "questionnaire":
"http://example.org/Questionnaire/MSKPGICQuestionnaire",
    "status": "completed",
    "subject": {
        "reference": "Patient/PatientExample"
    },
    "authored": "2026-01-15T10:30:00Z",
    "author": {
        "reference": "Patient/PatientExample"
    },
    "item": [
        {
            "linkId": "pgic-1",
            "text": "Since beginning treatment, how would you
describe your overall change in pain, function, and quality of life?",
            "answer": [
                {
                    "valueCoding": {
                        "extension": [
                            {
                                "url":
"http://hl7.org/fhir/StructureDefinition/ordinalValue",
                                "valueDecimal": 1
                            }
                        ],
                        "display": "Very much improved"
                    }
                }
            ]
        }
    ]
},
{
    "fullUrl":
"http://example.org/QuestionnaireResponse/KOOSJRExample",
    "resource": {
        "resourceType": "QuestionnaireResponse",
        "id": "KOOSJRExample",
        "meta": {
            "profile": [
                "http://hl7.org/fhir/us/core/StructureDefinition/us-
core-questionnaireresponse|6.1.0"
            ]
        },
        "language": "en",
        "text": {
            "status": "generated",
            "div": "\u003Cdiv xml:lang=\"en\"
xmlns=\"http://www.w3.org/1999/xhtml\" lang=\"en\" \u003E\u003Cp\u003E
KOOS

```

```

JR QuestionnaireResponse for John Doe. Total interval score:
50.012.\u003C/p\u003E\u003C/div\u003E"
    },
    "questionnaire":
"http://example.org/Questionnaire/KOOSJRQuestionnaire",
    "status": "completed",
    "subject": {
        "reference": "Patient/PatientExample"
    },
    "authored": "2026-01-15T10:30:00Z",
    "author": {
        "reference": "Patient/PatientExample"
    },
    "item": [
        {
            "linkId": "koosjr-1",
            "text": "How severe is your knee stiffness after first
wakening in the morning?",
            "answer": [
                {
                    "valueCoding": {
                        "extension": [
                            {
                                "url":
"http://hl7.org/fhir/StructureDefinition/ordinalValue",
                                "valueDecimal": 2
                            }
                        ],
                        "system": "http://loinc.org",
                        "code": "LA6751-7",
                        "display": "Moderate"
                    }
                }
            ]
        },
        {
            "linkId": "koosjr-2",
            "text": "Twisting/pivoting on your knee",
            "answer": [
                {
                    "valueCoding": {
                        "extension": [
                            {
                                "url":
"http://hl7.org/fhir/StructureDefinition/ordinalValue",
                                "valueDecimal": 2
                            }
                        ],
                        "system": "http://loinc.org",
                        "code": "LA6751-7",
                        "display": "Moderate"
                    }
                }
            ]
        }
    ],
},

```

```

{
  "linkId": "koosjr-3",
  "text": "Straightening knee fully",
  "answer": [
    {
      "valueCoding": {
        "extension": [
          {
            "url":
"http://hl7.org/fhir/StructureDefinition/ordinalValue",
            "valueDecimal": 1
          }
        ],
        "system": "http://loinc.org",
        "code": "LA6752-5",
        "display": "Mild"
      }
    }
  ],
},
{
  "linkId": "koosjr-4",
  "text": "Going up or down stairs",
  "answer": [
    {
      "valueCoding": {
        "extension": [
          {
            "url":
"http://hl7.org/fhir/StructureDefinition/ordinalValue",
            "valueDecimal": 3
          }
        ],
        "system": "http://loinc.org",
        "code": "LA6750-9",
        "display": "Severe"
      }
    }
  ],
},
{
  "linkId": "koosjr-5",
  "text": "Standing upright",
  "answer": [
    {
      "valueCoding": {
        "extension": [
          {
            "url":
"http://hl7.org/fhir/StructureDefinition/ordinalValue",
            "valueDecimal": 2
          }
        ],
        "system": "http://loinc.org",
        "code": "LA6751-7",

```

```

        "display": "Moderate"
    }
}
],
},
{
    "linkId": "koosjr-6",
    "text": "Rising from sitting",
    "answer": [
        {
            "valueCoding": {
                "extension": [
                    {
                        "url":
"http://hl7.org/fhir/StructureDefinition/ordinalValue",
                        "valueDecimal": 2
                    }
                ],
                "system": "http://loinc.org",
                "code": "LA6751-7",
                "display": "Moderate"
            }
        }
    ]
},
{
    "linkId": "koosjr-7",
    "text": "Bending to floor/picking up an object",
    "answer": [
        {
            "valueCoding": {
                "extension": [
                    {
                        "url":
"http://hl7.org/fhir/StructureDefinition/ordinalValue",
                        "valueDecimal": 3
                    }
                ],
                "system": "http://loinc.org",
                "code": "LA6750-9",
                "display": "Severe"
            }
        }
    ]
},
{
    "linkId": "total-interval-score",
    "answer": [
        {
            "valueDecimal": 50.012
        }
    ]
}
]
}
}

```

```

    },
    {
      "fullUrl":
"http://example.org/QuestionnaireResponse/HOOSJRExample",
      "resource": {
        "resourceType": "QuestionnaireResponse",
        "id": "HOOSJRExample",
        "meta": {
          "profile": [
            "http://hl7.org/fhir/us/core/StructureDefinition/us-
core-questionnaireresponse|6.1.0"
          ]
        },
        "language": "en",
        "text": {
          "status": "generated",
          "div": "\u003Cdiv xml:lang=\"en\"
xmlns=\"http://www.w3.org/1999/xhtml\" lang=\"en\" \u003E\u003Cp\u003EHOOS
JR QuestionnaireResponse for John Doe. Total interval score:
55.985.\u003C/p\u003E\u003C/div\u003E"
        },
        "questionnaire":
"http://example.org/Questionnaire/HOOSJRQuestionnaire",
        "status": "completed",
        "subject": {
          "reference": "Patient/PatientExample"
        },
        "authored": "2026-01-15T10:30:00Z",
        "author": {
          "reference": "Patient/PatientExample"
        },
        "item": [
          {
            "linkId": "hoosjr-1",
            "text": "Going up or down stairs",
            "answer": [
              {
                "valueCoding": {
                  "extension": [
                    {
                      "url":
"http://hl7.org/fhir/StructureDefinition/ordinalValue",
                      "valueDecimal": 2
                    }
                  ]
                },
                "system": "http://loinc.org",
                "code": "LA6751-7",
                "display": "Moderate"
              }
            ]
          },
          {
            "linkId": "hoosjr-2",
            "text": "Walking on an uneven surface",

```

```

      "answer": [
        {
          "valueCoding": {
            "extension": [
              {
                "url":
"http://hl7.org/fhir/StructureDefinition/ordinalValue",
                "valueDecimal": 2
              }
            ],
            "system": "http://loinc.org",
            "code": "LA6751-7",
            "display": "Moderate"
          }
        }
      ],
    },
    {
      "linkId": "hoosjr-3",
      "text": "Rising from sitting",
      "answer": [
        {
          "valueCoding": {
            "extension": [
              {
                "url":
"http://hl7.org/fhir/StructureDefinition/ordinalValue",
                "valueDecimal": 2
              }
            ],
            "system": "http://loinc.org",
            "code": "LA6751-7",
            "display": "Moderate"
          }
        }
      ]
    },
    {
      "linkId": "hoosjr-4",
      "text": "Bending to the floor/picking up an object",
      "answer": [
        {
          "valueCoding": {
            "extension": [
              {
                "url":
"http://hl7.org/fhir/StructureDefinition/ordinalValue",
                "valueDecimal": 3
              }
            ],
            "system": "http://loinc.org",
            "code": "LA6750-9",
            "display": "Severe"
          }
        }
      ]
    }
  ]
}

```

```

    ]
  },
  {
    "linkId": "hoosjr-5",
    "text": "Lying in bed",
    "answer": [
      {
        "valueCoding": {
          "extension": [
            {
              "url":
"http://hl7.org/fhir/StructureDefinition/ordinalValue",
              "valueDecimal": 1
            }
          ],
          "system": "http://loinc.org",
          "code": "LA6752-5",
          "display": "Mild"
        }
      }
    ]
  },
  {
    "linkId": "hoosjr-6",
    "text": "Sitting",
    "answer": [
      {
        "valueCoding": {
          "extension": [
            {
              "url":
"http://hl7.org/fhir/StructureDefinition/ordinalValue",
              "valueDecimal": 1
            }
          ],
          "system": "http://loinc.org",
          "code": "LA6752-5",
          "display": "Mild"
        }
      }
    ]
  },
  {
    "linkId": "total-interval-score",
    "answer": [
      {
        "valueDecimal": 55.985
      }
    ]
  }
]
}
},
{

```



```

        "fullUrl":
"http://example.org/QuestionnaireResponse/PROMISNRSEExample",
        "resource": {
            "resourceType": "QuestionnaireResponse",
            "id": "PROMISNRSEExample",
            "meta": {
                "profile": [
                    "http://hl7.org/fhir/us/core/StructureDefinition/us-
core-questionnaireresponse|6.1.0"
                ]
            },
            "language": "en",
            "text": {
                "status": "generated",
                "div": "\u003Cdiv xml:lang=\"en\"
xmlns=\"http://www.w3.org/1999/xhtml\"
lang=\"en\"\u003E\u003Cp\u003EPROMIS Pain Intensity NRS
QuestionnaireResponse for John Doe. Average pain (0-10):
6.\u003C/p\u003E\u003C/div\u003E"
            },
            "questionnaire":
"http://example.org/Questionnaire/PROMISNRSQuestionnaire",
            "status": "completed",
            "subject": {
                "reference": "Patient/PatientExample"
            },
            "authored": "2026-01-15T10:30:00Z",
            "author": {
                "reference": "Patient/PatientExample"
            },
            "item": [
                {
                    "linkId": "nrs-1",
                    "text": "In the past 7 days, how would you rate your
pain on average? (0-10)",
                    "answer": [
                        {
                            "valueInteger": 6
                        }
                    ]
                }
            ]
        }
    ]
}

```

## Response:

```
HTTP/1.1 202 Accepted
Content-Location: https://[base]/access/Patient/$submission-status/sub-123456
```

## Step 2: Extract Submission URL and Wait

- Extract submission URL from Content-Location header
- Wait 5 seconds before first poll

## Step 3: First Status Check (after 5 seconds)

```
GET https://[base]/access/Patient/$submission-status/sub-123456
```

## Response:

```
HTTP/1.1 202 Accepted
```

(Still processing - no body)

## Step 4: Second Status Check (after another 15 seconds)

```
GET https://[base]/access/Patient/$submission-status/sub-123456
```

## Response:

```
HTTP/1.1 200 OK
Content-Type: application/fhir+json

{
  "resourceType": "Parameters",
  "parameter": [
    {
      "name": "result",
      "valueCodeableConcept": {
        "coding": [
          {
            "system":
"https://globalalliantinc.com/access/CodeSystem/ACCESSReportDataResultCS"
,
            "code": "success",
            "display": "Success"
          }
        ],
        "text": "Data submission was successfully processed and recorded
for ACCESS Model evaluation."
      }
    }
  ]
}
```

```

    }
  ]
}

```

## Step 5: Process Result

- Parse the result code: "success"
- Display the text to the user
- Log successful submission for record-keeping

## BH Track Workflow Example

The following example demonstrates a complete data reporting workflow for a BH track patient:

### Step 1: Submit BH Data Report

POST [https://\[base\]/access/Patient/\\$report-data?entityId=ACCES67890](https://[base]/access/Patient/$report-data?entityId=ACCES67890)  
 Content-Type: application/fhir+json

```

{
  "resourceType": "Parameters",
  "id": "BHReportDataRequestExample",
  "meta": {
    "profile": [
      "https://dsacms.github.io/cmml-access-
model/StructureDefinition/access-report-data-in"
    ]
  },
  "language": "en",
  "parameter": [
    {
      "name": "participantID",
      "valueIdentifier": {
        "system": "https://dsacms.github.io/cmml-access-
model/participant-id",
        "value": "ACCES12345"
      }
    },
    {
      "name": "payerID",
      "valueIdentifier": {
        "type": {
          "coding": [
            {
              "system": "http://hl7.org/fhir/us/carinn-
bb/CodeSystem/C4BBIIdentifierType",
              "code": "payerid",

```

```

        "display": "Payer ID"
    }
    ],
    },
    "system": "urn:oid:2.16.840.1.113883.3.221.5",
    "value": "12345"
}
},
{
    "name": "track",
    "valueCodeableConcept": {
        "coding": [
            {
                "system": "https://dsacms.github.io/cmmi-access-
model/CodeSystem/ACCESSTrackCS",
                "code": "BH",
                "display": "Behavioral Health track"
            }
        ]
    }
},
},
{
    "name": "reportType",
    "valueCodeableConcept": {
        "coding": [
            {
                "system": "https://dsacms.github.io/cmmi-access-
model/CodeSystem/ACCESSReportTypeCS",
                "code": "baseline",
                "display": "Baseline Data Report"
            }
        ]
    }
},
},
{
    "name": "dataBundle",
    "resource": {
        "resourceType": "Bundle",
        "id": "BHReportDataBundleExample",
        "meta": {
            "profile": [
                "https://dsacms.github.io/cmmi-access-
model/StructureDefinition/access-data-reporting-bundle"
            ]
        },
    },
    "language": "en",
    "identifier": {
        "system": "http://example.org/data-bundle-id",
        "value": "bundle-bh-001"
    },
    "type": "document",
    "timestamp": "2026-01-26T12:00:00Z",
    "entry": [
        {

```

```

        "fullUrl":
"http://example.org/Composition/BHReportDataCompositionExample",
        "resource": {
            "resourceType": "Composition",
            "id": "BHReportDataCompositionExample",
            "meta": {
                "profile": [
                    "https://dsacms.github.io/cmmi-access-
model/StructureDefinition/access-data-reporting-composition"
                ]
            },
            "language": "en",
            "text": {
                "status": "generated",
                "div": "\u003Cdiv xml:lang=\"en\"
xmlns=\"http://www.w3.org/1999/xhtml\"
lang=\"en\" \u003E\u003Ch3\u003EACCESS BH Track Data
Report\u003C/h3\u003E\u003Cp\u003EPatient: John Doe (MBI:
1EG4TE5MK73)\u003C/p\u003E\u003Cp\u003EReport Date: 2026-01-
26\u003C/p\u003E\u003Cp\u003EAuthor: Jane
Smith\u003C/p\u003E\u003Cdiv\u003E"
            },
            "status": "final",
            "type": {
                "coding": [
                    {
                        "system": "http://loinc.org",
                        "code": "74465-6",
                        "display": "Questionnaire response Document"
                    }
                ]
            },
            "subject": {
                "reference": "Patient/PatientExample"
            },
            "date": "2026-01-26T12:00:00Z",
            "author": [
                {
                    "reference": "Practitioner/PractitionerExample"
                }
            ],
            "title": "ACCESS BH Track Data Report",
            "custodian": {
                "reference": "Organization/OrganizationExample"
            },
            "section": [
                {
                    "title": "Data reporting for BH track",
                    "code": {
                        "coding": [
                            {
                                "system": "https://dsacms.github.io/cmmi-access-
model/CodeSystem/ACCESSTrackCS",
                                "code": "BH"
                            }
                        ]
                    }
                }
            ]
        }
    ]
}

```

```

    ]
  },
  "text": {
    "status": "generated",
    "div": "\u003Cdiv xml:lang=\"en\"
xmlns=\"http://www.w3.org/1999/xhtml\"
lang=\"en\"\u003E\u003Cp\u003EBehavioral Health Track Data Reporting
including depression and anxiety assessments and patient-reported
outcomes.\u003C/p\u003E\u003C/div\u003E"
  },
  "section": [
    {
      "title": "Depression (PHQ-9)",
      "code": {
        "coding": [
          {
            "system": "http://loinc.org",
            "code": "44249-1"
          }
        ]
      },
      "entry": [
        {
          "reference":
"QuestionnaireResponse/DepressionPHQ9Example"
        }
      ]
    },
    {
      "title": "Anxiety (GAD-7)",
      "code": {
        "coding": [
          {
            "system": "http://loinc.org",
            "code": "69737-5"
          }
        ]
      },
      "entry": [
        {
          "reference":
"QuestionnaireResponse/AnxietyGAD7Example"
        }
      ]
    },
    {
      "title": "Overall Function (WHODAS 2.0)",
      "code": {
        "coding": [
          {
            "system": "https://dsacms.github.io/cmml-
access-model/CodeSystem/ACCESSReportDataCompositionSectionCS",
            "code": "WHODAS"
          }
        ]
      }
    }
  ]
}

```

```

        },
        "entry": [
            {
                "reference":
"QuestionnaireResponse/WHODASExample"
            }
        ]
    },
    {
        "title": "Patient's Global Impression of Change",
        "code": {
            "coding": [
                {
                    "system": "https://dsacms.github.io/cmmi-
access-model/CodeSystem/ACCESSReportDataCompositionSectionCS",
                    "code": "PGIC"
                }
            ]
        },
        "entry": [
            {
                "reference":
"QuestionnaireResponse/BHPGICExample"
            }
        ]
    }
]
}
}
}
},
{
    "fullUrl": "http://example.org/Patient/PatientExample",
    "resource": {
        "resourceType": "Patient",
        "id": "PatientExample",
        "meta": {
            "profile": [
                "http://hl7.org/fhir/us/core/StructureDefinition/us-
core-patient|6.1.0"
            ]
        },
        "language": "en",
        "text": {
            "status": "generated",
            "div": "\u003Cdiv xml:lang=\"en\"
xmlns=\"http://www.w3.org/1999/xhtml\" lang=\"en\" \u003EPatient John Doe,
male, born 1950-01-01, Medicare ID: 1EG4TE5MK73\u003C/div\u003E"
        },
        "identifier": [
            {
                "type": {
                    "coding": [
                        {

```

```

        "system":
"http://terminology.hl7.org/CodeSystem/v2-0203",
        "code": "MC"
    }
]
},
"system":
"http://terminology.hl7.org/NamingSystem/cmsMBI",
"value": "1EG4TE5MK73"
}
],
"name": [
{
    "family": "Doe",
    "given": [
        "John"
    ]
}
],
"gender": "male",
"birthDate": "1950-01-01"
}
},
{
    "fullUrl":
"http://example.org/Practitioner/PractitionerExample",
    "resource": {
        "resourceType": "Practitioner",
        "id": "PractitionerExample",
        "meta": {
            "profile": [
                "http://hl7.org/fhir/us/core/StructureDefinition/us-
core-practitioner|6.1.0"
            ]
        },
        "language": "en",
        "text": {
            "status": "generated",
            "div": "\u003Cdiv xml:lang=\"en\"
xmlns=\"http://www.w3.org/1999/xhtml\" lang=\"en\" \u003EPractitioner Jane
Smith, NPI: 1234567893\u003C/div\u003E"
        },
        "identifier": [
            {
                "system": "http://hl7.org/fhir/sid/us-npi",
                "value": "1234567893"
            }
        ],
        "name": [
            {
                "family": "Smith",
                "given": [
                    "Jane"
                ]
            }
        ]
    }
}

```



```

    ]
  }
},
{
  "fullUrl":
"http://example.org/Organization/OrganizationExample",
  "resource": {
    "resourceType": "Organization",
    "id": "OrganizationExample",
    "meta": {
      "profile": [
        "http://hl7.org/fhir/us/core/StructureDefinition/us-
core-organization|6.1.0"
      ]
    },
    "language": "en",
    "text": {
      "status": "generated",
      "div": "\u003Cdiv xml:lang=\"en\"
xmlns=\"http://www.w3.org/1999/xhtml\" lang=\"en\" \u003EExample
Healthcare Organization, ACCESS Participant ID:
ACCES12345\u003C/div\u003E"
    },
    "identifier": [
      {
        "system": "https://dsacms.github.io/cmmi-access-
model/participant-id",
        "value": "ACCES12345"
      }
    ],
    "active": true,
    "name": "Example Healthcare Organization"
  }
},
{
  "fullUrl":
"http://example.org/QuestionnaireResponse/DepressionPHQ9Example",
  "resource": {
    "resourceType": "QuestionnaireResponse",
    "id": "DepressionPHQ9Example",
    "meta": {
      "profile": [
        "http://hl7.org/fhir/us/core/StructureDefinition/us-
core-questionnaireresponse|6.1.0"
      ]
    },
    "language": "en",
    "text": {
      "status": "generated",
      "div": "\u003Cdiv xml:lang=\"en\"
xmlns=\"http://www.w3.org/1999/xhtml\" lang=\"en\" \u003E\u003Cp\u003E
PHQ-9 QuestionnaireResponse for John Doe. Total score:
7.\u003C/p\u003E\u003C/div\u003E"
    },
  },

```

```

      "questionnaire":
      "http://example.org/Questionnaire/PHQ9Questionnaire",
      "status": "completed",
      "subject": {
        "reference": "Patient/PatientExample"
      },
      "authored": "2026-01-15T10:30:00Z",
      "author": {
        "reference": "Patient/PatientExample"
      },
      "item": [
        {
          "linkId": "phq9-1",
          "text": "Little interest or pleasure in doing things",
          "answer": [
            {
              "valueCoding": {
                "extension": [
                  {
                    "url":
"http://hl7.org/fhir/StructureDefinition/ordinalValue",
                    "valueDecimal": 1
                  }
                ],
              "system": "http://loinc.org",
              "code": "LA6569-3",
              "display": "Several days"
            }
          ]
        },
        {
          "linkId": "phq9-2",
          "text": "Feeling down, depressed, or hopeless",
          "answer": [
            {
              "valueCoding": {
                "extension": [
                  {
                    "url":
"http://hl7.org/fhir/StructureDefinition/ordinalValue",
                    "valueDecimal": 1
                  }
                ],
              "system": "http://loinc.org",
              "code": "LA6569-3",
              "display": "Several days"
            }
          ]
        },
        {
          "linkId": "phq9-3",
          "text": "Trouble falling or staying asleep, or sleeping
too much",

```

```

      "answer": [
        {
          "valueCoding": {
            "extension": [
              {
                "url":
"http://hl7.org/fhir/StructureDefinition/ordinalValue",
                "valueDecimal": 2
              }
            ],
            "system": "http://loinc.org",
            "code": "LA6570-1",
            "display": "More than half the days"
          }
        }
      ],
    },
    {
      "linkId": "phq9-4",
      "text": "Feeling tired or having little energy",
      "answer": [
        {
          "valueCoding": {
            "extension": [
              {
                "url":
"http://hl7.org/fhir/StructureDefinition/ordinalValue",
                "valueDecimal": 1
              }
            ],
            "system": "http://loinc.org",
            "code": "LA6569-3",
            "display": "Several days"
          }
        }
      ],
    },
    {
      "linkId": "phq9-5",
      "text": "Poor appetite or overeating",
      "answer": [
        {
          "valueCoding": {
            "extension": [
              {
                "url":
"http://hl7.org/fhir/StructureDefinition/ordinalValue",
                "valueDecimal": 0
              }
            ],
            "system": "http://loinc.org",
            "code": "LA6568-5",
            "display": "Not at all"
          }
        }
      ],
    }
  ]
}

```

```

    ]
  },
  {
    "linkId": "phq9-6",
    "text": "Feeling bad about yourself – or that you are a
failure or have let yourself or your family down",
    "answer": [
      {
        "valueCoding": {
          "extension": [
            {
              "url":
"http://hl7.org/fhir/StructureDefinition/ordinalValue",
              "valueDecimal": 1
            }
          ],
          "system": "http://loinc.org",
          "code": "LA6569-3",
          "display": "Several days"
        }
      }
    ]
  },
  {
    "linkId": "phq9-7",
    "text": "Trouble concentrating on things, such as
reading the newspaper or watching television",
    "answer": [
      {
        "valueCoding": {
          "extension": [
            {
              "url":
"http://hl7.org/fhir/StructureDefinition/ordinalValue",
              "valueDecimal": 1
            }
          ],
          "system": "http://loinc.org",
          "code": "LA6569-3",
          "display": "Several days"
        }
      }
    ]
  },
  {
    "linkId": "phq9-8",
    "text": "Moving or speaking so slowly that other people
could have noticed? Or the opposite – being so fidgety or restless that
you have been moving around a lot more than usual",
    "answer": [
      {
        "valueCoding": {
          "extension": [
            {

```

```

        "url":
"http://hl7.org/fhir/StructureDefinition/ordinalValue",
        "valueDecimal": 0
    }
    ],
    "system": "http://loinc.org",
    "code": "LA6568-5",
    "display": "Not at all"
}
}
],
},
{
    "linkId": "phq9-9",
    "text": "Thoughts that you would be better off dead, or
of hurting yourself in some way",
    "answer": [
        {
            "valueCoding": {
                "extension": [
                    {
                        "url":
"http://hl7.org/fhir/StructureDefinition/ordinalValue",
                        "valueDecimal": 0
                    }
                ],
                "system": "http://loinc.org",
                "code": "LA6568-5",
                "display": "Not at all"
            }
        }
    ]
},
{
    "linkId": "phq9-10",
    "text": "How difficult have these made it for you to do
your work, take care of things at home, or get along with other people?",
    "answer": [
        {
            "valueCoding": {
                "extension": [
                    {
                        "url":
"http://hl7.org/fhir/StructureDefinition/ordinalValue",
                        "valueDecimal": 1
                    }
                ],
                "system": "http://loinc.org",
                "code": "LA6573-5",
                "display": "Somewhat difficult"
            }
        }
    ]
},
{

```

```

        "linkId": "total-score",
        "answer": [
            {
                "valueDecimal": 7
            }
        ]
    }
}
},
{
    "fullUrl":
"http://example.org/QuestionnaireResponse/AnxietyGAD7Example",
    "resource": {
        "resourceType": "QuestionnaireResponse",
        "id": "AnxietyGAD7Example",
        "meta": {
            "profile": [
                "http://hl7.org/fhir/us/core/StructureDefinition/us-
core-questionnaireresponse|6.1.0"
            ]
        },
        "language": "en",
        "text": {
            "status": "generated",
            "div": "\u003Cdiv xml:lang=\"en\"
xmlns=\"http://www.w3.org/1999/xhtml\" lang=\"en\" \u003E\u003Cp\u003EGAD-
7 QuestionnaireResponse for John Doe. Total score:
6.\u003C/p\u003E\u003C/div\u003E"
        },
        "questionnaire":
"http://example.org/Questionnaire/GAD7Questionnaire",
        "status": "completed",
        "subject": {
            "reference": "Patient/PatientExample"
        },
        "authored": "2026-01-15T10:30:00Z",
        "author": {
            "reference": "Patient/PatientExample"
        },
        "item": [
            {
                "linkId": "gad7-1",
                "text": "Feeling nervous, anxious, or on edge",
                "answer": [
                    {
                        "valueCoding": {
                            "extension": [
                                {
                                    "url":
"http://hl7.org/fhir/StructureDefinition/ordinalValue",
                                    "valueDecimal": 1
                                }
                            ]
                        },
                        "system": "http://loinc.org",

```

```

        "code": "LA6569-3",
        "display": "Several days"
    }
}
],
},
{
    "linkId": "gad7-2",
    "text": "Not being able to stop or control worrying",
    "answer": [
        {
            "valueCoding": {
                "extension": [
                    {
                        "url":
"http://hl7.org/fhir/StructureDefinition/ordinalValue",
                        "valueDecimal": 1
                    }
                ],
                "system": "http://loinc.org",
                "code": "LA6569-3",
                "display": "Several days"
            }
        }
    ]
},
{
    "linkId": "gad7-3",
    "text": "Worrying too much about different things",
    "answer": [
        {
            "valueCoding": {
                "extension": [
                    {
                        "url":
"http://hl7.org/fhir/StructureDefinition/ordinalValue",
                        "valueDecimal": 1
                    }
                ],
                "system": "http://loinc.org",
                "code": "LA6569-3",
                "display": "Several days"
            }
        }
    ]
},
{
    "linkId": "gad7-4",
    "text": "Trouble relaxing",
    "answer": [
        {
            "valueCoding": {
                "extension": [
                    {

```

```

        "url":
"http://hl7.org/fhir/StructureDefinition/ordinalValue",
        "valueDecimal": 1
    }
    ],
    "system": "http://loinc.org",
    "code": "LA6569-3",
    "display": "Several days"
}
}
],
},
{
    "linkId": "gad7-5",
    "text": "Being so restless that it is hard to sit
still",
    "answer": [
        {
            "valueCoding": {
                "extension": [
                    {
                        "url":
"http://hl7.org/fhir/StructureDefinition/ordinalValue",
                        "valueDecimal": 0
                    }
                ],
                "system": "http://loinc.org",
                "code": "LA6568-5",
                "display": "Not at all"
            }
        }
    ]
},
{
    "linkId": "gad7-6",
    "text": "Becoming easily annoyed or irritable",
    "answer": [
        {
            "valueCoding": {
                "extension": [
                    {
                        "url":
"http://hl7.org/fhir/StructureDefinition/ordinalValue",
                        "valueDecimal": 1
                    }
                ],
                "system": "http://loinc.org",
                "code": "LA6569-3",
                "display": "Several days"
            }
        }
    ]
},
{
    "linkId": "gad7-7",

```



```

        "text": "Feeling afraid as if something awful might
happen",
        "answer": [
            {
                "valueCoding": {
                    "extension": [
                        {
                            "url":
"http://hl7.org/fhir/StructureDefinition/ordinalValue",
                            "valueDecimal": 1
                        }
                    ],
                    "system": "http://loinc.org",
                    "code": "LA6569-3",
                    "display": "Several days"
                }
            }
        ],
        {
            "linkId": "total-score",
            "answer": [
                {
                    "valueDecimal": 6
                }
            ]
        }
    ],
    },
    {
        "fullUrl":
"http://example.org/QuestionnaireResponse/WHODASExample",
        "resource": {
            "resourceType": "QuestionnaireResponse",
            "id": "WHODASExample",
            "meta": {
                "profile": [
                    "http://hl7.org/fhir/us/core/StructureDefinition/us-
core-questionnaireresponse|6.1.0"
                ]
            },
            "language": "en",
            "text": {
                "status": "generated",
                "div": "\u003Cdiv xml:lang=\"en\"
xmlns=\"http://www.w3.org/1999/xhtml\"
lang=\"en\"\u003E\u003Cp\u003EWHODAS 2.0 (12-item) QuestionnaireResponse
for John Doe. Total score: 26.\u003C/p\u003E\u003C/div\u003E"
            },
            "questionnaire":
"http://example.org/Questionnaire/WHODASQuestionnaire",
            "status": "completed",
            "subject": {
                "reference": "Patient/PatientExample"
            }
        }
    }
]

```

```

    },
    "authored": "2026-01-15T10:30:00Z",
    "author": {
      "reference": "Patient/PatientExample"
    },
    "item": [
      {
        "linkId": "whodas-s1",
        "text": "Standing for long periods such as 30 minutes",
        "answer": [
          {
            "valueCoding": {
              "extension": [
                {
                  "url":
"http://hl7.org/fhir/StructureDefinition/ordinalValue",
                  "valueDecimal": 2
                }
              ],
              "display": "Mild"
            }
          ]
        },
      },
      {
        "linkId": "whodas-s2",
        "text": "Taking care of household responsibilities",
        "answer": [
          {
            "valueCoding": {
              "extension": [
                {
                  "url":
"http://hl7.org/fhir/StructureDefinition/ordinalValue",
                  "valueDecimal": 3
                }
              ],
              "display": "Moderate"
            }
          ]
        },
      },
      {
        "linkId": "whodas-s3",
        "text": "Learning a new task, for example learning how
to get to a new place",
        "answer": [
          {
            "valueCoding": {
              "extension": [
                {
                  "url":
"http://hl7.org/fhir/StructureDefinition/ordinalValue",
                  "valueDecimal": 2
                }
              ]
            }
          ]
        },
      }
    ]
  }
}

```

```

        ],
        "display": "Mild"
    }
}
],
},
{
    "linkId": "whodas-s4",
    "text": "How much of a problem did you have joining in
community activities (for example, festivities, religious or other
activities) in the same way as anyone else can?",
    "answer": [
        {
            "valueCoding": {
                "extension": [
                    {
                        "url":
"http://hl7.org/fhir/StructureDefinition/ordinalValue",
                        "valueDecimal": 2
                    }
                ],
                "display": "Mild"
            }
        }
    ]
},
{
    "linkId": "whodas-s5",
    "text": "How much of a problem did you have with being
emotionally affected by your health problems?",
    "answer": [
        {
            "valueCoding": {
                "extension": [
                    {
                        "url":
"http://hl7.org/fhir/StructureDefinition/ordinalValue",
                        "valueDecimal": 3
                    }
                ],
                "display": "Moderate"
            }
        }
    ]
},
{
    "linkId": "whodas-s6",
    "text": "Concentrating on doing something for ten
minutes?",
    "answer": [
        {
            "valueCoding": {
                "extension": [
                    {

```

```

        "url":
"http://hl7.org/fhir/StructureDefinition/ordinalValue",
        "valueDecimal": 2
    }
    ],
    "display": "Mild"
}
}
],
},
{
    "linkId": "whodas-s7",
    "text": "Walking a long distance such as a kilometer
[or equivalent]?",
    "answer": [
        {
            "valueCoding": {
                "extension": [
                    {
                        "url":
"http://hl7.org/fhir/StructureDefinition/ordinalValue",
                        "valueDecimal": 3
                    }
                ],
                "display": "Moderate"
            }
        }
    ]
},
{
    "linkId": "whodas-s8",
    "text": "Washing your whole body?",
    "answer": [
        {
            "valueCoding": {
                "extension": [
                    {
                        "url":
"http://hl7.org/fhir/StructureDefinition/ordinalValue",
                        "valueDecimal": 1
                    }
                ],
                "display": "None"
            }
        }
    ]
},
{
    "linkId": "whodas-s9",
    "text": "Getting dressed?",
    "answer": [
        {
            "valueCoding": {
                "extension": [
                    {

```

```

        "url":
"http://hl7.org/fhir/StructureDefinition/ordinalValue",
        "valueDecimal": 1
    }
    ],
    "display": "None"
}
}
],
},
{
    "linkId": "whodas-s10",
    "text": "Dealing with people you do not know",
    "answer": [
        {
            "valueCoding": {
                "extension": [
                    {
                        "url":
"http://hl7.org/fhir/StructureDefinition/ordinalValue",
                        "valueDecimal": 2
                    }
                ],
                "display": "Mild"
            }
        }
    ]
},
{
    "linkId": "whodas-s11",
    "text": "Maintaining a friendship",
    "answer": [
        {
            "valueCoding": {
                "extension": [
                    {
                        "url":
"http://hl7.org/fhir/StructureDefinition/ordinalValue",
                        "valueDecimal": 2
                    }
                ],
                "display": "Mild"
            }
        }
    ]
},
{
    "linkId": "whodas-s12",
    "text": "Your day-to-day work",
    "answer": [
        {
            "valueCoding": {
                "extension": [
                    {

```

```

        "url":
"http://hl7.org/fhir/StructureDefinition/ordinalValue",
        "valueDecimal": 3
    }
    ],
    "display": "Moderate"
}
}
],
},
{
    "linkId": "whodas-h1",
    "text": "Overall, in the past 30 days, how many days
were these difficulties present?",
    "answer": [
        {
            "valueDecimal": 3
        }
    ]
},
{
    "linkId": "whodas-h2",
    "text": "In the past 30 days, for how many days were
you totally unable to carry out your usual activities or work because of
any health condition?",
    "answer": [
        {
            "valueDecimal": 3
        }
    ]
},
{
    "linkId": "whodas-h3",
    "text": "In the past 30 days, not counting the days
that you were totally unable, for how many days did you cut back or
reduce your usual activities or work because of any health condition?",
    "answer": [
        {
            "valueDecimal": 3
        }
    ]
},
{
    "linkId": "total-score",
    "answer": [
        {
            "valueDecimal": 26
        }
    ]
}
]
}
},
{

```

```

        "fullUrl":
"http://example.org/QuestionnaireResponse/BHPGICExample",
        "resource": {
            "resourceType": "QuestionnaireResponse",
            "id": "BHPGICExample",
            "meta": {
                "profile": [
                    "http://hl7.org/fhir/us/core/StructureDefinition/us-
core-questionnaireresponse|6.1.0"
                ]
            },
            "language": "en",
            "text": {
                "status": "generated",
                "div": "\u003Cdiv xml:lang=\"en\"
xmlns=\"http://www.w3.org/1999/xhtml\" lang=\"en\" \u003E\u003Cp\u003E
PGIC
QuestionnaireResponse for John Doe. Response: Very much
improved.\u003C/p\u003E\u003C/div\u003E"
            },
            "questionnaire":
"http://example.org/Questionnaire/BHPGICQuestionnaire",
            "status": "completed",
            "subject": {
                "reference": "Patient/PatientExample"
            },
            "authored": "2026-01-15T10:30:00Z",
            "author": {
                "reference": "Patient/PatientExample"
            },
            "item": [
                {
                    "linkId": "pgic-1",
                    "text": "Since beginning treatment, how would you
describe your overall change in activity limitations, symptoms, emotions,
and quality of life?",
                    "answer": [
                        {
                            "valueCoding": {
                                "extension": [
                                    {
                                        "url":
"http://hl7.org/fhir/StructureDefinition/ordinalValue",
                                        "valueDecimal": 1
                                    }
                                ],
                                "display": "Very much improved"
                            }
                        ]
                    }
                ]
            ]
        }
    ]
}

```

```

    }
  ]
}

```

## Response:

```

HTTP/1.1 202 Accepted
Content-Location: https://[base]/access/Patient/$submission-status/sub-789012

```

**Step 2-5:** Follow the same polling pattern as MSK example above until receiving success response.

### API-Specific Error Scenarios

For general error handling guidance and common error scenarios (missing parameters, invalid tokens, server errors, etc.), see the Error Handling section in General Guidance.

The following errors are specific to the Data Reporting API:

#### Patient Not Aligned

**Scenario:** Attempting to report data for a patient who is not currently aligned to this provider.

```

GET https://[base]/access/Patient/$submission-status/sub-123456

HTTP/1.1 200 OK
Content-Type: application/fhir+json

{
  "resourceType": "Parameters",
  "parameter": [
    {
      "name": "result",
      "valueCodeableConcept": {
        "coding": [
          {
            "system":
"https://globalalliantinc.com/access/CodeSystem/ACCESSReportDataResultCS"
,
            "code": "patient-not-aligned",
            "display": "Patient not aligned"
          }
        ],
        "text": "Patient is not currently aligned to this participant in
the specified track. Data reporting is only accepted for aligned
patients."
      }
    }
  ]
}

```



### Missing Required Data Bundle Parameter

**Scenario:** Data reporting request submitted without the required dataBundle parameter.

POST https://[base]/access/Patient/\$report-data?entityId=ACCES67890

HTTP/1.1 400 Bad Request

Content-Type: application/fhir+json

```
{
  "resourceType": "OperationOutcome",
  "issue": [
    {
      "severity": "error",
      "code": "required",
      "details": {
        "text": "Missing required parameter: dataBundle"
      },
      "diagnostics": "At least one 'dataBundle' parameter containing an
ACCESS Data Reporting Bundle is required. The bundle must conform to the
ACCESSDataReportingBundle profile."
    }
  ]
}
```

### Invalid Composition Structure

**Scenario:** Data reporting bundle contains an invalid Composition structure.

HTTP/1.1 400 Bad Request

Content-Type: application/fhir+json

```
{
  "resourceType": "OperationOutcome",
  "issue": [
    {
      "severity": "error",
      "code": "structure",
      "details": {
        "text": "Invalid Composition structure in data reporting bundle"
      },
      "diagnostics": "The Composition resource must include the dateTime
data element."
    }
  ]
}
```

### Missing Required Track Section

**Scenario:** Composition is missing required section for the specified track.

HTTP/1.1 400 Bad Request

Content-Type: application/fhir+json

```
{
  "resourceType": "OperationOutcome",
  "issue": [
    {
      "severity": "error",
      "code": "required",
      "details": {
        "text": "Missing required track section in Composition"
      },
      "diagnostics": "For submitting data to the BH track, the
Composition must include the BH section."
    }
  ]
}
```

### Invalid Resource Reference

**Scenario:** Composition references resources that are not included in the bundle.

HTTP/1.1 400 Bad Request

Content-Type: application/fhir+json

```
{
  "resourceType": "OperationOutcome",
  "issue": [
    {
      "severity": "error",
      "code": "invariant",
      "details": {
        "text": "Invalid resource reference in Composition"
      },
      "diagnostics": "Composition.section[blood-pressure].entry
references 'Observation/bp-reading-123' but this resource is not included
in the Bundle. All referenced resources must be included in the bundle."
    }
  ]
}
```

### Multiple Track Sections Submitted

**Scenario:** Composition has more than one track section specified.

HTTP/1.1 400 Bad Request

Content-Type: application/fhir+json

```
{
  "resourceType": "OperationOutcome",
  "issue": [
    {
      "severity": "error",
      "code": "invariant",
      "details": {
```

```

        "text": "Invalid resource reference in Composition"
      },
      "diagnostics": "Composition contains data for both the CKM and BH
tracks. Only data for one track can be provided.
    }
  ]
}

```

### Duplicate Data Submission

**Scenario:** Attempting to submit data that was already reported in a previous submission.

HTTP/1.1 200 OK

Content-Type: application/fhir+json

```

{
  "resourceType": "Parameters",
  "parameter": [
    {
      "name": "result",
      "valueCodeableConcept": {
        "coding": [
          {
            "system":
"https://globalalliantinc.com/access/CodeSystem/ACCESSReportDataResultCS"
,
            "code": "duplicate",
            "display": "Duplicate submission"
          }
        ],
        "text": "Data submission is a duplicate of a previously submitted
report for this reporting period."
      }
    }
  ]
}

```

### Incomplete Data Submission

**Scenario:** Attempting to submit data that was already reported in a previous submission.

HTTP/1.1 200 OK

Content-Type: application/fhir+json

```

{
  "resourceType": "Parameters",
  "parameter": [
    {
      "name": "result",
      "valueCodeableConcept": {
        "coding": [
          {
            "system":

```

```

"https://globalalliantinc.com/access/CodeSystem/ACCESSReportDataResultCS"
,
    "code": "incomplete-data",
    "display": "Data submission is missing required observations
or measurements for the specified track (see the 'issues' parameter for
specific issues and details).",
    }
],
"text": "PHQ-9 questionnaire response has more than two missing
answers."
}
},

{
    "resourceType": "OperationOutcome",
    "issue": [
        {
            "severity": "warning",
            "code": "incomplete",
            "expression": "Composition.section.where(title =
'BH').section.where(title = 'PHQ-9')",
        }
    ]
}
]
}

```

**NOTE:** These scenarios represent a mix of business logic results (HTTP 200 with specific result codes) and technical validation errors (HTTP 400). Always check the HTTP status code first to determine the type of response.

## Subscriptions Guidance

The ACCESS Model provides automatic email notifications for important events during the patient ACCESS Model lifecycle.

### Overview

Upon successful patient alignment, participants are automatically enrolled to receive email notifications about upcoming deadlines and events that require action or awareness. These email notifications are automatically stopped when a patient is unaligned, either manually or through the ACCESS Model system.

### Event Types

The following event types trigger email notifications to ACCESS participants:

| Event Name                            | Event Code                 | Description                                                                                                                             | When Triggered                                  |
|---------------------------------------|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|
| <b>Provider Lock-In Period Ending</b> | provider-lock-ending       | The 90-day provider lock-in period is ending soon, allowing the patient to switch to a different provider if desired                    | Before the lock-in period ends                  |
| <b>Control Group Period Ending</b>    | control-group-ending       | The 12-month control group assignment period is ending soon, allowing the patient to be reconsidered for alignment in the ACCESS Model  | Before control group period ends                |
| <b>Alignment Renewal Due</b>          | alignment-renewal          | The initial alignment period is ending and renewal action is required to continue services under the ACCESS Model                       | Before initial alignment period expires         |
| <b>Baseline Reporting Due</b>         | baseline-reporting-window  | The baseline data report is due soon. If baseline reporting is not received within 60 days of alignment, the patient will be unaligned. | When baseline reporting deadline is approaching |
| <b>Quarterly Reporting Due</b>        | quarterly-reporting-window | The quarterly data report is due soon. Quarterly reporting must be submitted 70 to 110 days after the prior data submission.            | When quarterly reporting window opens           |

| Event Name                       | Event Code           | Description                                                                                                                                                               | When Triggered                                       |
|----------------------------------|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------|
| <b>End-of-Year Reporting Due</b> | eoy-reporting-window | The end-of-year data report is due soon. End-of-year reporting must be submitted no later than 425 days from the date of alignment (365 days plus an additional 60 days). | When end-of-year reporting window opens              |
| <b>Unalignment Notice</b>        | unalignment          | The patient has been unaligned from the ACCESS Model due to a change in eligibility status.                                                                               | When patient is unaligned by the ACCESS Model system |

### Email Notification Enrollment

Email notifications are automatically configured during the participant onboarding process. The email addresses registered during onboarding will receive notifications for all relevant events related to aligned patients.

#### Key Points:

- Email addresses are configured during participant onboarding
- Notifications are sent automatically for all events
- No additional configuration needed during alignment

#### What Happens After Alignment:

Upon successful alignment (HTTP 200 with result code `aligned` or `aligned-switch-approved`), the system automatically:

1. Begins monitoring for upcoming events that require notification
2. Sends email notifications to registered addresses when events occur
3. Continues notifications throughout the alignment period
4. Stops notifications when the patient is unaligned

You do not need to manually configure or manage these notifications - they are handled automatically by the system based on your registered email addresses.

### Email Notification Content

Email notifications include:

- **Event Type:** Clear description of what is happening (e.g., "Participant Lock-In Period Ending")
- **Submission ID:** Submission ID from the successful alignment for the patient, provided in the Beneficiary Alignment Report

- **Participant ID:** Your ACCESS participant identifier
- **Deadline Date:** When action is required or event occurs
- **Action Required:** Specific steps you need to take
- **Additional Information:** Relevant details and guidance

### *Example Email Notification*

Subject: ACCESS Model Notification - Participant Lock-In Period Ending

Dear ACCESS Participant,

Event: Participant Lock-In Period Ending

Submission ID: sub-123456

Participant ID: ACCES12345

Deadline Date: May 1, 2026

Action Required:

The 90-day participant lock-in period for this patient ends on May 1, 2026.

After this date, the patient may switch to another ACCESS participant if they choose.

Additional Information:

Please contact the patient to discuss their care options and ensure they are aware of this upcoming change.

If you have questions, please contact ACCESS Model support at [ACCESSModelTeam@cms.hhs.gov](mailto:ACCESSModelTeam@cms.hhs.gov).

## Notification Lifecycle

### *Enrollment*

- Automatic upon successful alignment
- Uses email addresses from participant onboarding
- Notifications sent for all relevant events

### *Active Period*

- Remains active for duration of alignment
- Triggers email notifications at appropriate times

### *Cancellation*

- Automatic upon successful unalignment
- No further notifications sent for that patient
- Participant continues to receive notifications for other aligned patients

## Implementation Guidance

### *Receiving Notifications*

Your email system should:

- Accept emails from ACCESS Model notification service
- Configure spam filters to allow ACCESS emails
- Set up forwarding rules if needed for team distribution
- Monitor inbox regularly for time-sensitive notifications

### *Processing Notifications*

When you receive a notification email:

1. **Review Event Details** - Understand what is happening
2. **Check Deadline** - Note when action is required
3. **Update Internal Systems** - Record event in your system
4. **Take Appropriate Action** - Follow up with patient if needed
5. **Complete Required Actions** - Perform required tasks before deadline

### *Security Considerations*

- Email notifications contain limited patient information for privacy
- Use secure email systems with encryption
- Train staff on handling Personally Identifiable Information (PII) and/or Protected Health Information (PHI) in emails
- Follow your organization's email security policies
- Do not forward notification emails outside your organization

### *Related Artifacts*

- `ACCESSEventTypeCS` CodeSystem
- `ACCESSEventTypeVS` ValueSet



## Testing Guide

This page provides detailed testing guidance to ensure implementations conform to ACCESS Model requirements before production deployment.

### Testing Checklist

Before deploying to production, implementers should complete the following validation steps:

#### FHIR Conformance Validation:

- All resource instances validate against their declared profiles
- All Must Support elements are properly populated when data is available
- All required elements are present in submissions
- Value set bindings use codes from the correct code systems
- Patient resources include valid MBI in correct format

#### Operation Testing:

- All four operations (`$check-eligibility`, `$align`, `$unalign`, `$report-data`) successfully submit requests
- `Content-Location` header is correctly parsed and used for polling
- `$submission-status` correctly returns HTTP 202 Accepted while processing
- `$submission-status` correctly returns HTTP 200 OK with result when complete
- All result codes are correctly handled

#### Authentication & Security:

- OAuth 2.0 token acquisition succeeds
- Token is correctly included in Authorization header
- Token refresh works before expiration
- TLS 1.3 is enforced
- MBI and PII/PHI are encrypted in transit and storage

#### Error Handling:

- HTTP 400 errors are handled without retry
- HTTP 401 errors trigger token refresh
- HTTP 500/503 errors implement exponential backoff retry
- Network errors implement retry logic
- Timeout scenarios are handled gracefully

#### Asynchronous Pattern:

- Polling implements appropriate wait intervals (5-30 seconds)
- Polling includes timeout mechanism
- Submission IDs are stored and associated with original requests
- Polling continues after network interruption

## Test Scenarios by API

### Eligibility API Test Scenarios

| Scenario          | Input                                   | Expected Result                                 | Validation                         |
|-------------------|-----------------------------------------|-------------------------------------------------|------------------------------------|
| Eligible Patient  | Valid patient with qualifying diagnosis | HTTP 200, result "eligible"                     | Can proceed to alignment           |
| Control Group     | Patient randomized to control           | HTTP 200, result "not-eligible-control-group"   | 12-month wait documented           |
| Already Aligned   | Patient aligned to another provider     | HTTP 200, result "not-eligible-already-aligned" | Cannot align                       |
| Missing Parameter | Request without patient                 | HTTP 400, OperationOutcome                      | Error identifies missing parameter |
| Invalid Track     | Track code not in value set             | HTTP 400, OperationOutcome                      | Error identifies invalid track     |

### Alignment API Test Scenarios

| Scenario                | Input                                       | Expected Result                                | Validation                          |
|-------------------------|---------------------------------------------|------------------------------------------------|-------------------------------------|
| Successful Alignment    | Eligible patient with conditions            | HTTP 200, result "aligned"                     | Subscriptions created               |
| Provider Switch         | Alignment with switch consent after 90 days | HTTP 200, result "aligned-switch-approved"     | Previous subscriptions cancelled    |
| Lock-In Active          | Switch attempt within 90 days               | HTTP 200, result "not-aligned-already-aligned" | Switch rejected                     |
| Missing Conditions      | Alignment without conditions                | HTTP 400, OperationOutcome                     | Error identifies missing conditions |
| No Qualifying Diagnosis | Conditions don't match track                | HTTP 200, result "not-aligned-diagnoses"       | Specific requirements noted         |

### Unalignment API Test Scenarios

| Scenario               | Input                               | Expected Result                              | Validation                  |
|------------------------|-------------------------------------|----------------------------------------------|-----------------------------|
| Successful Unalignment | Valid reason (geographic-relocated) | HTTP 200, result "unaligned"                 | Subscriptions cancelled     |
| Manual Review          | Unalignment requiring review        | HTTP 200, result "unalignment-pending"       | Subscriptions remain active |
| Not Aligned            | Patient not currently aligned       | HTTP 200, result "not-unaligned-not-aligned" | Cannot unalign              |

|                       |                               |                               |                                    |
|-----------------------|-------------------------------|-------------------------------|------------------------------------|
| <b>Missing Reason</b> | Unalignment<br>without reason | HTTP 400,<br>OperationOutcome | Error identifies<br>missing reason |
|-----------------------|-------------------------------|-------------------------------|------------------------------------|

## Validation Tools and Resources

### FHIR Validation:

- Online validator: <https://validator.fhir.org/>